

CLINICAL OBSTETRICS

NOTES

FOURTH EDITION

PRE-SUMMARIZED
READY-TO-STUDY
HIGH-YIELD NOTES

FOR THE TIME-POOR
MEDICAL, PRE-MED,
USMLE OR PA STUDENT



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A Message From Our Team

Studying medicine or any health-related degree can be stressful; believe us, we know from experience! The human body is an incredibly complex organism, and finding a way to streamline your learning is crucial to succeeding in your exams and future profession. Our goal from the outset has been to create the greatest educational resource for the next generation of medical students, and to make them as affordable as possible.

In this fourth edition of our notes we have made a number of text corrections, formatting updates, and figure updates which we feel will enhance your study experience. We have also endeavoured to use only open-source images and/or provide attribution where possible.

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What's included: Ready-to-study anatomy, physiology and pathology notes of the relevant clinical obstetrics topics presented in succinct, intuitive and richly illustrated downloadable PDF documents. Once downloaded, you may choose to either print and bind them, or make annotations digitally on your iPad or tablet PC.

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OBSTETRIC DEFINITIONS

OBSTETRIC DEFINITIONS

Births, Miscarriages & Abortions

Abortion	Termination of pregnancy (Spontaneous/Intentional) @ <20wks gestation OR <500g
Birth	Complete expulsion of the foetus >20wks OR >500g (Irrespective of placenta)
Miscarriages	Missed Miscarriage: Asymptomatic Intrauterine Foetal Death (IUFD) Threatened Miscarriage: Uterine bleeding +/- Contractions (Note: WITHOUT Labour) Inevitable Miscarriage: Uterine bleeding + Contractions + Dilation (Ie: WITH labour) Incomplete Miscarriage: An ACTIVE miscarriage with retained products Complete Miscarriage: An INACTIVE miscarriage with complete expulsion
Neonatal Death	Early Neonatal Death: Death of baby within 1wk of birth Late Neonatal Death: Death of baby within 4wks of birth
Still Birth	A "Birth" (Ie: >20wks &/or >500g) of a baby showing no signs of life. (As opposed to Live Birth – Has spontaneous breathing/heartbeat/movement)
TOP – Termination of Pregnancy	Medical TOP – Stat Dose Mifepristone/RU486 → 48hrs later – Vaginal Misoprostol Surgical TOP – Dilation & Curettage / Manual evacuation of foetus.
Viability	Gestation of >24wks Or birthweight of >500g

Timing:

Trimesters	T1: 0-12wks T2: 12-28wks T3: 28-40wks
Pre-Term	Pre-Term: <37wks
Term	Term: 37-42wks
Post-Term	Post Term: >42wks
Perinatal Period	Time within 28days of a "Birth". (Ie: Does NOT include Abortions)

Counting Babies:

Gravidity	Number of pregnancies (including current pregnancy) (Nulligravida – Never been pregnant) (Primigravida – First pregnancy)
Parity	Number of births @ >20wks gestation (Including Stillborns) (Nullipara – Never carried a pregnancy to >20wks) (Primipara – 1 previous "birth") (Multipara - >1 previous "births") (Grand-Multipara - >5 previous "births") (Great-Grand-Multipara - >10 previous "births")
Twins – Chorionicity & Amnionicity	Chorionicity = # of Placentas (Monochorionic = Single; Dichorionic = Double; etc) Amnionicity = # of Amnionic Sacs (Monoamnionic = Single; Diamnionic = Double)

Antenatal Screening Anomalies:

AFI – Amniotic Fluid Index	Sum of the amniotic fluid depth (cm) of the largest vertical pockets in each of the 4 uterine quadrants. (N=8-24cm) (<8= Oligohydramnios = Not Enough Amniotic Fluid) (>24= Polyhydramnios = Too Much Amniotic Fluid)
GBS	Group B Streptococcus
GDM	Gestational Diabetes Mellitus

Growth Anomalies:

CPD – Cephalo-Pelvic Disproportion	Foetal head is too big for the maternal pelvis
IUGR – Intrauterine Growth Restriction	Failure to reach genetic growth potential. (Including A plateau/decline in growth velocity) Note: Not necessarily “small for gestational age” (SGA)
LBW – Low Birth Weight	LBW: Low Birth Weight <2500g VLBW: Very low birth weight <1500g ELBW: Extremely low birth weight <1000g
SGA/SFD	Small for Gestational Age/Small for Dates = Foetus <10 th percentile for Gestational Age. (Note: NOT necessarily IUGR)

Bleeding:

APH – Antepartum Haemorrhage	PV bleeding after 24wks gestation (Including During Labour).
PPH: Post-Partum Haemorrhage	Primary PPH: PV Bleeding >500mL WITHIN 24hrs of Delivery Secondary PPH: PV Bleeding WITHIN 6wks of Delivery
Placenta Praevia	A persistent low-lying placenta after 24wks gestation. (<24wks, it is a “Low-lying placenta”)

Lie, Presentation, Attitude & Position:

Foetal Lie	Longitudinal or Transverse (Occasionally Oblique, but will → either Long/Trans during labour)
Foetal Presentation	The presenting part of the foetal body @ the birth canal. - Cephalic or Breech (If Longitudinal Lie) - Shoulder (If Transverse Lie)
Foetal Attitude/Posture	Relation of the Foetal parts to each other
Foetal Position	Right or Left side of Uterus
Breech Presentation	1. Frank Breech – Bum first, legs extended (“Pike”) 2. Complete Breech – Bum first, legs flexed (“Tuck”) 3. Footling Breech – Foot first (one or two) – (le: At least one hip extended)

Membrane Rupture:

ARM	Artificial rupture of membranes (using Amniohook)
PROM/PPROM	PROM: Premature Rupture of Membranes = ROM @>37wks, >24hrs <i>prior</i> to Onset of Labour PPROM: Preterm Premature Rupture of Membranes = PROM @ <37wks without onset of labour.
SROM	Spontaneous Rupture of Membranes (with or without labour) (Note: Can be a PROM or PPROM)

Labour/Delivery:

Engagement	Descent of the presenting part of the foetus into the mother's pelvis. Eg: If Cephalic, the head may be 4/5ths above the brim.
Labour	Regular Painful Contractions + Cervical Dilatation/Effacement + Descent of the Foetus - Stage 1: Onset of labour → Fully Dilated Cervix - Stage 2: Fully Dilated Cervix → Birth of Foetus - Stage 3: Birth → Delivery of Placenta & Membranes
Pre-Term Term Post-Term	Pre-Term: <37wks Term: 37-42wks Post Term: >42wks
LUCS	Lower-Uterine Caesarean Section (Lower horizontal incision) (Cf. Classical Caesarean – Midline Incision)
SVD/SVB	Spontaneous Vaginal Delivery/Birth
Trial of Scar/VBAC (Vaginal Birth After Caesar)	Trial of Scar = Attempt at vaginal delivery after Caesarean Section VBAC = <i>Successful</i> Vaginal Delivery after CS

Perinatal:

IRDS – Infant Respiratory Distress Syndrome	(AKA: “Hyaline Membrane Disease”) Signs of increased respiratory effort in a baby due to insufficient surfactant &/or structural immaturity of the lungs.
Perinatal Period	Time within 28days of a “Birth”. (Ie: Does NOT include Abortions)

EMBRYONIC SEXUAL DEVELOPMENT

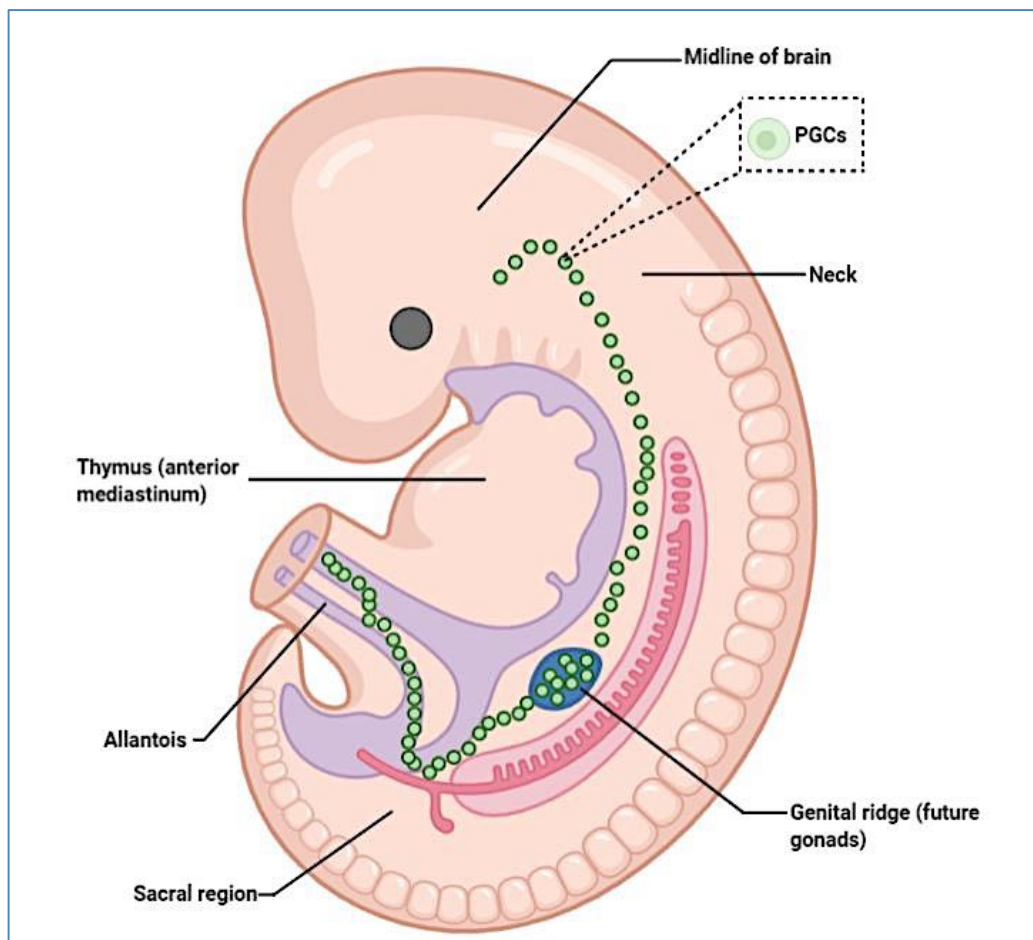
EMBRYONIC SEXUAL DEVELOPMENT

Gametogenesis:

- The formation of gametes (spermatogenesis / oogenesis)
- Largely regulated by hormonal/endocrine system
- A process of **meiosis**
- **Male:** Mitotic division of Spermatogonia and entry to meiosis is continuous from onset at puberty until death
- **Female:** Meiotic division is discontinuous – begins in embryo, lies dormant once born & is completed upon fertilisation by a sperm
 - Events are cyclic between puberty and menopause – **menstrual cycle**
- Gametes are *haploid* cells ($\frac{1}{2}$ Chromosome number – 23 –)
 - 22 pairs of somatic chromosomes
 - 1 pair of sex chromosomes (X, Y)
 - Sperm contains X & Y
 - Egg only contains X & X
 - **Combinations:**
 - XY = produces a male
 - XX = produces a female
 - **Y is the sex determining chromosome**

Embryo Sexual Differentiation:

- During Wk 5 of embryonic growth gonadal tissue develops into a **gonadal ridge**
 - Primordial germ cells migrate into gonadal ridge which then develop into ovaries or testes depending on DNA
 - Mesonephric (Wolffian) ducts = future male ducts
 - Paramesonephric (Mullerian) ducts = future female ducts
 - Both empty into a common chamber = the **cloaca**
 - At this stage the embryo is said to be sexually indifferent as the gonadal ridge can develop either way

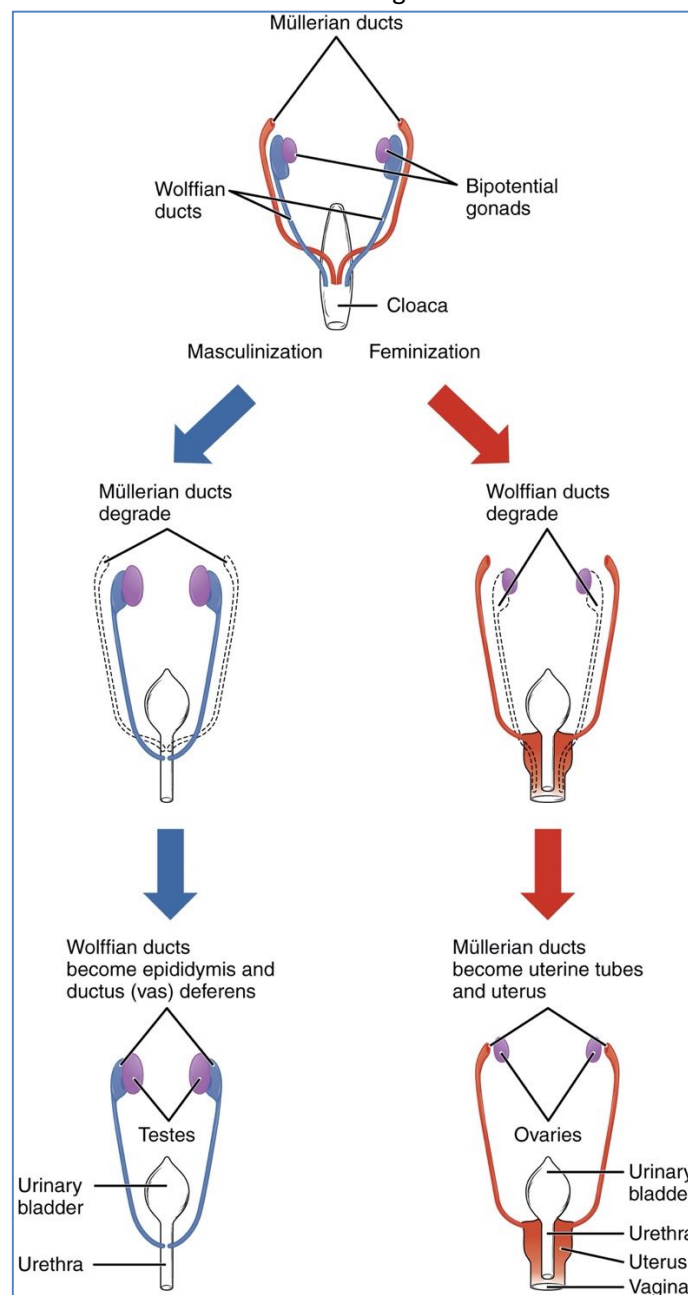


- **Male:**

- Y = The **sex determining** chromosome - causes the gonadal ridge to develop into testes (Ie: Females are the default sex)
- Testes then produce testosterone
 - Testosterone causes the Mesonephric (Wolffian) duct to develop into the male tract
- Also produces AMH (anti-mullerian hormone)/MIH (mullerian inhibiting hormone) which causes the Paramesonephric (Mullerian) duct to degenerate
- External genitalia:
 - Penis develops from the genital tubercle
 - Scrotum develops from the labiascrotal swellings

- **Female:**

- By default, if there is no 'Y' chromosome → no testes, the female tract will develop
 - The gonadal ridge develops into immature ovaries
 - Without testosterone, the Mesonephric (Wolffian) ducts will degenerate
 - The cortical (outer) part of the immature ovaries forms follicles
 - Without AMH/MIH, the paramesonephric ducts differentiate into structures of the female duct system
- External genitalia:
 - glans clitoris develops from the genital tubercle
 - Labia minora forms from the urethral groove

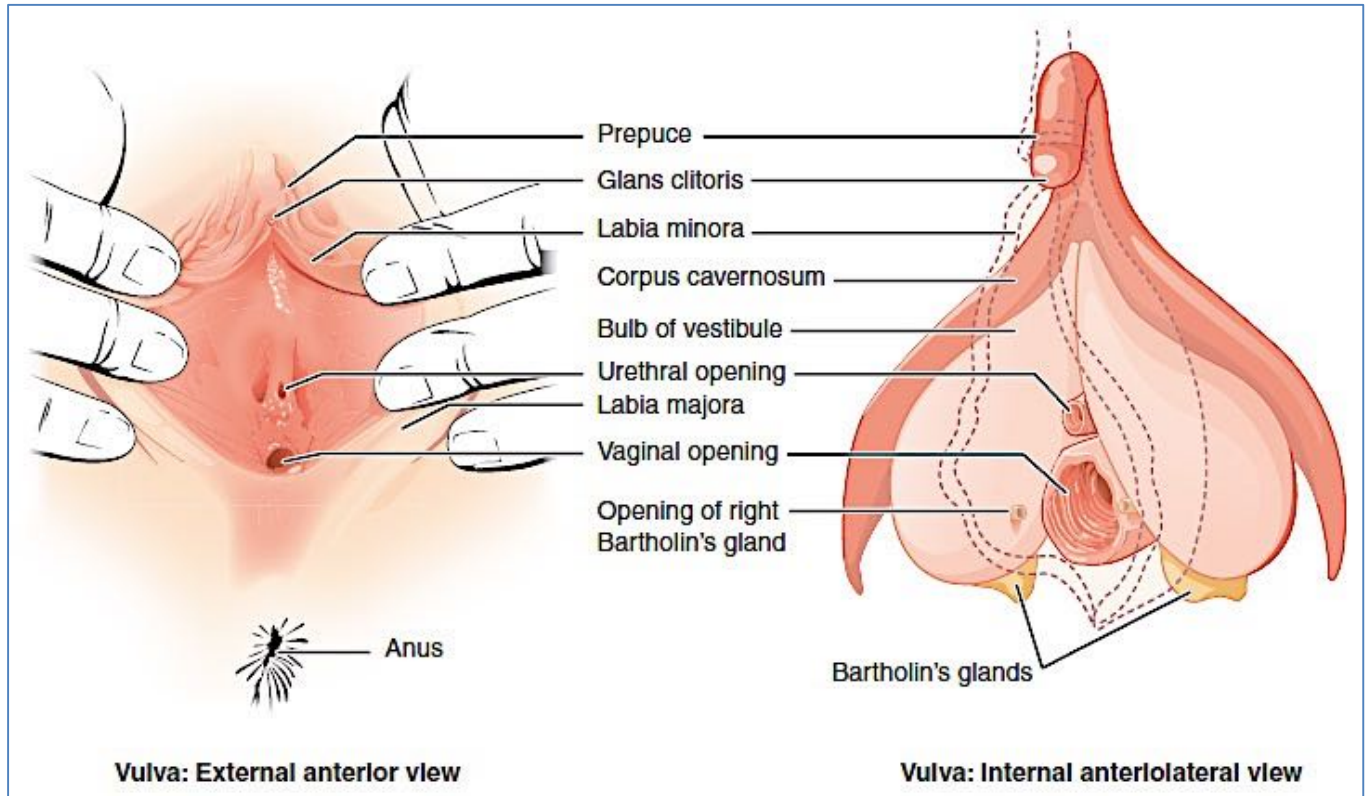


REVIEW OF BASIC FEMALE REPRODUCTIVE ANATOMY

REVIEW OF BASIC FEMALE REPRODUCTIVE ANATOMY

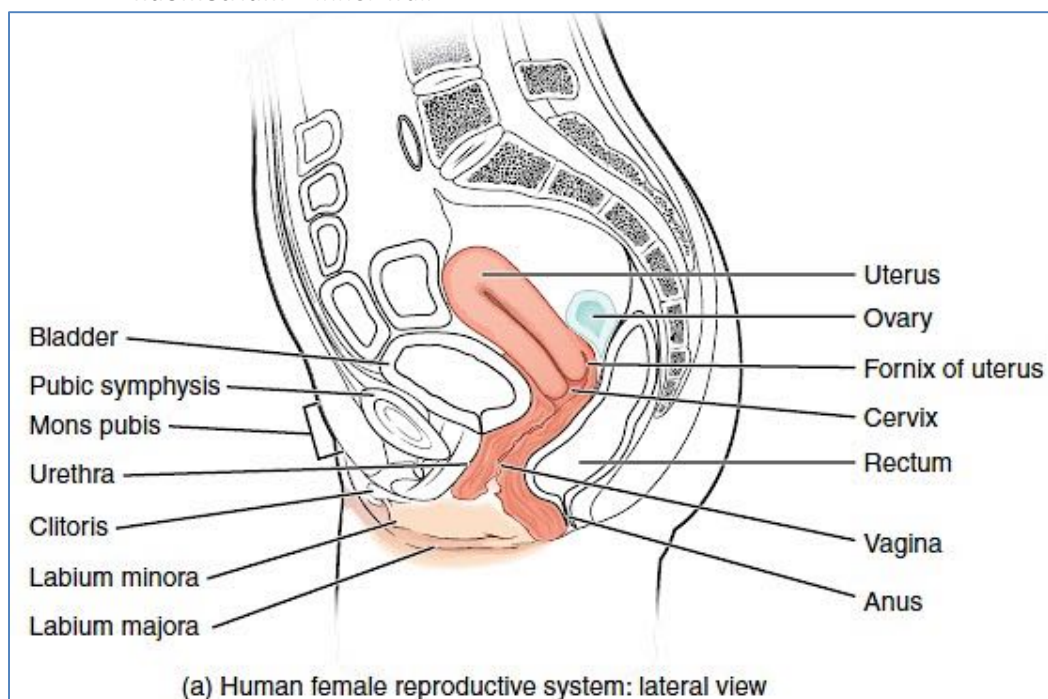
Review of Female Reproductive Structures:

- **Anatomy:**
 - **Vagina/Vulva:**
 - Labia Majora & Minora
 - Clitoris & prepuce of clitoris
 - Urethral orifice

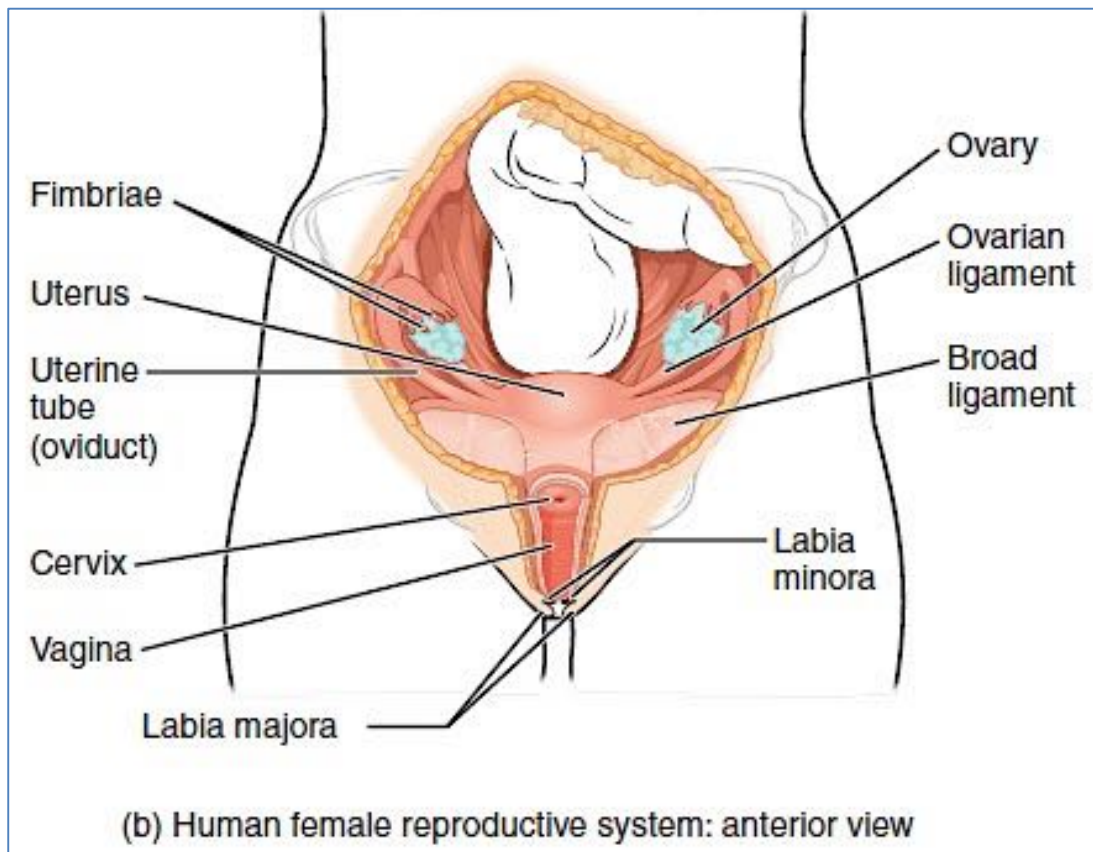


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- **Uterus** - Fundus (top / head), Body, Cervix (external os, canal, internal os), Lumen (internal cavity)
 - **Perimetrium** – Outer wall
 - **Myometrium** – Middle of wall
 - **Endometrium** – Inner wall

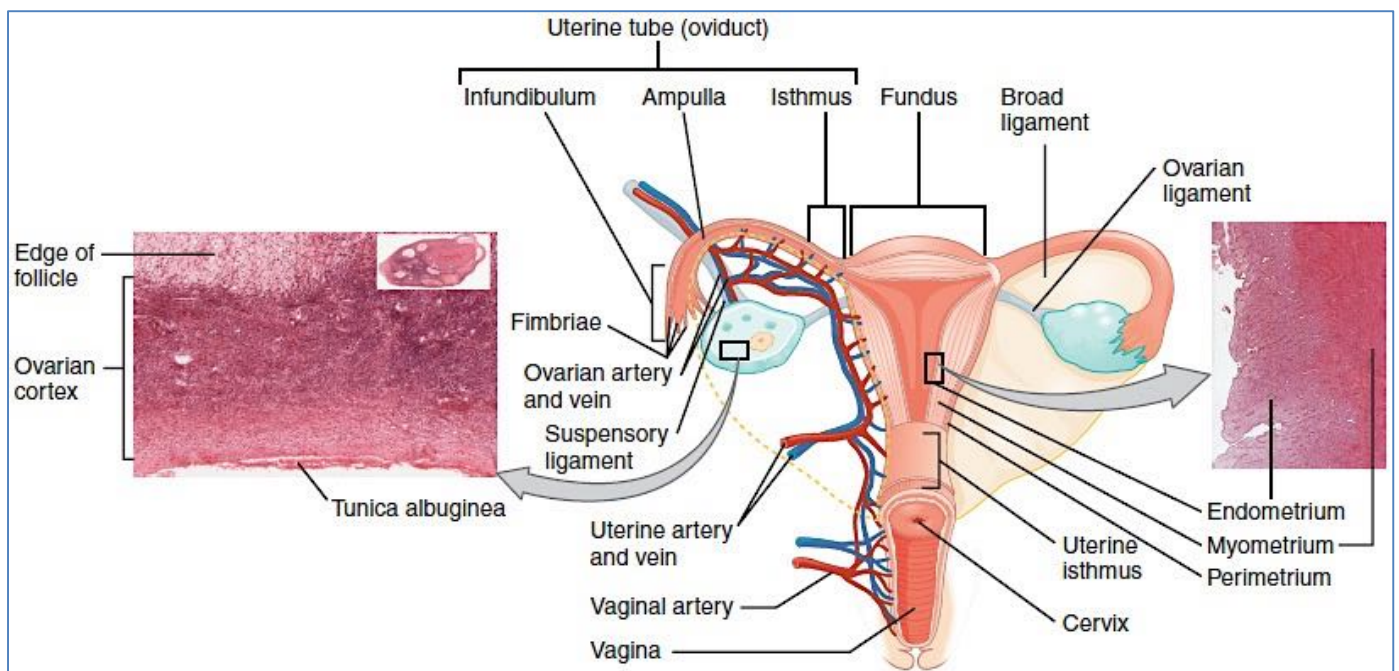


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- **Uterine (fallopian) Tubes**
 - Common site of fertilisation
 - Infundibulum – projections = fimbriae (closest to ovary) → Receives oocyte
- **Ovaries (gonads)**
 - Produce female gametes (oocytes)
 - Secrete female sex hormones – (Oestrogen & Progesterone)
 - Held in place by ligaments & muscles

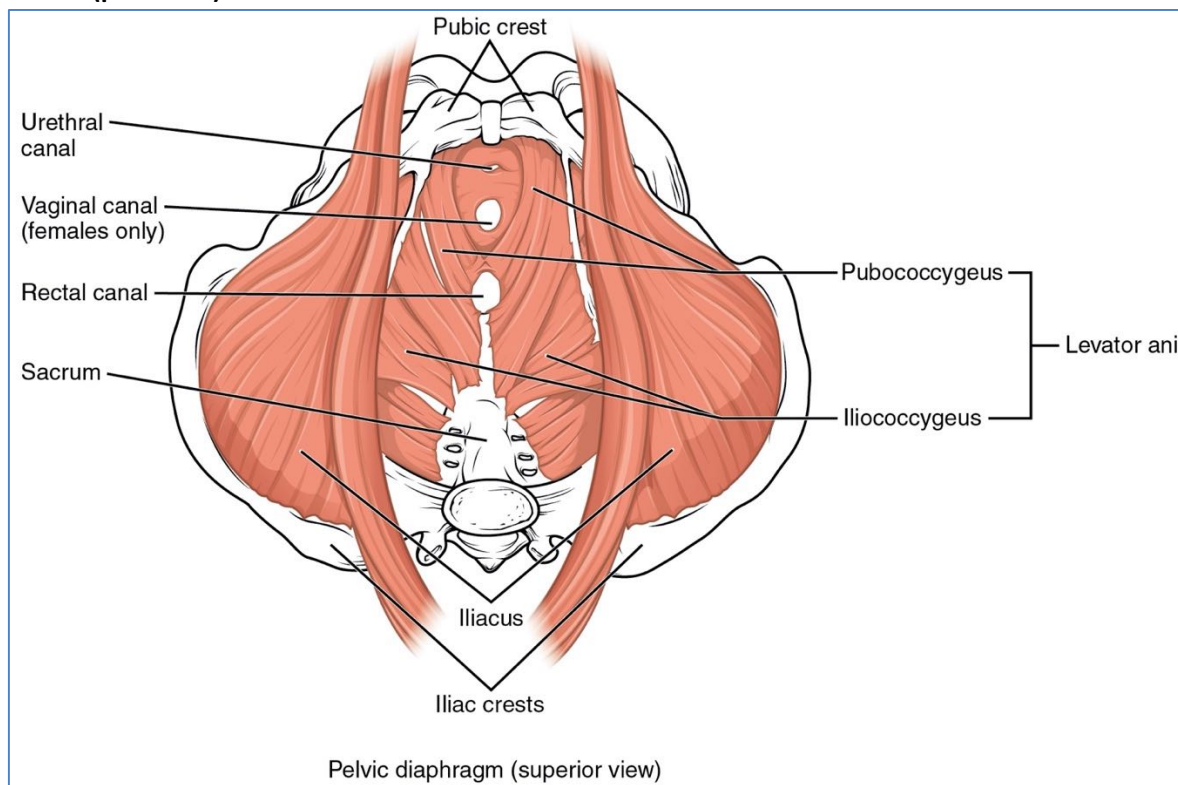


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- **Blood Supply:**
 - **Internal iliac artery:**
 - Branches from common iliac artery
 - Uterine Artery
 - Vaginal Artery
 - To external genitalia
 - **Ovarian Artery:**
 - To ovaries, uterine tubes and uterus

- **Pelvic Diaphragm:**

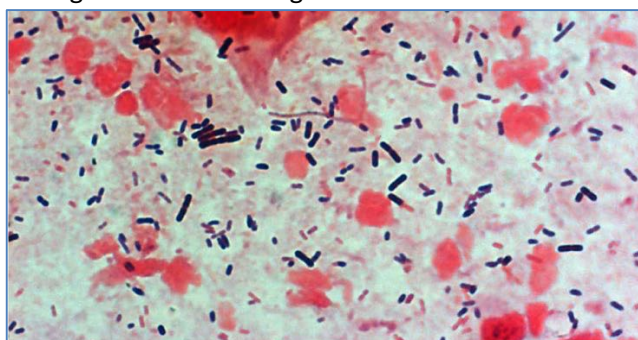
- **Levator Ani (anterior half)**
 - Iliococcygeus
 - Pubococcygeus
- **(posterior) Coccygeus** (ischiococcygeus)
- **(posterior) Piriformis**



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Normal Flora of the Genital Tract

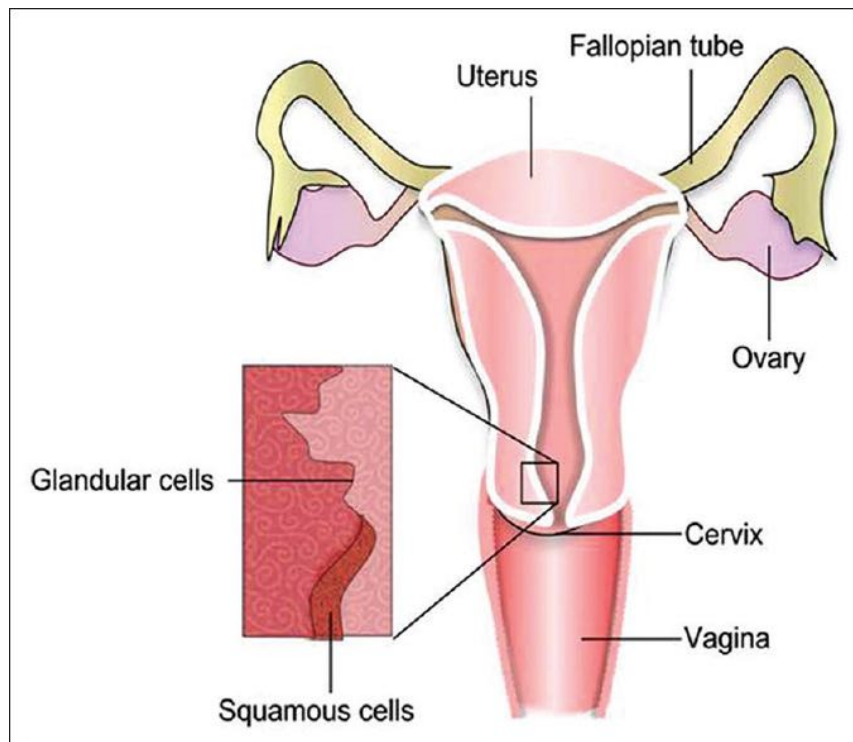
- **Male:**
 - **Urethra** – Few Organisms - (*Staph epidermidis*, *Streptococci*, *Uroplasma urealyticum*)
- **Female:**
 - **Vagina** – High Numbers of Bacteria – (*Lactobacillus* - Blue Gram Positive Rods, + Some Anaerobes)
 - → Produce lactic acid
 - → Protects against Bacterial Vaginosis & Yeast Infections



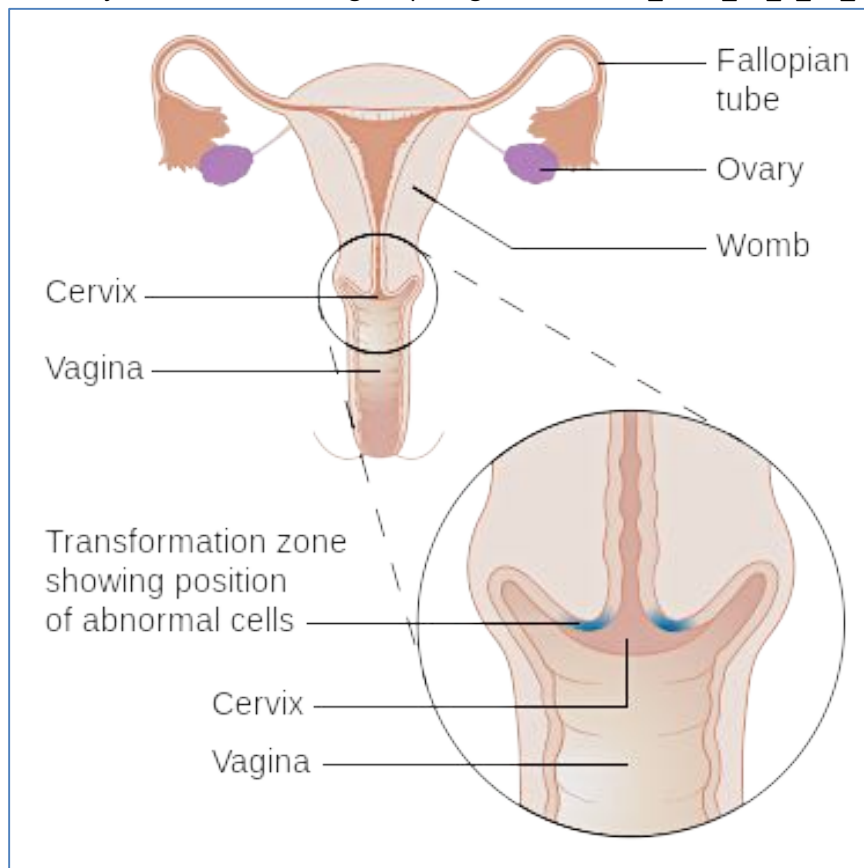
https://embryology.med.unsw.edu.au/embryology/index.php?title=File:Bacteria_-_gram-stained_vaginal_smear_05.jpg

Background Information on the Cervix:

- **Note: The Transformation Zone – Commonest location of Cervical Cancer**
 - TZ = The location of Transition from Squamous to Columnar Epithelium
 - **Note:** During puberty, Columnar Epithelium Migrates out of the os → Exposed to Vaginal Acidity → Metaplasia to Squamous Epithelium
 - This is the area Predisposed to Cancer

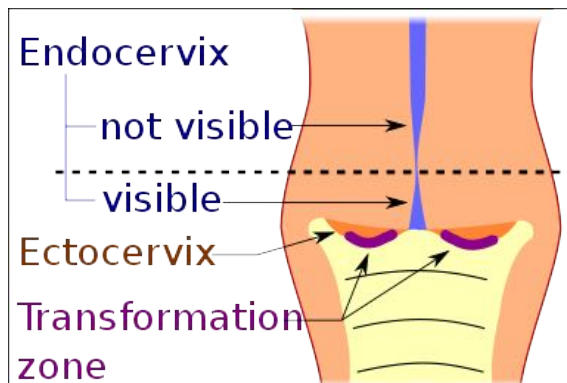


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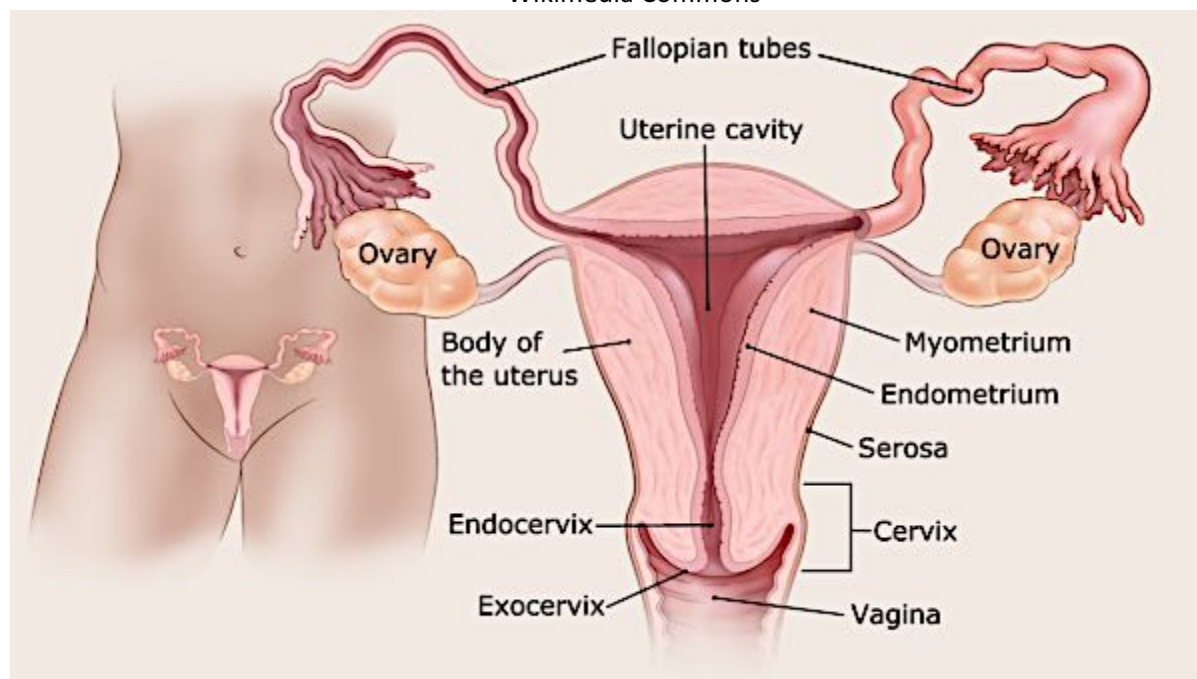


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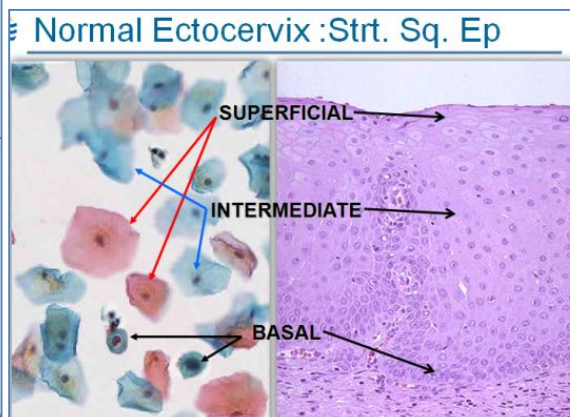
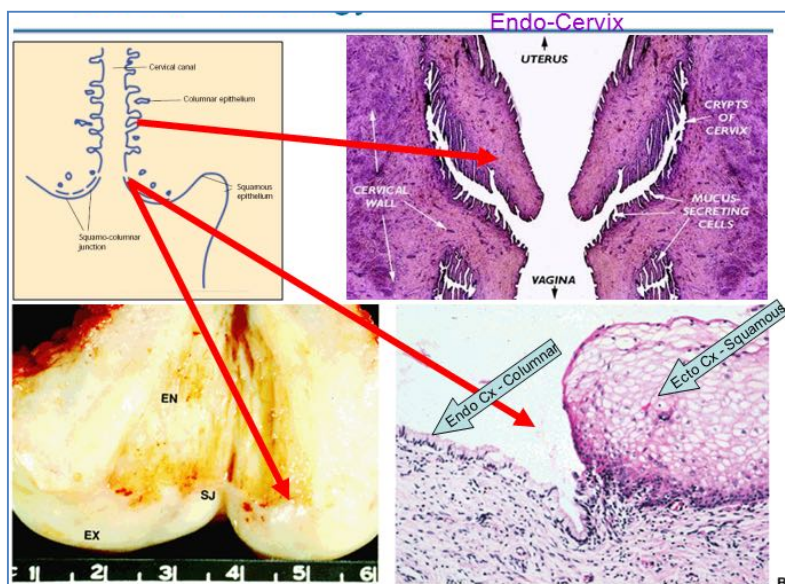
- **Note: The Normal Cervix – Anatomy & Histology:**
 - **Endocervix** = Simple Columnar Epithelium
 - **Ectocervix** = Stratified Squamous Epithelium



Mikael Häggström, M.D. - Author info - Reusing images- Conflicts of interest: None Mikael Häggström, CC0, via Wikimedia Commons

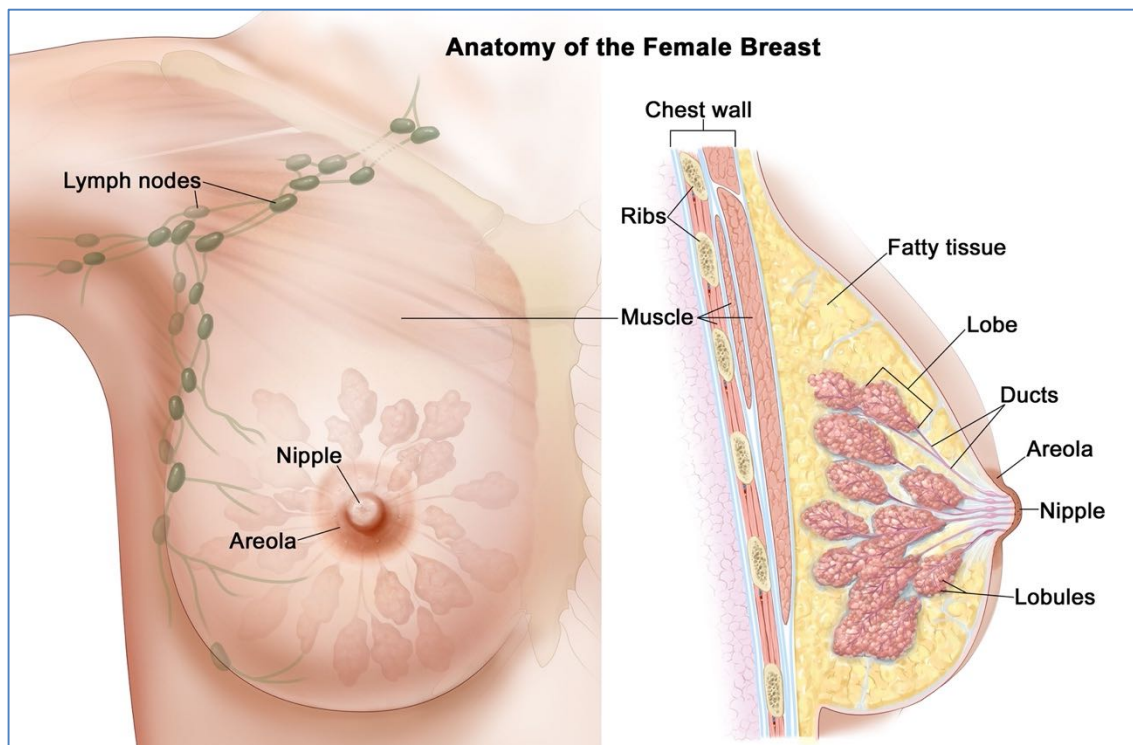


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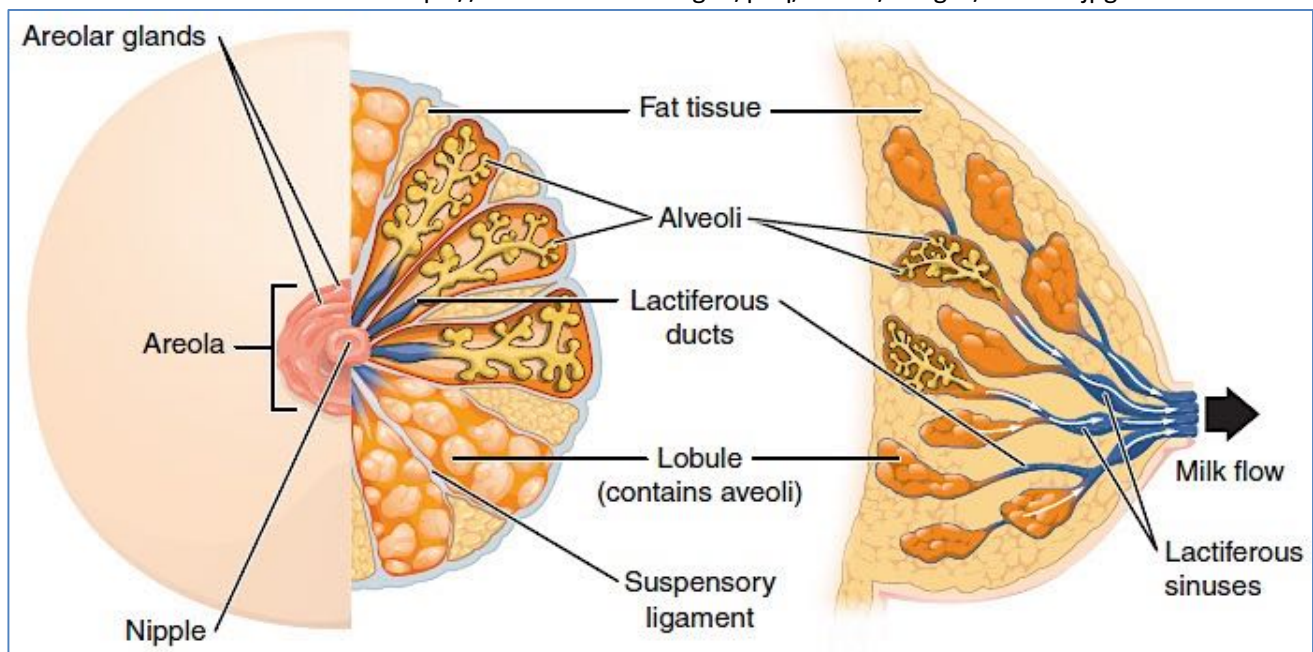


Overview of The Breast:

- **Mammary glands:**
 - o Exist in both sexes – only functional in females
 - o Contained within the breast – within the hypodermis (superficial fascia), anterior to pectoral muscles of the thorax
- **Areola** – ring of pigmented skin surrounding nipple – contains large sebaceous glands (stop chapping)
- **Nipple** – protrudes from centre of areola
- Attached to Pec-Major by Suspensory Ligaments
- **Glandular Breast Tissue:**
 - o Approx 20 lobes/lobules → Converge to Lactiferous Ducts → Lactiferous Sinuses → Nipple
 - o Padded and separated from each other by connective tissue (suspensory ligaments) and fat
 - o Within the lobes are smaller lobules – containing glandular alveoli – produce milk during lactation
 - o Compound alveolar glands pass milk into the lactiferous ducts → accumulates in a lactiferous sinus
- **Lymphatic Drainage:**
 - o Supraclavicular, Infraclavicular, Parasternal, Pectoral, Axillary, Central, Subscapular



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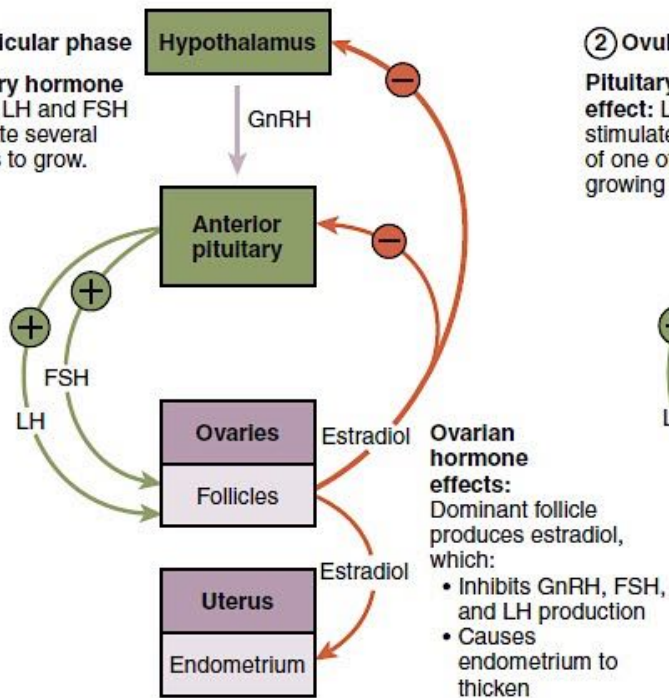
Review of Female Reproductive Physiology:

- **Puberty:**
 - A gradual series of events that transform a child into a sexually mature adult
 - **Female:** Marked by first menstrual period (average age 13)
 - **(Male:** Marked by physical development of Male Sex Characteristics)
- **Initiation:**
 - Activation of **Hypothalamo-Pituitary-Gonadal Axis** → establishes regulation of gonadal function
 - **At puberty** → ↓ Sensitivity of the hypothalamus to Inhibitory Steroid Hormones → ↑ GnRH → ↑ FSH & LH → ↑ Gonadal Testosterone/Oestrogen/Progesterone → Sexual Maturation
- **The Female Reproductive Cycle:**
 - The monthly series of events associated with the maturation of an egg
 - Typically 28 days long
 - **Days 1-5: *Menstruation*:**
 - **Shedding of the Endometrium**
 - Low levels of all hormones (FSH, LH, Oestrogen & Progesterone)
 - **Days 5-14: The Follicular/Proliferative Phase:**
 - **Follicular Recruitment & Growth**
 - **+ Endometrial Proliferation**
 - Rising levels of Oestrogen as Follicle/s get larger
 - **Day 14 (Mid-Cycle): *Ovulation*:**
 - Surge of FSH & LH → Ovulation into peritoneal cavity → Oocyte enters Fallopian Tubes
 - **FERTILE**
 - **Days 14-28: The Luteal Phase:**
 - **Transformation of Follicle → Corpus Luteum**
 - Corpus Luteum Secretes Mainly Progesterone (& Some Oestrogen)
 - Degenerates (Unless pregnancy occurs → Corpus Luteum persists until the placenta can take over)
 - **FERTILE**
 - **Day 28: End of Cycle:**
 - **Corpus Luteum Degenerates** → No Oestrogen/Progesterone to sustain Thick Endometrium → Endometrial Arteries become Spastic & Tortuous → Menstruation

(Diagram over the page)

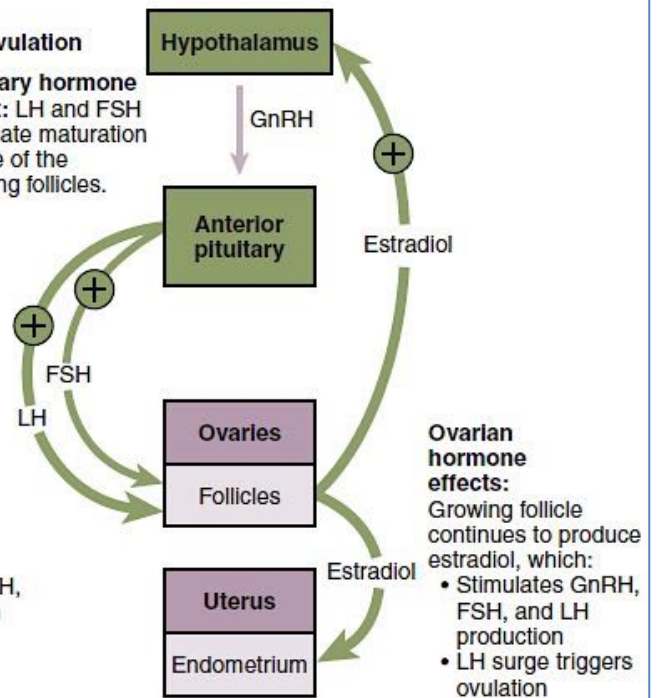
① Follicular phase

Pituitary hormone effect: LH and FSH stimulate several follicles to grow.



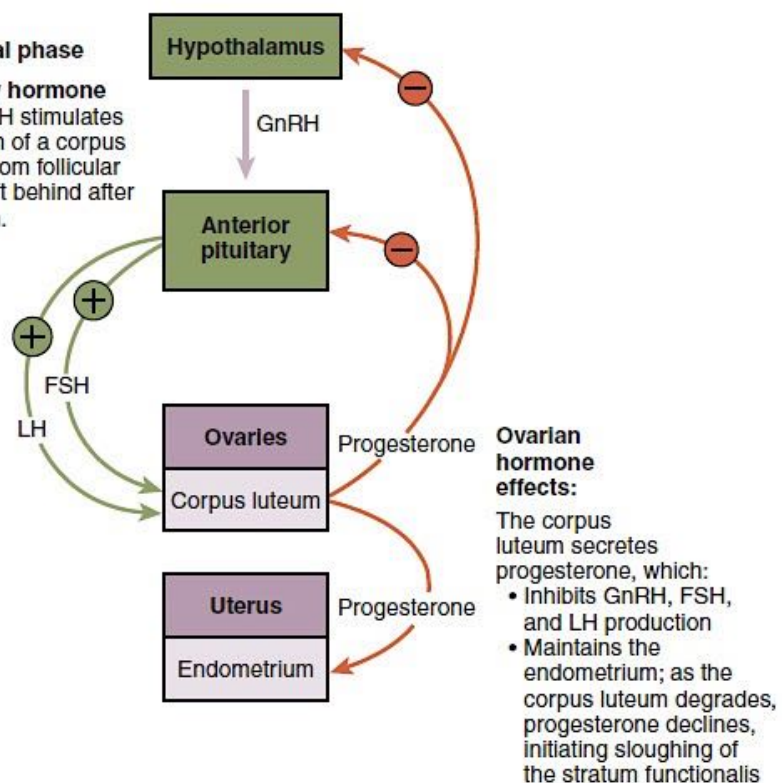
② Ovulation

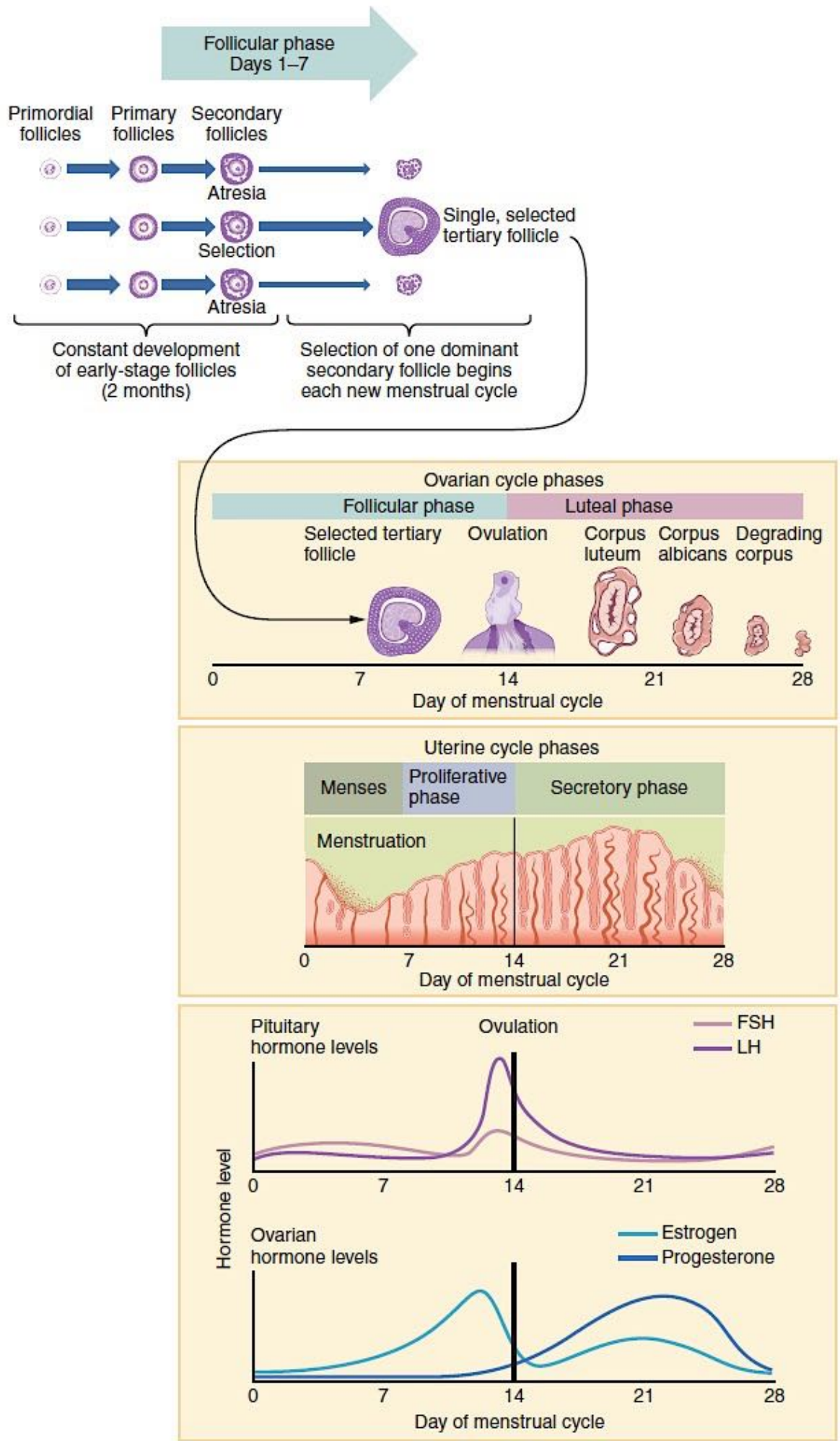
Pituitary hormone effect: LH and FSH stimulate maturation of one of the growing follicles.



③ Luteal phase

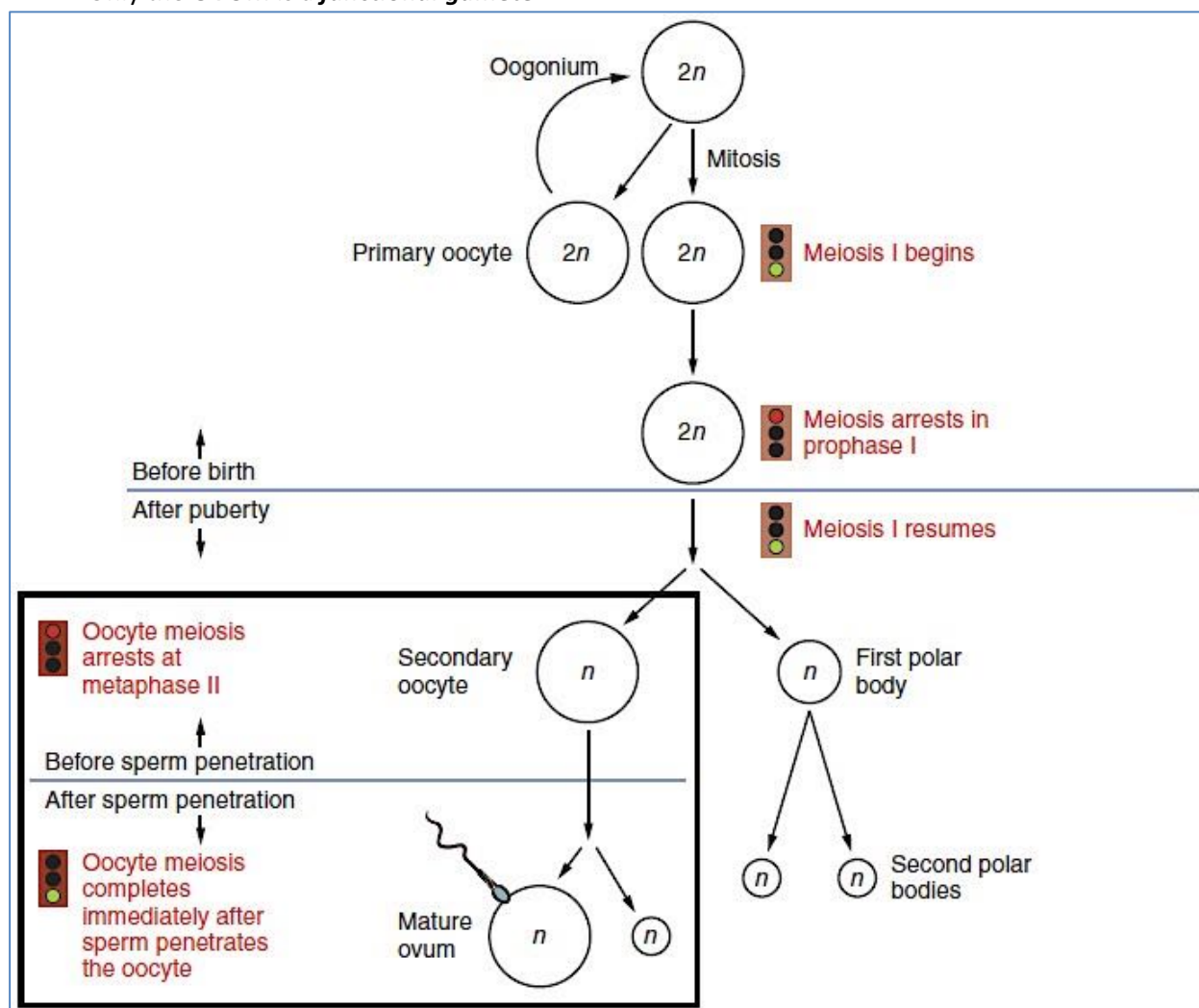
Pituitary hormone effect: LH stimulates formation of a corpus luteum from follicular tissue left behind after ovulation.



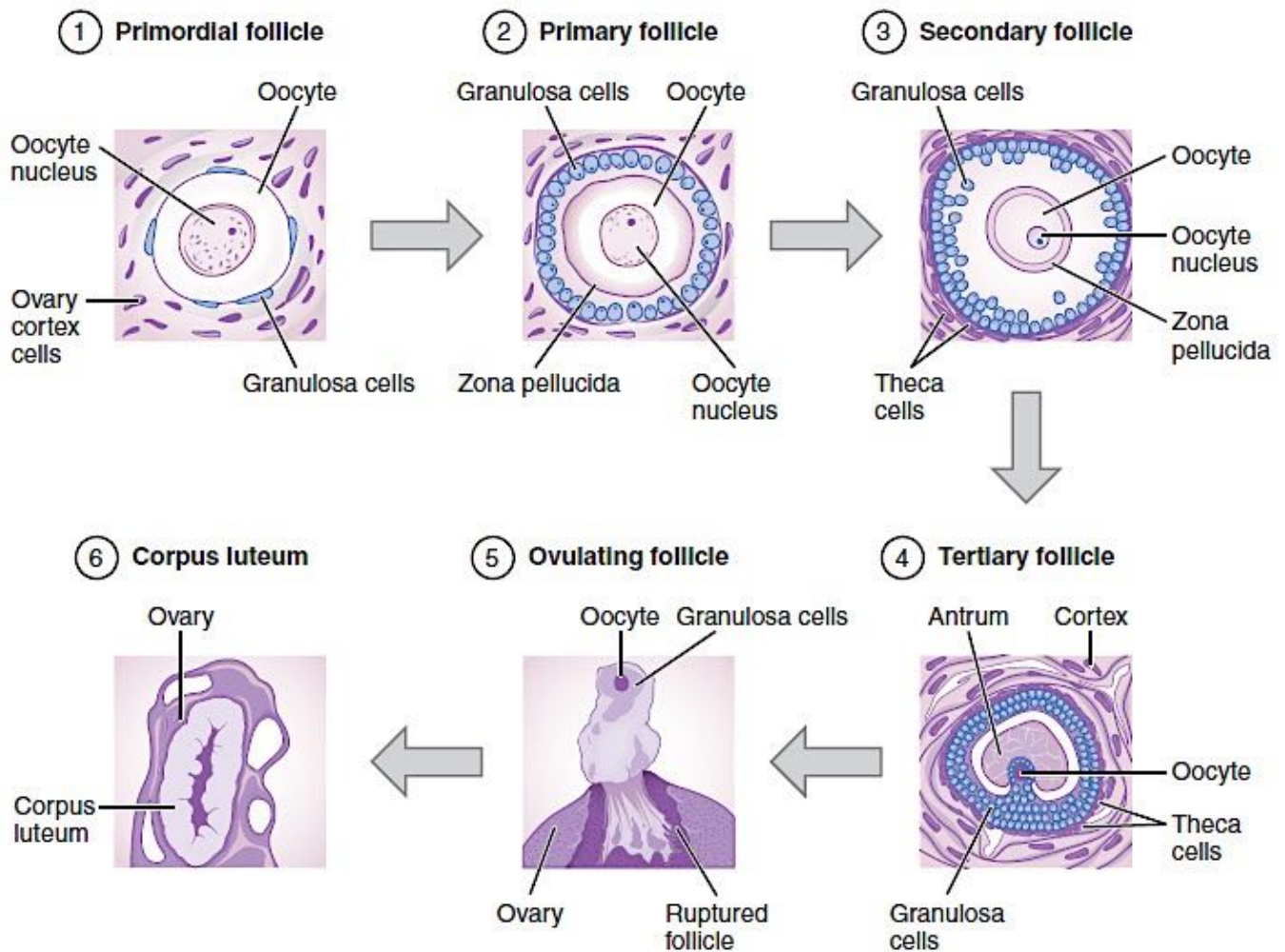


Meiosis (Female) – Oogenesis:

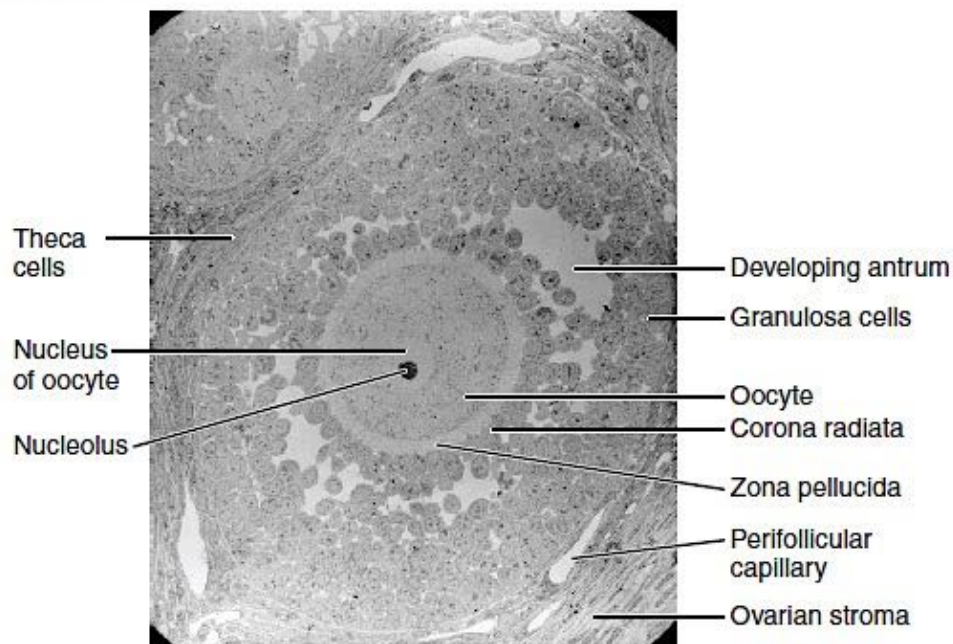
- It is thought that in general, the total number of eggs in a female is predetermined at birth
 - **Female gamete production = Oogenesis**
 - o Done through **meiosis**
 - Specialized cell division
 - Usually produces 4 haploid cells
- 1) **Foetal period** - the, **Oogonia** (diploid ovarian stem cells) multiply rapidly by mitosis, then enter a growth phase and lay in nutrient reserves as **Primary Oocytes**
 - 2) These **Primary Oocytes** then become **surrounded by** a single layer of **Follicle Cells** forming a **Primordial Follicle**
 - 3) **Primary Oocytes** (of the primordial follicles) then begin the **first meiotic division**. However, they are **arrested in prophase I**
 - 4) Female is born with approximately 2million primary oocytes. By puberty, 250000 primary oocytes are left.
 - 5) **Puberty—Menopause**: Each month, a small number of **primary oocytes** are recruited in response to the LH surge midway through the menstrual cycle. (Luteinising Hormone) As these **primary oocytes** prepare to divide, a spindle forms on its edge, creating a small “nipple” where half of the chromosomes will be cast during division
 - 6) Only **one of the primary oocytes** is selected to **continue meiosis I**. Produces **2 haploid cells** (23 chromosomes each) **dissimilar in size**. The smaller cell is the “**first polar body**” (little->no cytoplasm) and the larger cell is the **secondary oocyte**. → The **secondary oocyte** is then arrested in **metaphase II** and **OVULATED**. (unequal Cytoplasmic divisions ensure that a fertilised egg has ample nutrients for its week-journey to the uterus.)
 - 7) The **ovulated secondary oocyte MUST be penetrated by a SPERM** for it to **complete MEIOSIS II**, yielding one large **OVUM** and a “**Second polar body**”
- *Note:** - The potential products of oogenesis are 3 small polar bodies and one large ovum. (3 polar bodies aren't always formed – first polar body often perishes before meiosis II)
-Only the **OVUM** is a **functional gamete**



(a) Stages of Folliculogenesis



(b) A Secondary Follicle



PREGNANCY

PREGNANCY

Aetiology:

- Unprotected Intercourse
- IVF (In Vitro Fertilisation)
- AI (Artificial insemination)

Clinical Features:

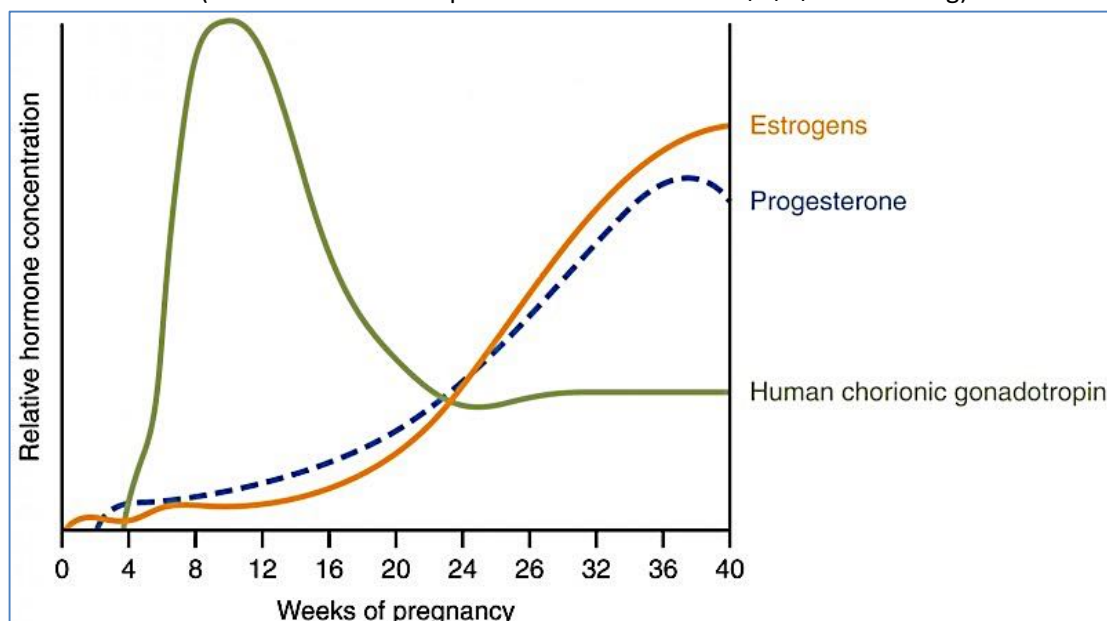
- **Symptoms:**
 - o Amenorrhoea
 - o Nausea/Vomiting
 - o Heartburn/Reflux
 - o Breast Tenderness
 - o Urinary Frequency
 - o Constipation
 - o Fatigue
- **Signs:**
 - o Softening of the cervix (4-6wks)
 - o "Chadwick's Sign" - Bluish Hue of Vagina (Engorgement of pelvic vasculature)
 - o Uterine Enlargement
 - o Breast Enlargement & Areolar Darkening

Diagnosis of Pregnancy

- **B-HCG:**
 - o +ve in *Serum* @ 9days post-conception
 - o +ve in *Urine* @ 28days post-conception
- **Transvaginal USS:**
 - o 5 wks: Gestational Sac Visible
 - o 6 wks: Foetal Pole Visible
 - o 7-8wks: Foetal Heartbeat Detectable
- **Trans-Abdominal USS:**
 - o >6 wks: Pregnancy Detectable

Physiological Amenorrhoea:

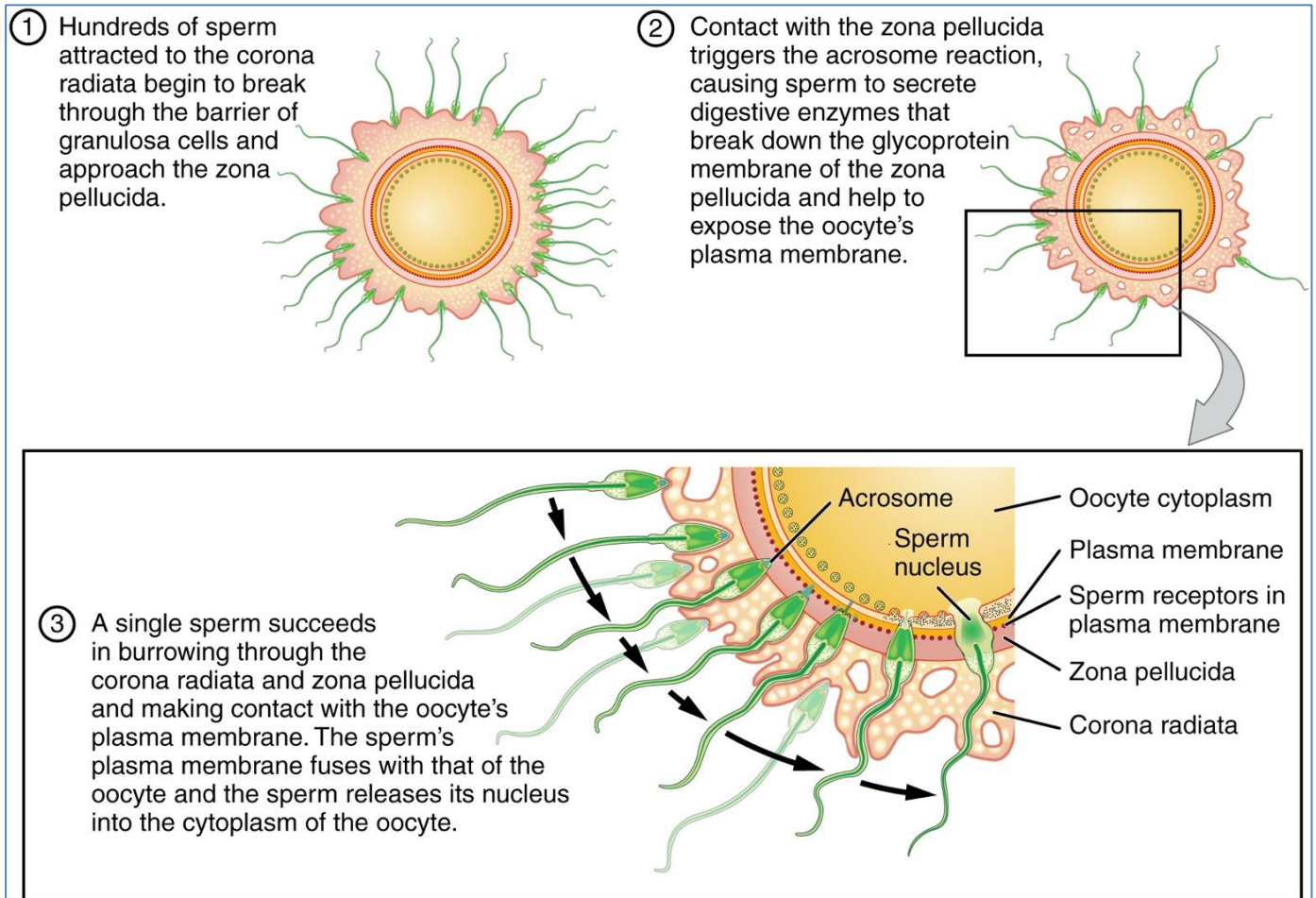
- Trophoblasts secrete B-HCG (Human Chorionic Gonadotropin) → Promotes Corpus Luteum to continue secreting Oestrogen & Progesterone (Ie: Prevent Menstruation)
- Trophoblasts → "Chorion" (continues secreting B-HCG)
 - o **(Note: B-HCG *Below* expected for dates could = Ectopic/Abortion/Wrong Dates)**
- Chorion → Placenta (Takes over from Corpus Luteum → Secretes ↑↑↑ Oest & Prog)



FERTILISATION & IMPLANTATION

From Egg (Oocyte) → Zygote:

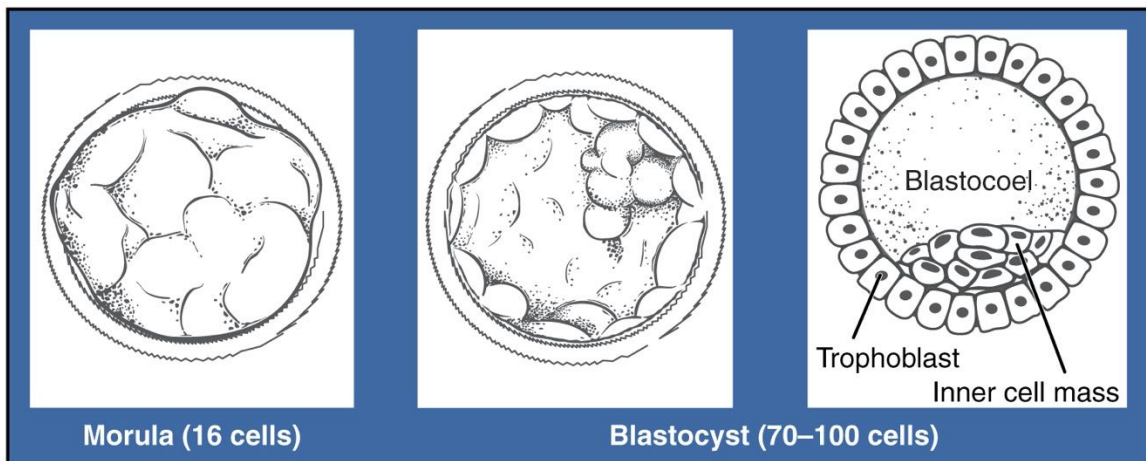
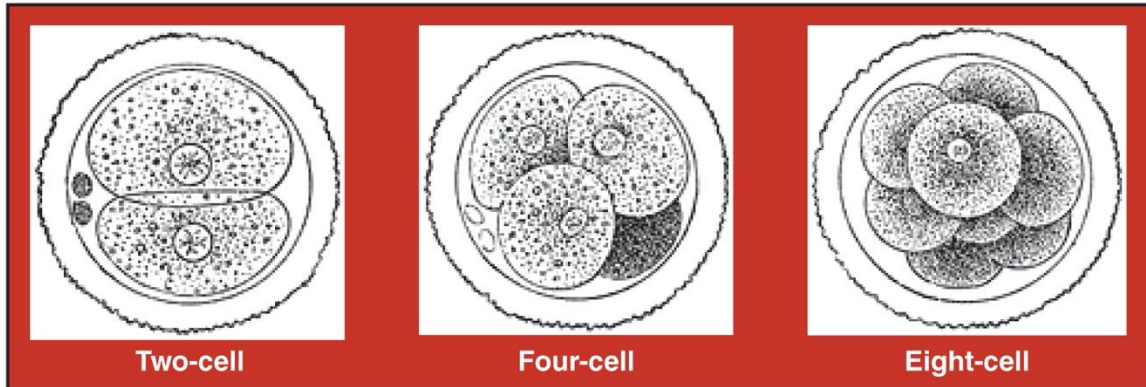
- **Viability:**
 - Oocyte is viable up to 24 hrs after ovulation
 - Sperm are viable inside female for up to 2 days
 - 3 day viable window = Sex between 2 days before ovulation and 1 day after
- **Fertilisation:**
 - Only a few thousand sperm reach the uterine tubes (site of fertilisation)
 - Sperm must be capacitated (increased mobility + fragile acrosomal membranes) before they can penetrate the oocyte (caused by secretions of female tract)
 - Oocyte is encapsulated in a "corona radiata" and the deeper "zona pellucida", which both need to be breached before the oocyte itself can be penetrated by a sperm
- **Steps:**
 1. Sperm penetrates the corona radiata, assisted by hyaluronidase, and corona radiata cells fall away from the oocyte
 2. Sperm head binds to zona pellucida, which triggers the release of acrosomal enzymes → digest holes through the zona pellucida (hundreds of sperm are needed to expose the oocyte membrane)
 3. Single sperm fuses with oocyte membrane, its nucleus is pulled into the oocyte cytoplasm
 4. Ca^{2+} is released by oocyte ER, causing cortical reaction → cortical granules in oocyte membrane release enzymes (zonal inhibiting proteins) that destroy remaining sperm receptors + detach all sperm still bound
 5. Secondary oocyte completes meiosis II, forming ovum pronucleus + 2nd polar body
 6. The lucky sperm loses its tail + midpiece, & its nucleus swells to 5x → forms male pronucleus
 7. These 2 pronuclei fuse and form the zygote (moment of fertilisation)
 8. Mitosis of conceptus begins



From Zygote → Blastocyst Implantation:

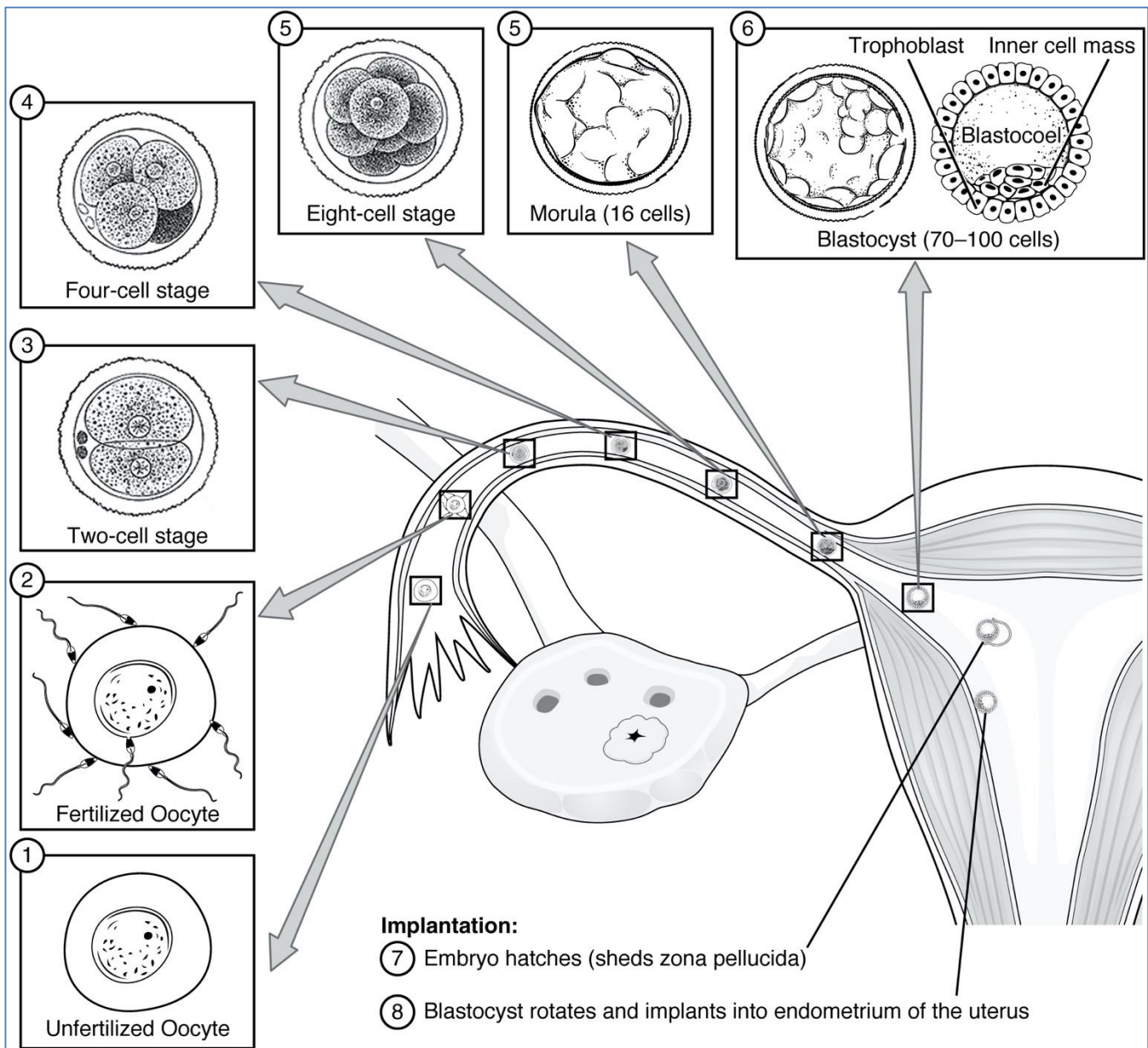
- Cleavage & Blastocyst Formation:

- Rapid series of mitotic divisions without intervening growth → produces many smaller cells collectively called the **morula**
- Cells of the morula compact (forming the inner cell mass), the zygote accumulates fluid and trophoblast cells develop around the inside of the zona pellucida (Early blastocyst)
- Zona pellucida starts to break down and the blastocyst “hatches” from it



■ Occurs in uterine tube ■ Occurs in uterus

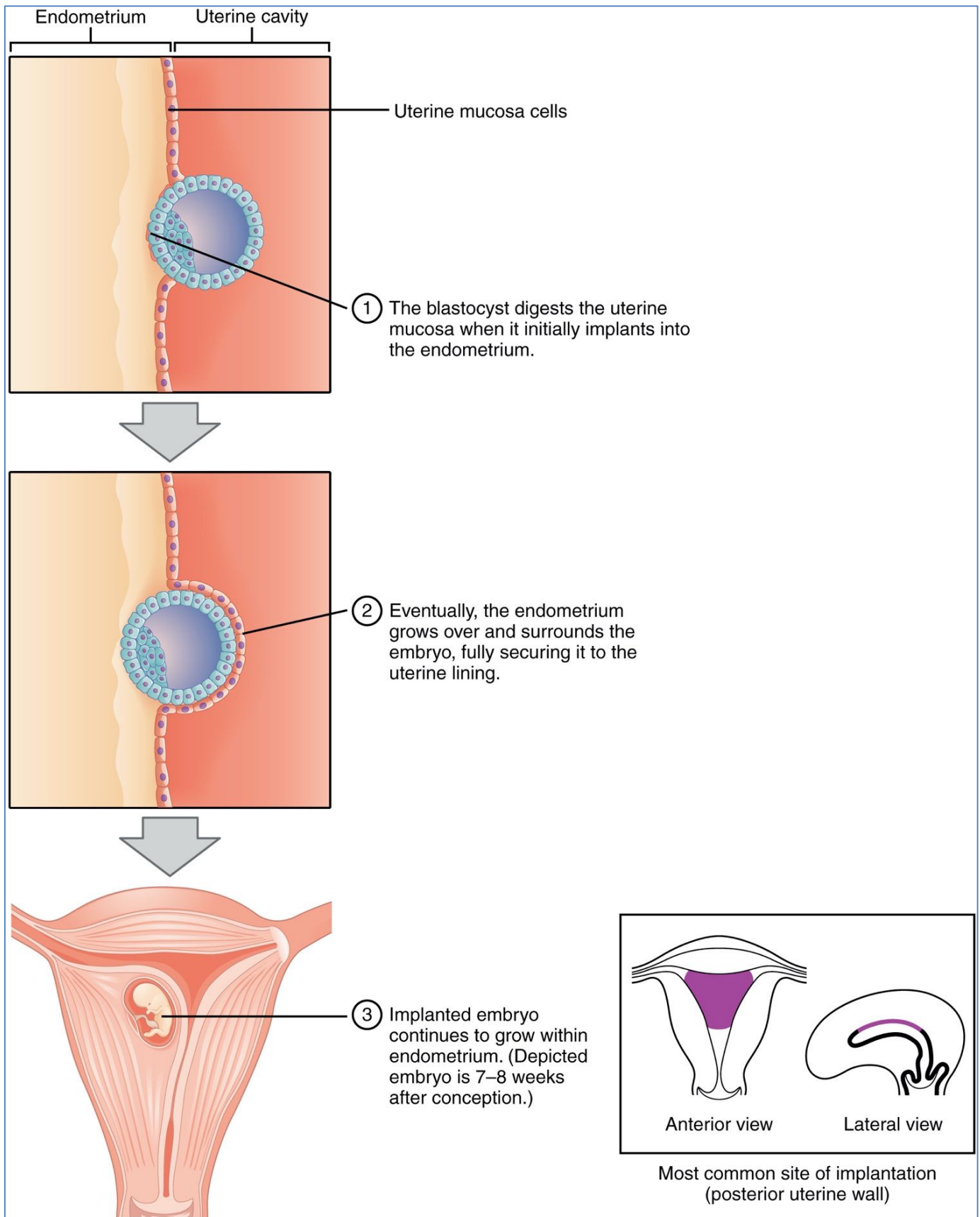
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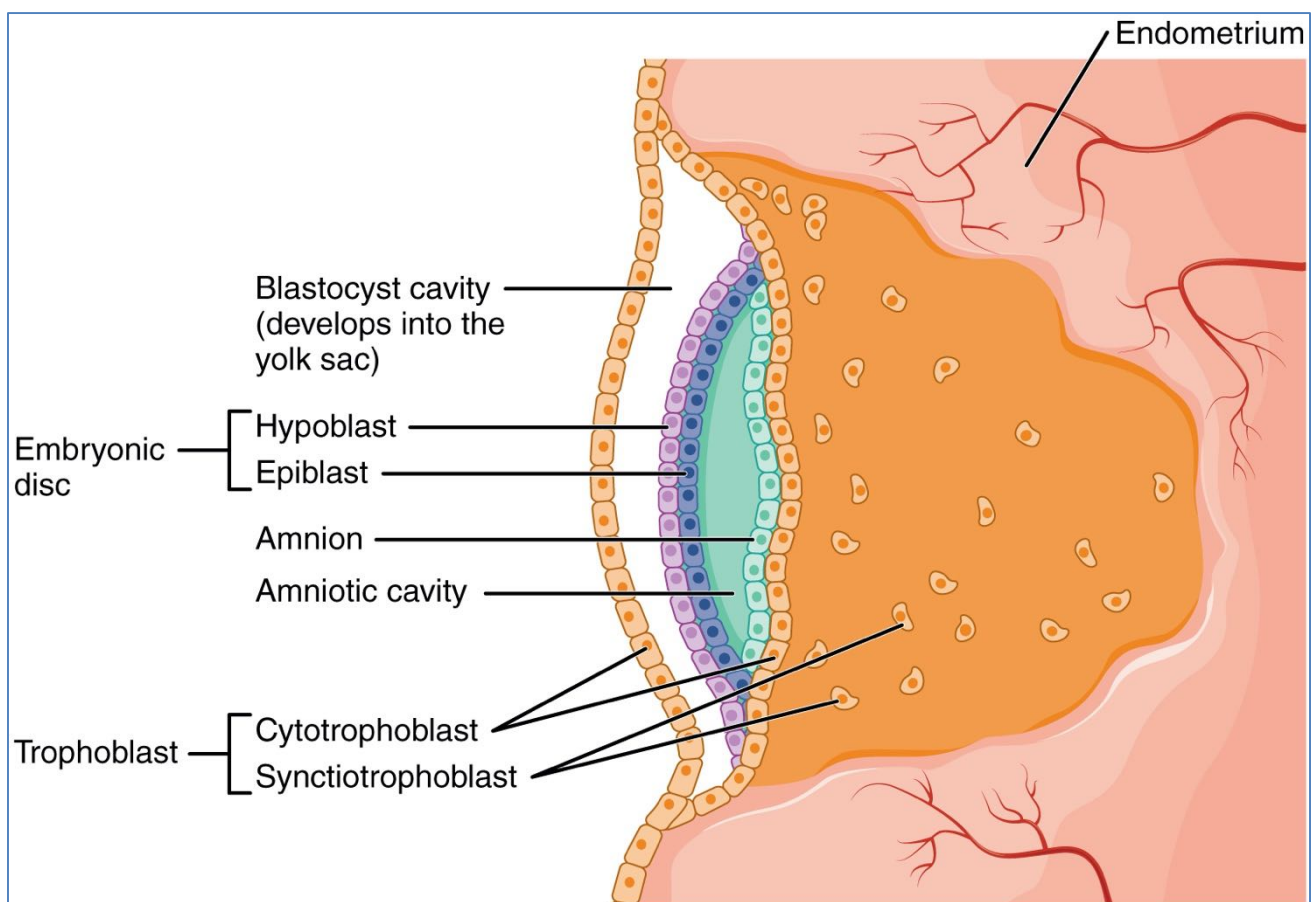
- **Implantation (6-7 days after ovulation)**

- Endometrium must be receptive (at the height of the secretory stage) (caused by surging estrogens and progesterone)
- Trophoblast cells bind to the extracellular matrix of endometrial cells on the inner uterine wall (implantation may be attempted several times where the blastocyst detaches from the immature Endometrium and floats south to try again)
- Endometrium thickens, blood vessels become more permeable/leaky and inflammatory cells invade the area
- Trophoblast cells proliferate, forming 2 distinct layers: **cytotrophoblast** (adjacent to inner cell mass) and the **syncytiotrophoblast** (multinucleated mass – invades the Endometrium by digesting uterine cells)
- At this stage implanted embryo obtains nutrition by digesting endometrial cells
- Blastocyst burrows through the Endometrium and is surrounded by a pool of blood (**lacuna**) Blastocyst is covered over and sealed off from uterine cavity by endometrial cells
- Successful implantation takes 5 days (12 days after ovulation – usually time of menstruation)
- To stop menses, trophoblast cells secrete hCG (Human Chorionic Gonadotropin) which promotes Corpus Luteum to continue secreting oestrogen and progesterone
 - Trophoblast cells differentiate into the **chorion** which continues secreting hCG
- Chorion later forms the **placenta**, which takes over from corpus luteum & produces shit loads of progesterone and oestrogen

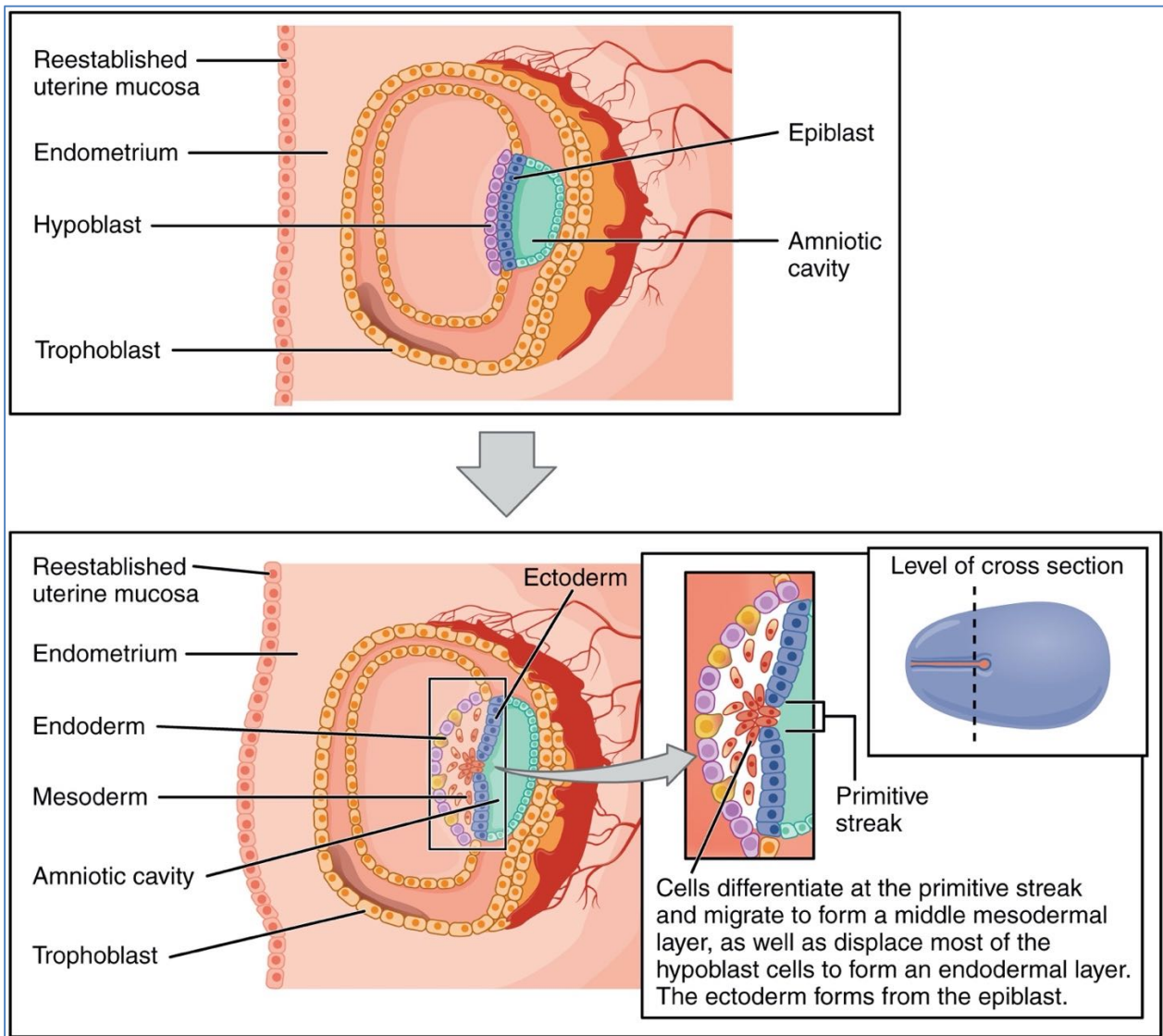


- **Placentation:**

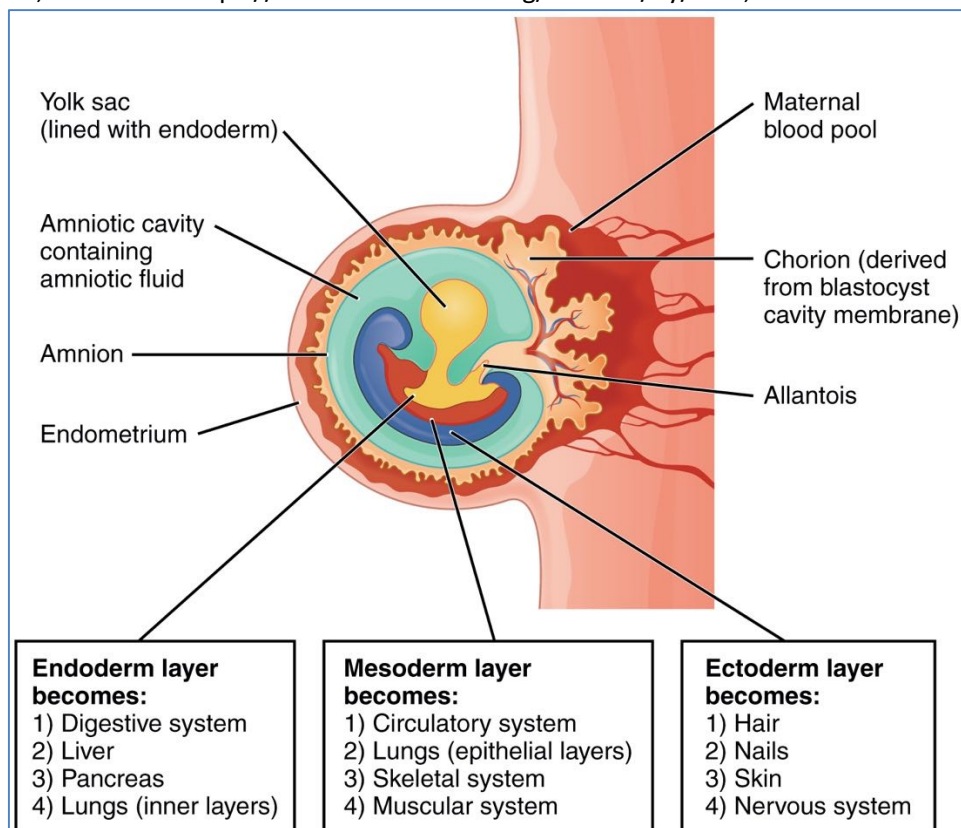
- Cells from original inner cell mass give rise to a layer of **extraembryonic mesoderm** which lines the inner surface of the cytotrophoblast. Together the **cytotrophoblast** and **extraembryonic mesoderm** form the **Chorion**
- **Germ layers form:** inner cell mass facing blastocyst cavity form a flat disc with 2 layers:
 - **1: Ectoderm:**
 - Forms nervous system and skin during gastrulation
 - – and the amnion formed from the ectoderm (later fills with fluid → amniotic sac)
 - **(Mesoderm:** formed later during gastrulation – forms heart, blood vessels, connective Tissue & everything else)
 - **2: Endoderm:**
 - Forms the mucosa (epithelial lining) of the GI tract, respiratory tract, and urogenital tract
 - - and the yolk sack formed from the endoderm (later forms part of the gut/digestive tube and produces the earliest blood cells & blood vessels)
 - – and the allantois – structural base of the umbilical chord
- **By 8 weeks all organ systems are formed**



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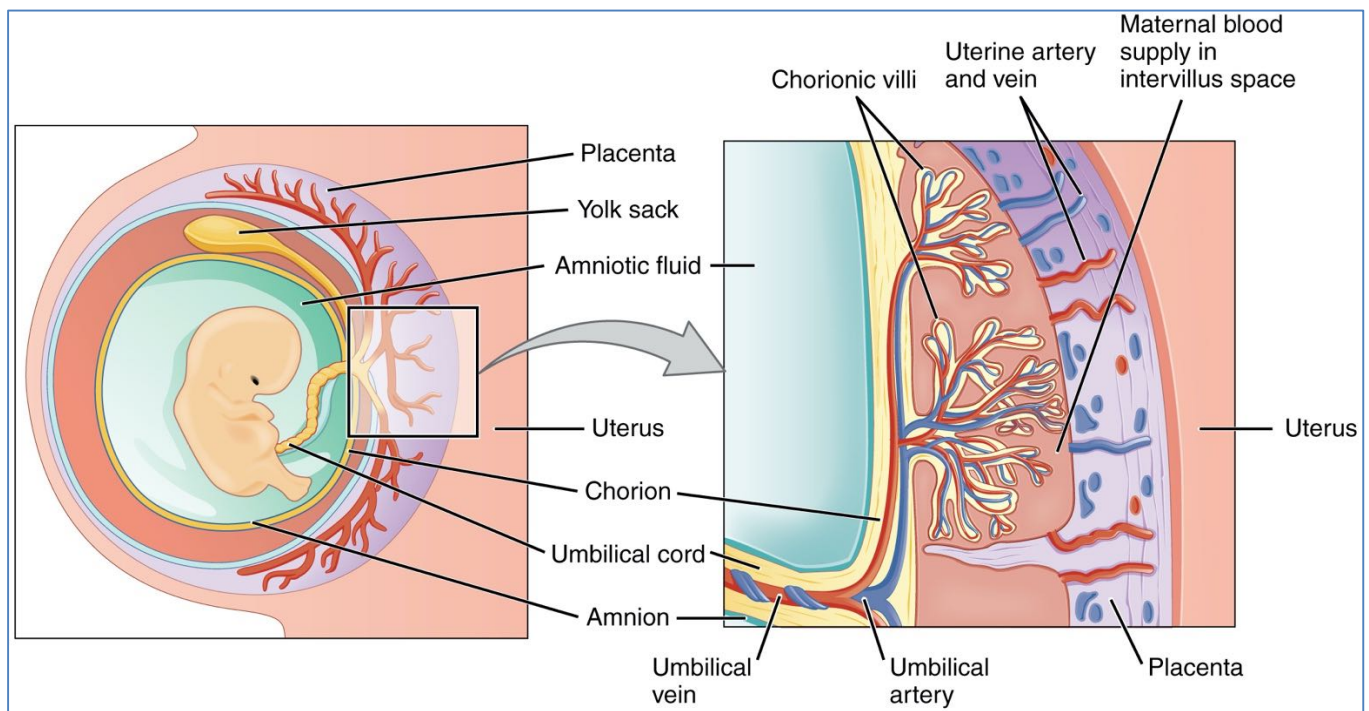


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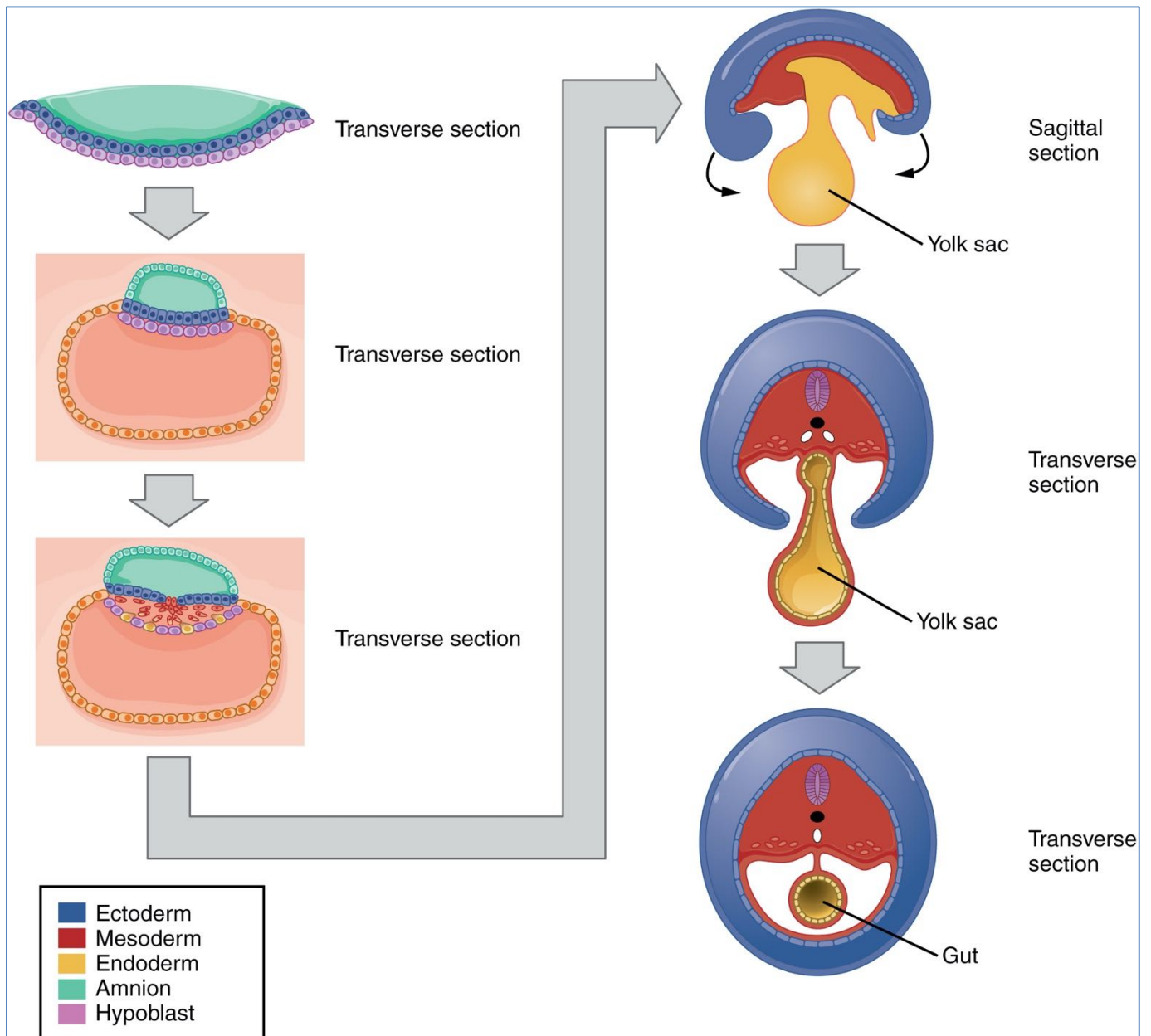
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- **Chorion** develops fingerlike **chorionic villi** which are in contact with maternal blood and are highly vascularised
- Large blood-filled **lacunae** form in the Endometrium and the villi become totally immersed in blood
- Around the chorionic villi, the Endometrium splits into 2 parts: **the decidua basalis** (basal side/ stratum basalis [side of umbilical cord]) and **the decidua capsularis** (luminal side of embryo)
 - Decidua capsularis expands to accommodate foetus & villi are compressed & degenerate
 - Decidua basalis forms the eventual placenta and villi increase in number and branch even more profusely
 - Placenta functions as a nutritive, respiratory, excretory and endocrine organ
 - Transports glucose, amino acids, fatty acids, O₂, CO₂ and other wastes

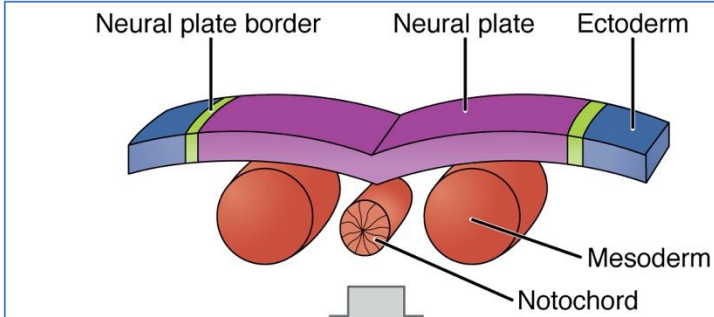


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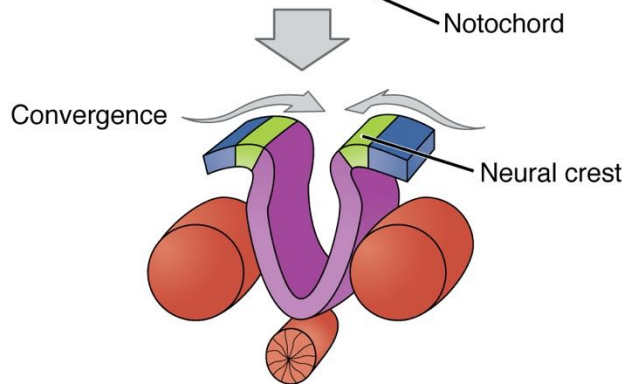
Embryonic Development



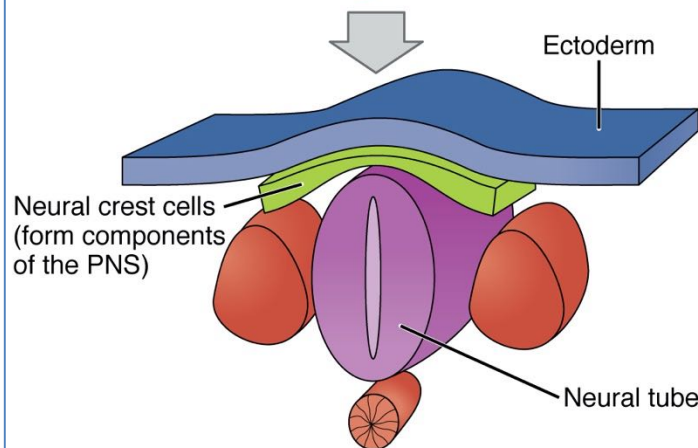
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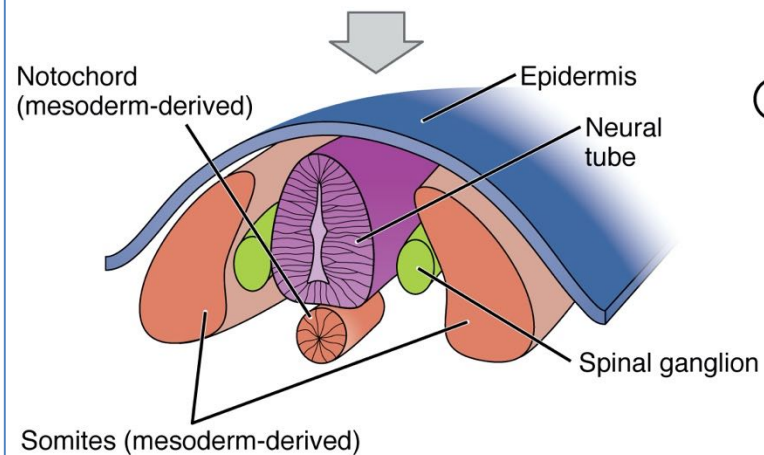
- 1 Neuroectodermal tissues differentiate from the ectoderm and thicken into the neural plate. The neural plate border separates the ectoderm from the neural plate.



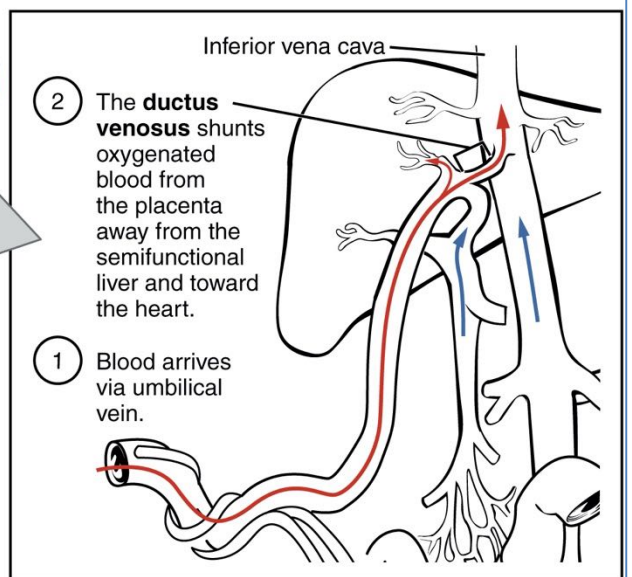
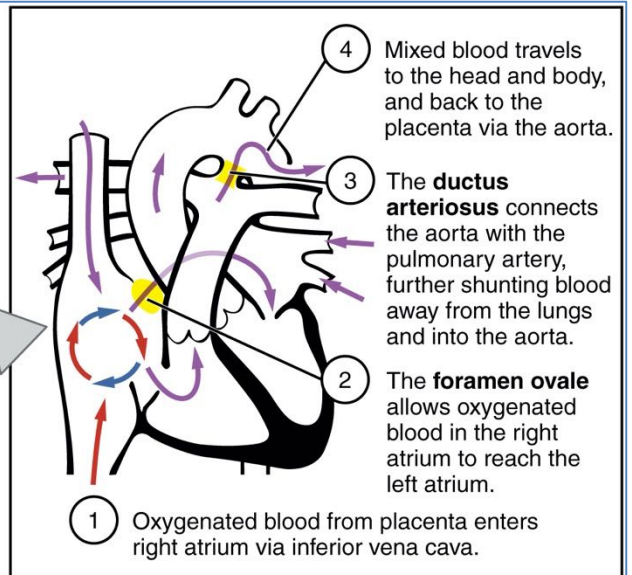
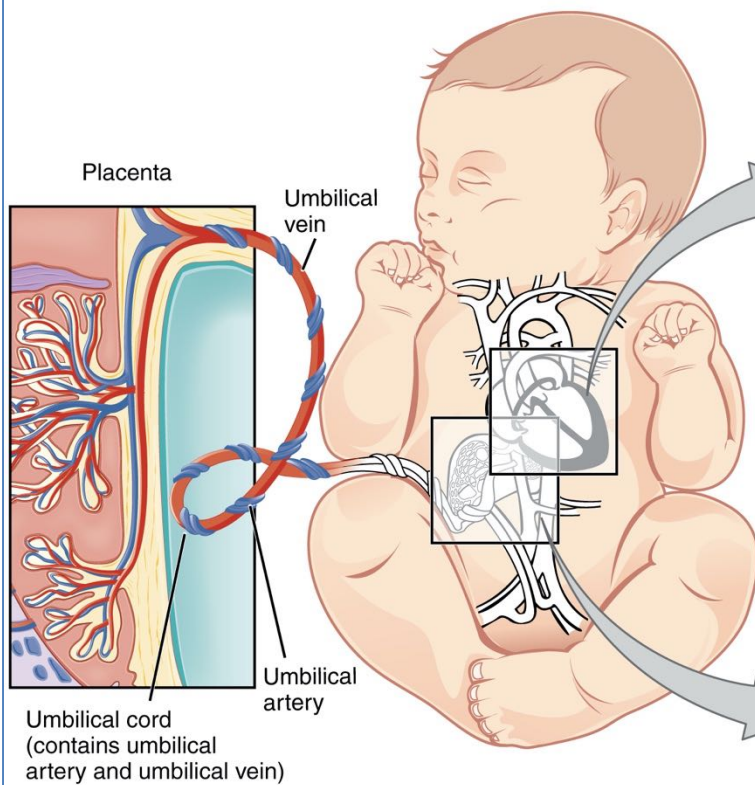
- 2 The neural plate bends dorsally, with the two ends eventually joining at the neural plate borders, which are now referred to as the neural crest.



- 3 The closure of the neural tube disconnects the neural crest from the epidermis. Neural crest cells differentiate to form most of the peripheral nervous system.



- 4 The notochord degenerates and only persists as the nucleus pulposus of the intervertebral discs. Other mesoderm cells differentiate into the somites, the precursors of the axial skeleton and skeletal muscle.

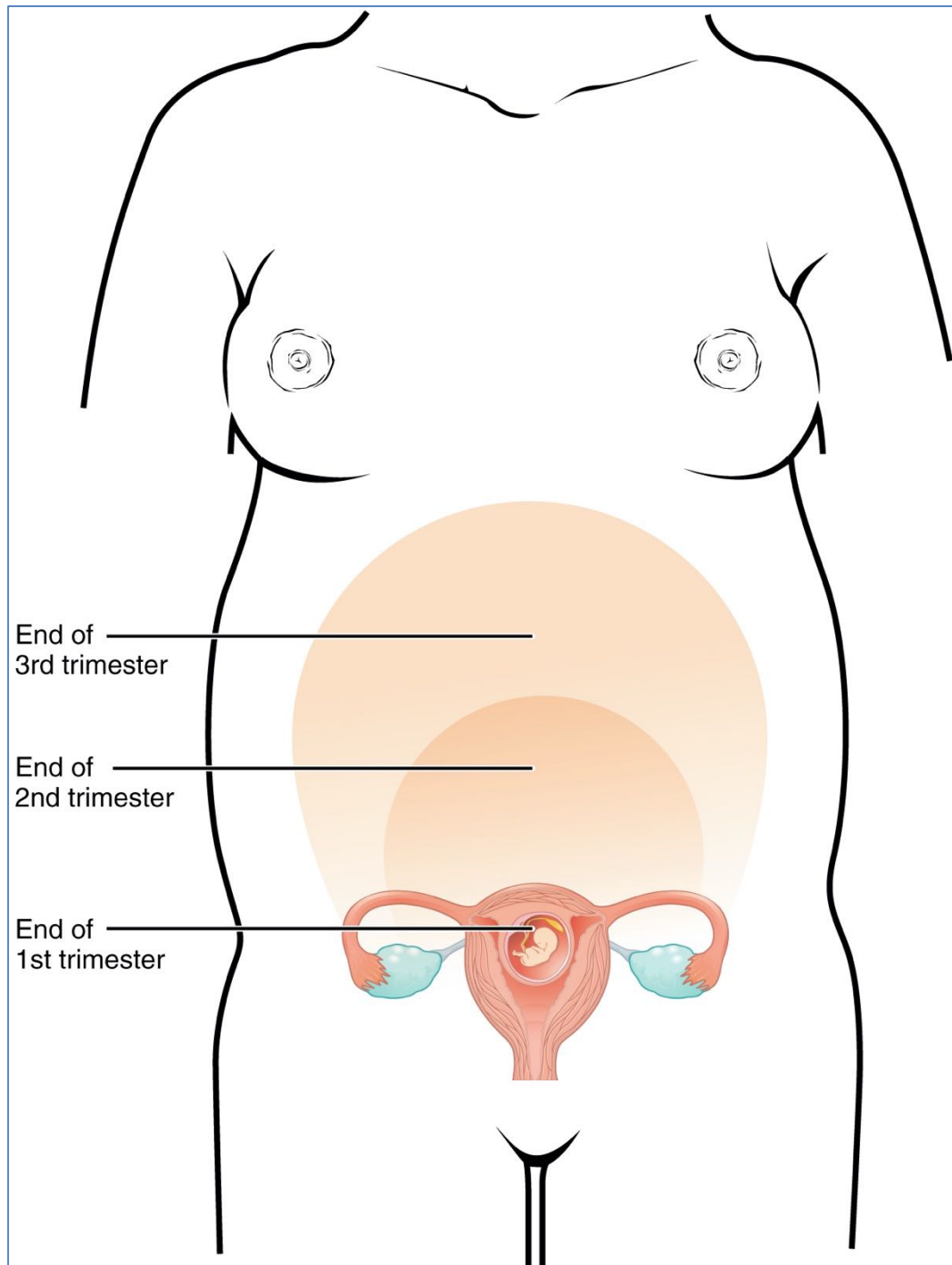


MATERNAL PHYSIOLOGIC ADAPTATIONS TO PREGNANCY

MATERNAL PHYSIOLOGIC ADAPTATIONS TO PREGNANCY

Reproductive System Changes:

- ↑Vascularity & Size of Reproductive Organs
 - o Vagina develops purplish hue (Chadwick's sign)
 - o Breasts Enlarge + Darken
- Placenta Secretes **Relaxin** → Relaxes Pelvic Ligaments + Pubic Symphysis
- Placenta Secretes **Human Chorionic Somatomammotropin (hCS)** →
 - o → Breast Maturation
 - o → ↑Maternal Insulin Resistance → Glucose-Sparing Effect → More Nutrients for Foetus (Note: Can cause Gestational Diabetes Mellitus)
- Uterus enlarges – pushes higher into abdominal cavity – abdominal organs intrude into thoracic cavity causing ribs to flare



Urinary System Changes:

- Bladder Compression → Decreased capacity of the bladder to store urine → Urinary Frequency
- ↓Sphincter Tone (Progesterone → SM Relaxation) → Incontinence & ↑UTIs
- Increased plasma volume → 50% ↑GFR → Polyuria
 - ↑ Urine Production – (To remove the extra foetal metabolic wastes)
 - → Decreased reabsorption of ions in nephron
 - Note: an increase in aldosterone compensates for the above (In the distal nephron)
- Glycosuria (Filtration outpaces Reabsorption)
- Dilation & Lengthening of the ureters (progesterone → SM relaxation)

Respiratory System Changes:

- 20% ↑Oxygen Consumption
- Progesterone increases the sensitivity of the respiratory centre to CO₂
 - Increased respiration (↑Tidal Volume & ↑Minute Ventilation Rate)
 - Functional Tachypnoea
- ↓Diaphragm Mobility → ↓Total Lung Capacity
 - Diaphragm also rises up to 4cm
 - ↑Rib Flaring → Diameter of thorax increases by 6-10cm
 - ↑Accessory Muscle Usage
- No Changes in ABG Parameters:
 - No changes in partial oxygen pressure in blood
 - No changes in partial CO₂ pressure in blood

Cardiovascular System Changes:

- Total body water rises
 - → blood volume increases 25-40% to accommodate foetus
 - → Also compensates for potential blood loss during birth
- ↑Cardiac Output Required to perfuse the extra body tissues (placenta, breasts, fat tissues)→ Hyperdynamic Circulation. Achieved by:
 - ↑HR
 - ↑Plasma Volume (25-40%) → Hypertension
- Uterus can compress on Iliac Veins → ↓Venous Return from Legs
 - → Varicose Veins
 - → Haemorrhoids
 - → Leg Oedema

Haematological System Changes:

- ↑Plasma Volume (25-40%) → Spurious Anaemia (due to Haemodilution)
- Hypercoagulable State (↑Almost ALL Coagulation Factors; + ↓Antithrombin-III)
 - → ↑DVT/PE Risk
- ↑Leukocyte Count (But Impaired Function) → Relative Immunodeficiency
- Gestational Thrombocytopenia (Consumptive)

Gastrointestinal System Changes:

- **Physiological:**
 - Morning Sickness – due to ↑Oestrogen & ↑Progesterone
 - Heartburn/Reflux – Increased Pressure on stomach by foetus
 - Constipation – Increased fluid uptake + ↓Bowel Motility (Due to Progesterone)
- **GI tract is displaced superiorly & laterally**
 - Important to know in case of appendicitis as the appendix will be higher than normal (usually experience pain in right flank)
- **Mouth:**
 - Saliva increases (keeps the mouth alkaline – keeps the pH up to prevent mouth ulcers & tooth decay)
- **Gastroesophageal sphincter is more relaxed (due to progesterone) & has more pressure to resist:**
 - Increased risk of aspiration if under anaesthetic or unconscious → Pregnant women must fast for longer than normal before anaesthetic
- **Stomach:**
 - Hydrochloric acid production decreases
 - Decreased gastric motility (due to progesterone) allows more time for digestion of stomach contents
- **Bowel:**
 - Decreased bowel motility (less peristalsis due to progesterone) → more time to absorb nutrition & water → Constipation
 - Constipation also caused by mechanical obstruction of the rectum by the uterus
- **Liver:**
 - No change in size
 - Works harder to produce enzymes but they don't increase (Eg: ALT). Raised liver enzymes are a sign of disorder
 - One enzyme will be increased = "Alkaline Phosphatase" – double/triple in pregnancy – Produced by the placenta & liver
- **Gallbladder:**
 - Progesterone → less contraction of the gallbladder → Flow of bile becomes slower → more time for the metabolism of fat/cholesterol
 - If Bile-flow is too slow → Gallstones risk is higher

Endocrine Changes:

- **Pancreas:**
 - ↑Insulin Secretion → To compensate for HCS-Mediated Insulin Resistance
 - But we still get hyperglycaemia. Why?:
 - Pregnancy induces insulin resistance to maintain the high blood sugar
 - Human placental lactogen & cortisone/cortisol & Progesterone competes with the insulin for the receptors → Insulin 'resistance' (Competitive inhibition)
 - (Note: Maternal Insulin does NOT cross the Placenta – Bub makes its own Insulin)
- **Thyroid:**
 - HCG → Stimulates Thyroid (In addition to Normal TSH)
 - ↑Size & Vascularity
 - ↑Thyroid Hormone Production – but may not have hyperthyroidism. Why? – Oestrogen increases thyroid binding globulin to decrease the 'free thyroxin' in the blood doesn't increase
 - (But normal T3/T4 levels due to ↑Thyroid-Binding Globulin)
 - **Human Chorionic Thyrotropin (hCT)** causes hypermetabolism and increase Ca^{+} concentration → ensuring mineralisation of foetus' bones
- **Parathyroid:**
 - ↑PTH → ↑Bone Ca-Resorption → ↑Ca Available to Foetus's Bones
- **Adrenals:**
 - ↑ACTH Secretion →
 - ↑Aldosterone → Water Retention → Oedema
 - ↑Cortisol → Relative Immunodeficiency (Protects Foetus from Mum)
- **Pituitary:**
 - ↑Prolactin Secretion → Lactation

EMERGENCY CONTRACEPTION & ABORTION

EMERGENCY CONTRACEPTION & ABORTION

Emergency Contraception:

- **Goal:**
 - Last-chance contraception if current contraceptive failed (Eg: Broken Condom)
- **Timing:**
 - ASAP after unprotected sex (<120hrs / 5days)
 - **Note: Completely ineffective Post-Implantation**
- **Methods:**
 - **#1 Progesterone-Only:** (AKA: **"Plan-B"**) Single Dose – 1.5mg
 - (Effective <72hrs & Fewer Side Effects)
 - **Note: NOT Effective After Implantation!!**
 - **Copper IUD:** Best efficacy; But inconvenient & invasive
 - **Still Effective After Implantation**

Abortion Pre-Requisites:

1. Counselling on Alternatives (Eg: Adoption)
2. Informed Consent (& any domestic legal prerequisites are met)
3. Comprehensive History
4. Discuss Contraception *After* Abortion
5. STI-Screen & Education
6. Antibiotic Prophylaxis Prior to Abortion

Abortion:

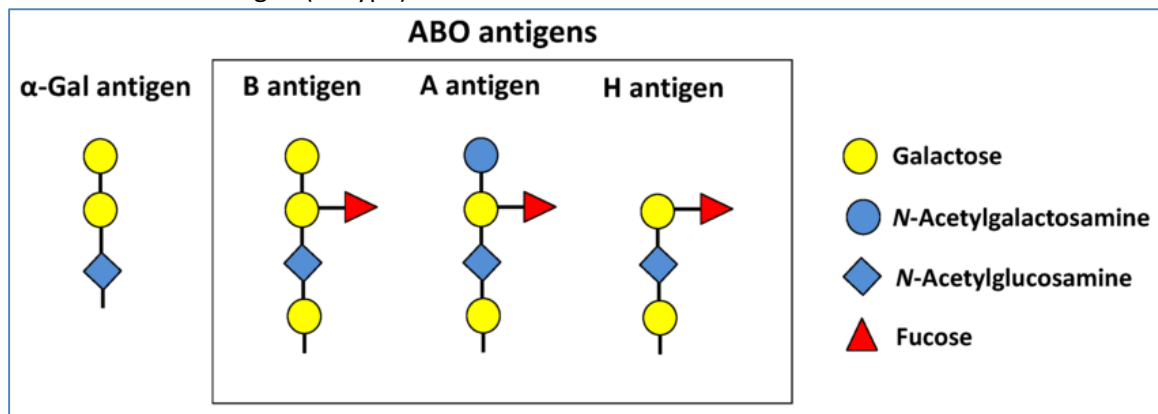
- **Early Medical Abortion (<6wks):**
 - **STAT DOSE - Mifepristone/RU486:**
 - **Progesterone Receptor Agonist** → Prevents Endometrium from supporting Fertilised Egg
 - Effective <63days (7wks) since last period
 - No need to come into Hospital
 - **Or Methotrexate – (Used more in Ectopic Pregnancies)**
- **Early Surgical Abortion (<14wks):**
 - **Dilation & Suction Curettage:**
 - Available up to 14wks Gestation
- **Late Medical Abortion (14-20wks):**
 - **STAT DOSE - Mifepristone/RU486:**
 - **Progesterone Receptor Agonist** → Prevents Endometrium from supporting Fertilised Egg
 - Effective <63days (7wks) since last period
 - **48HRS LATER – Vaginal Misoprostol:**
 - **Synthetic Prostaglandin** → Ripens Cervix & Induces Labour
 - + Analgesia, +/- Anti-Emetics, +/- Anti-D-Ig in Rh-Negative Mothers
 - Very Safe – Small risk of bleeding; <1:100000 risk of death
- (Note: In most jurisdictions, abortion beyond ~20wks is generally not permitted due to onset of legal rights of the foetus and are only considered in extreme circumstances)

RHESUS DISEASE

RHESUS DISEASE

BLOOD GROUP ANTIGENS:

- There are ≈400 known RBC Antigens
- We are only concerned with 2 categories; the **ABO & Rh Antigens**
- **1: 'ABO' Blood Group Antigens:**
 - Sugar Chains emanating from the RBC membrane
 - Determines the 'A/B/AB/O' blood types
 - A-Antigen
 - B-Antigen
 - A & B-Antigens
 - H-Antigen (O-Type)



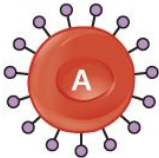
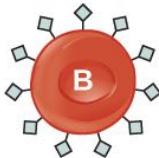
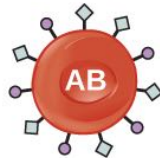
- Exist due to 3 allelic genes (A, B & O)
 - A & B alleles can show *CoDominance* (AB-Type)
 - A & B alleles are Dominant over the 'O' allele
 - Homozygous 'OO' is dominant over A or B alleles

ABO genotype in the offspring		ABO alleles inherited from the mother		
		A	B	O
ABO alleles inherited from the father	A	A	AB	A
	B	AB	B	B
	O	A	B	O

<https://www.ncbi.nlm.nih.gov/books/NBK2267/>

- **2: 'Rh' (Rhesus/Rh-D) Blood Group Antigens:**

- Membrane-Bound protein on RBC
- **Presence/Absence of the RhD'-Gene determines +ve/-ve blood type**
 - Presence of RhD → Positive
 - Absence of RhD → Negative

	Blood Type			
	A	B	AB	O
Red blood cell type				
Isohemagglutinins	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens on red blood cell	 A antigen	 B antigen	 A and B antigens	None

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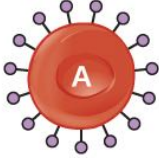
- **Relevance in Transfusions:**
 - **Rh-Positive Patients:** Can receive *either* Rh-Positive OR Rh-Negative Blood
 - **Rh-Negative Patients:** Should **ONLY** receive Rh-Negative Blood (except in extreme emergencies and Rh-Negative blood is unavailable)
- **Relevance in Pregnancy:**
 - **If the mother is Rh-Negative, but the foetus is (potentially) Rh-Positive →**
 - → 'Rh-Incompatibility'
 - (If father is Rh-Positive or father's Rh-status is unknown)
 - Normally, maternal and foetal blood don't mix, but sometimes a **sensitizing event** can occur, causing foetal blood to contact maternal blood
 - Eg: Abdominal trauma during pregnancy
 - Eg: Amniocentesis
 - Eg: Miscarriage
 - Eg: Ectopic pregnancy
 - Eg: Chorionic villus sampling
 - Eg: Bleeding during pregnancy
 - **If Rh-Negative mother gets sensitized to Rh-Positive Foetus →**
 - → Mother's immune system produces **Rh-Antibodies**
 - → **Rh-Antibodies** Cross the placenta
 - → Enter foetal bloodstream → Attack foetal RBC's → **Hemolytic Anemia**
 - **Note:** You can prevent an Rh-Negative mother from being sensitized by administering Rh-Immunoglobulin (aka: Anti-D-Antibodies) at strategic times during the pregnancy

Mother's Rh factor	Father's Rh factor	Baby's Rh factor	Precautions
Rh positive	Rh positive	Rh positive	None
Rh negative	Rh negative	Rh negative	None
Rh positive	Rh negative	Could be Rh positive or Rh negative	None
Rh negative	Rh positive	Could be Rh positive or Rh negative	Rh immune globulin injections

BLOOD GROUP ANTIBODIES:

- Anti-A / Anti-B Antibodies:

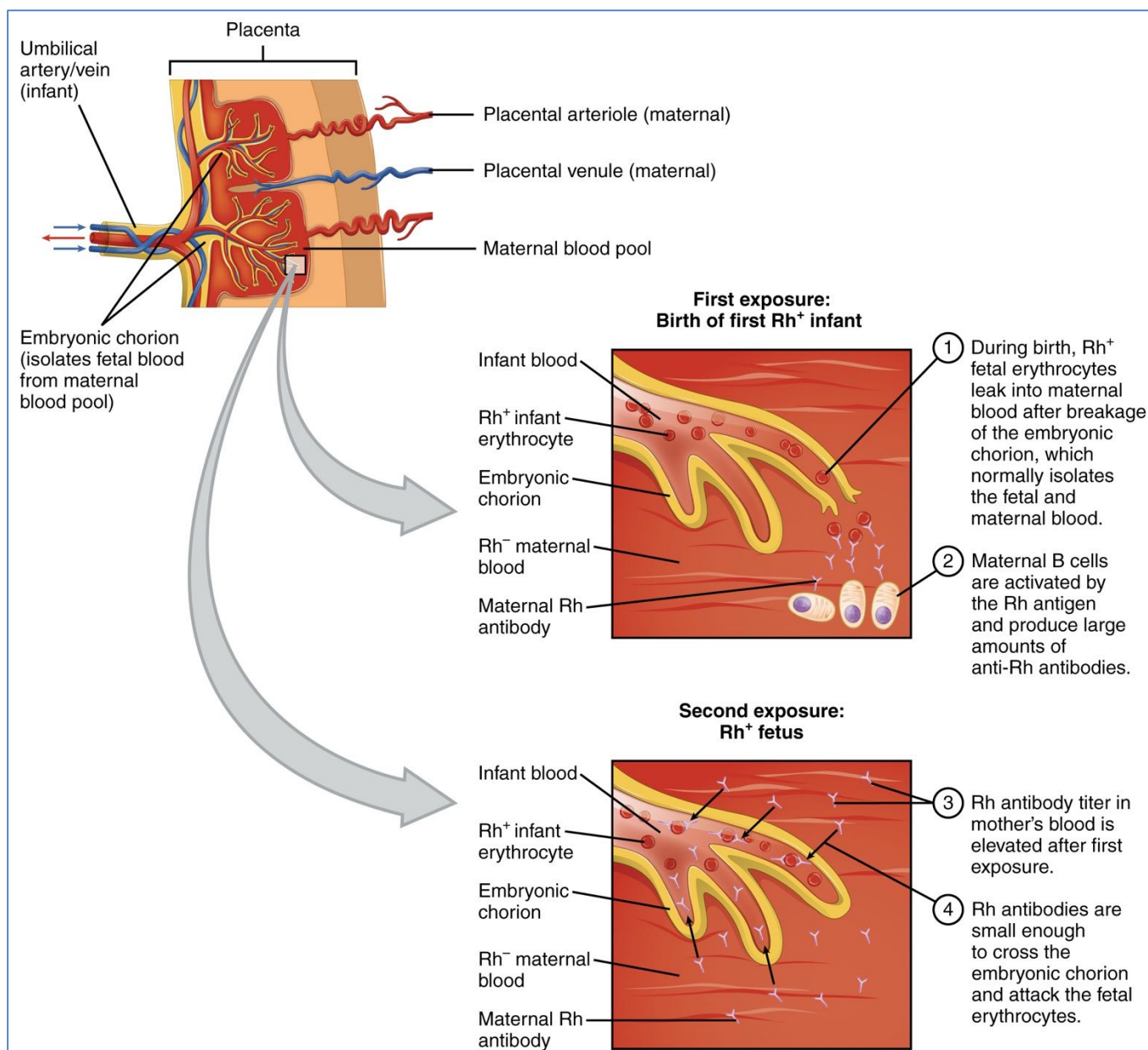
- **Are Naturally-Occurring Antibodies:**
 - Ie: Present at birth
 - Ie: Do not require an immune-sensitizing event
- **Are Immunoglobulins of *type*:**
 - IgM type antibodies
- **Are Present In *plasma* of people who lack the corresponding Antigen**
 - Eg: A-type individual will have 'Anti-B' Antibodies (against B-Antigens)
 - Eg: B-Type individual will have 'Anti-A' Antibodies (against A-Antigens)
 - Eg: O-type individual will have 'Anti-A' & 'Anti-B' Antibodies
- **If Antibodies contact their respective Antigen, A Haemolytic Reaction may occur**
- **Clinical significance:**
 - Determines a patient's ABO-compatibility when receiving transfusions

	Blood Type			
	A	B	AB	O
Red blood cell type				
Isohemagglutinins	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens on red blood cell	 A antigen	 B antigen	 A and B antigens	None

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- **Anti-D Antibodies:**

- **Are Immune Antibodies:**
 - Ie: Produced following an immune-sensitizing event
 - Eg: Via transfusion ...or
 - Eg: Trans-Placental Passage
- **Are Immunoglobulins of type:**
 - IgG type antibodies
 - Note: Only IgG-Ab's are capable of trans-placental passage
- **Most Important IgG = the 'Rh-Antibody' (Anti-D)**
- **Clinical Significance:**
 - Determines a foetus' risk of Haemolytic Disease of the Newborn
 - Determines a patient's absolute ability to receive a Rh-Positive transfusion



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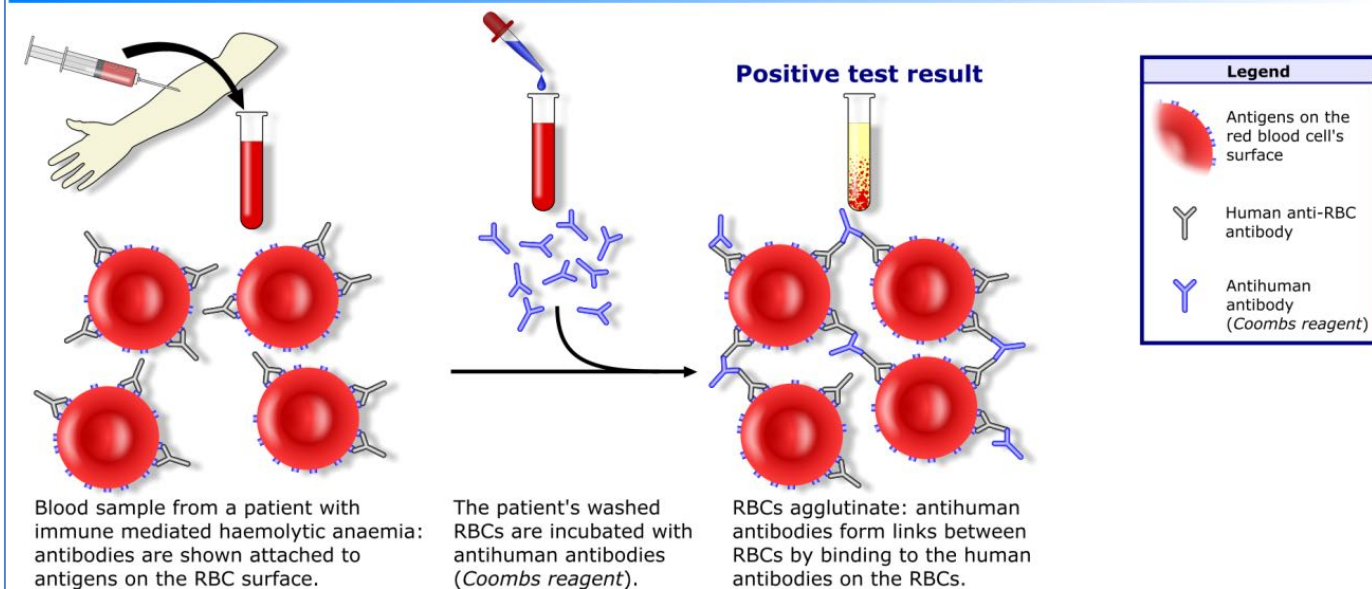
ANTIGLOBULIN TEST (COOMB'S TEST):

- 2 Clinical Blood Tests – Direct & Indirect

○ **Direct (DAT):**

- Detect if antibodies or complement have bound to RBC surface antigens *in vivo*
- Used clinically when immune-mediated *hemolytic anemia* (antibody-mediated destruction of RBCs) is suspected
- A Positive Result → means an immune mechanism is attacking the patient's RBC's

Direct Coombs test / Direct antiglobulin test

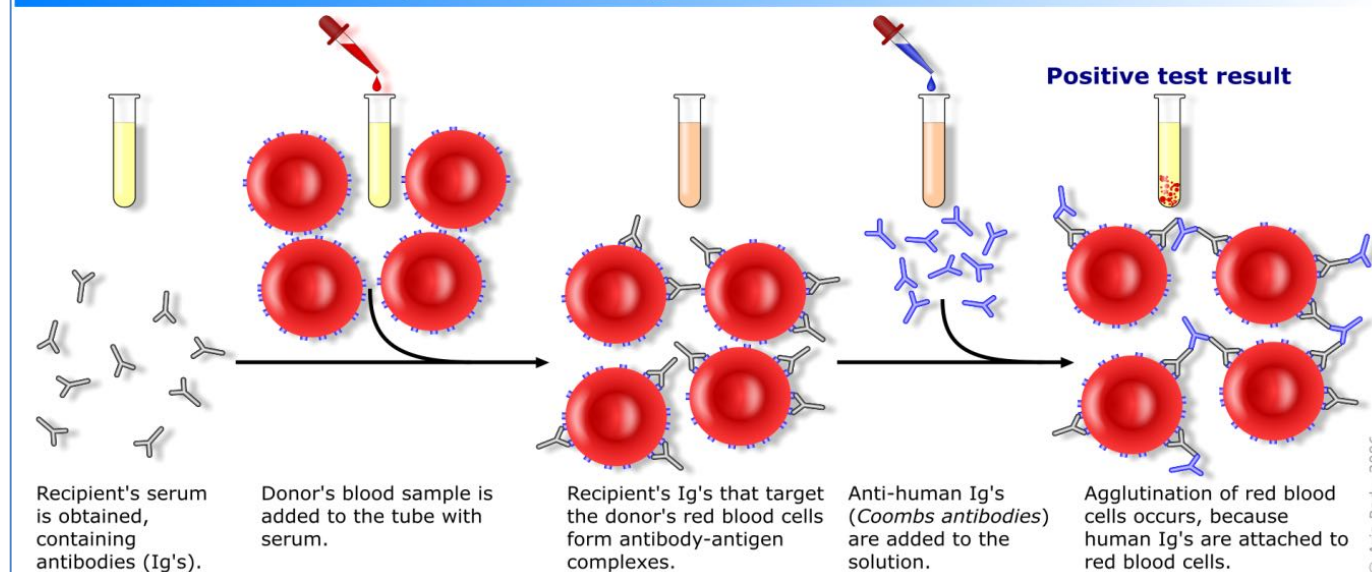


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○ **Indirect (IDAT):**

- Detects antibodies against RBCs present in the patient's serum
- Serum is extracted from the blood, and is incubated with RBCs of known antigenicity
- If agglutination occurs, the indirect Coombs test is positive
- It is used to detect very low concentrations of antibodies present in a patient's plasma/serum *prior to a blood transfusion*
- *In antenatal care*, the IAT is used to screen pregnant women for antibodies that may cause *haemolytic disease of the newborn*

Indirect Coombs test / Indirect antiglobulin test



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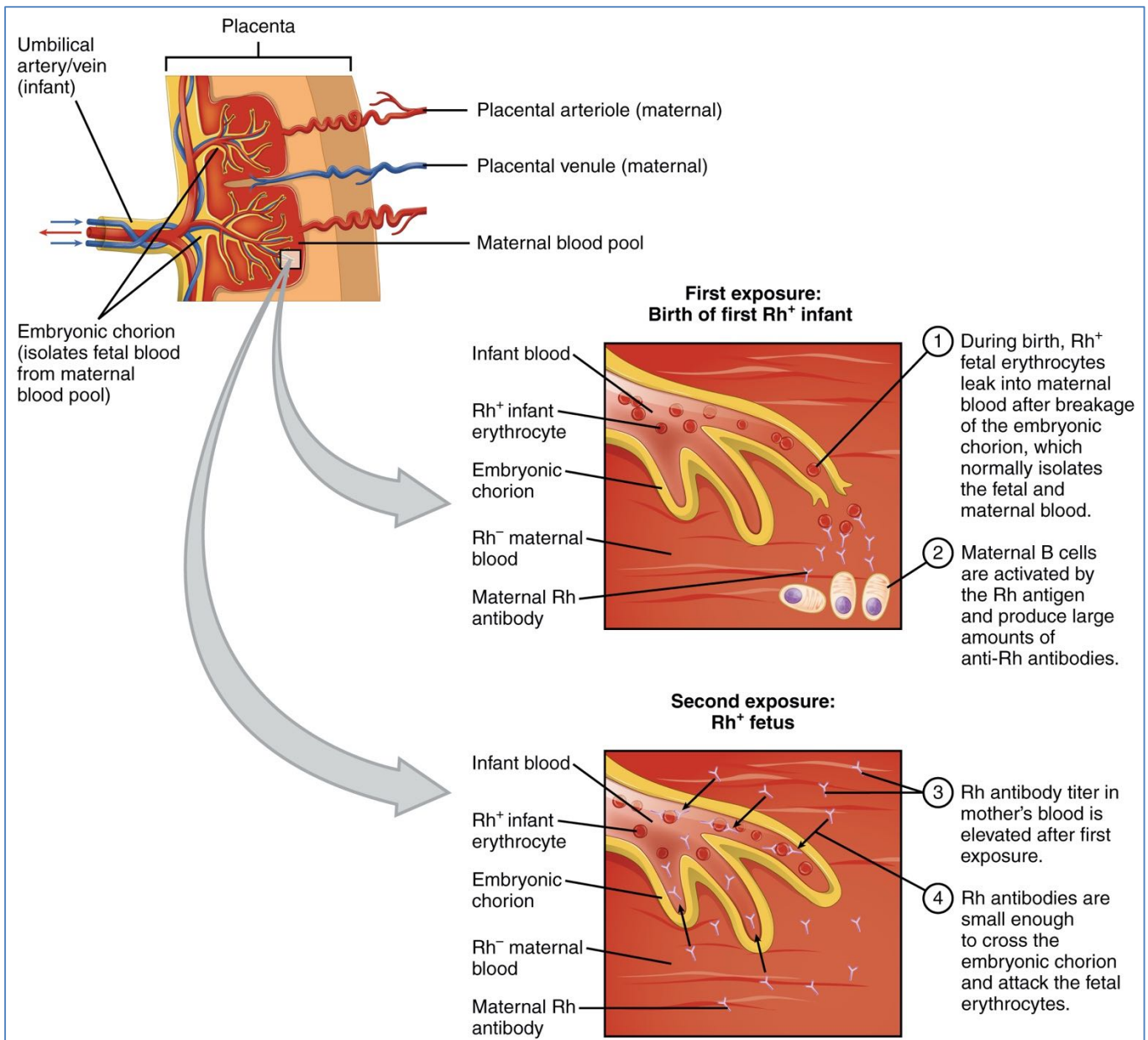
HAEMOLYTIC DISEASE OF THE FOETUS/NEWBORN:

- What is it?

- Condition that develops **in a foetus**, when the **Mother's IgG Anti-RhD Antibodies** crosses the placenta → Foetal Circulation
- → Attacks the red blood cells in the foetal circulation
- The RBC's are broken down and the foetus can develop reticulocytosis and anaemia → Death

- Pathogenesis:

- **Mother is usually O-Negative**
 - Will have Anti-A & Anti-B Antibodies
 - BUT unless she's been exposed to the RhD-Antigen, she won't have Anti-D Antibodies
- **During Pregnancy:**
 - 1: Sometimes Foetal Blood Mixes with Maternal Blood (Eg: Placental Injury/Amniocentesis)
 - 2: If Foetal Blood is *Positive*, The Mother's Immune System Sensitizes to the RhD-Antigen
 - 3: Maternal Immune system produces Anti-D Antibodies
 - 4: These Maternal Immune Antibodies ('IgG'-Ab's) can cross the Placenta → Foetus
 - 5: Antibodies Attack Foetal RBCs → Haemolysis
 - 6: Excess Haemolysis may lead to Jaundice (Heme → Bilirubin)
- **Note: Once Sensitized to RhD, ALL SUBSEQUENT PREGNANCIES ARE AT RISK!**



- **Lab Findings:**

○ **Cord Blood:**

- Rh-D Antigen Present (Positive)
- Positive Direct Coomb's Test (Ie: Mother's Anti-D Antibodies detected on Foetal RBCs)
- ↑ Bilirubin

○ **Maternal Blood:**

- Rh-D Antigen Absent (Negative)
- Positive Indirect Coomb's Test (Ie: High levels of maternal serum Anti-D Antibodies)

- **Prophylaxis Against Rh Sensitization:**

- Passive administration of Exogenous Rh-D-Antibodies into mother can prevent the Primary Immune Response from occurring in the first place
- Ie: The Exogenous Rh-D-Antibodies destroy any foetal Rh-Positive Blood cells (that cross the placenta) before the immune system has time to become sensitized

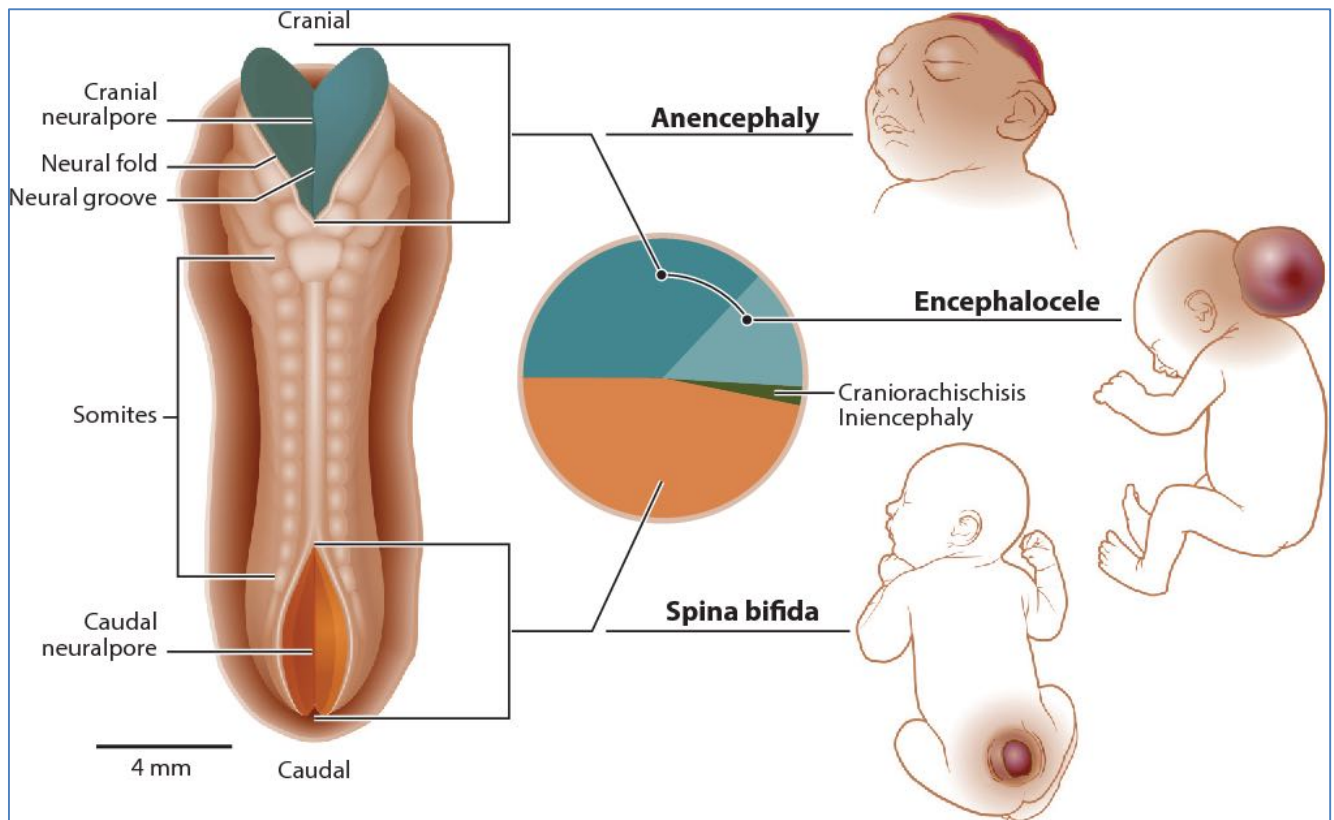
Mother's Rh factor	Father's Rh factor	Baby's Rh factor	Precautions
Rh positive	Rh positive	Rh positive	None
Rh negative	Rh negative	Rh negative	None
Rh positive	Rh negative	Could be Rh positive or Rh negative	None
Rh negative	Rh positive	Could be Rh positive or Rh negative	Rh immune globulin injections

ANTEPARTUM CARE

ANTEPARTUM CARE

Pre-Conception Counselling:

- **Folate Supplementation:**
 - **8-12 weeks pre-conception**
 - **& During first >3mths of pregnancy**
 - ****To prevent neural tube defects (reduce risk of Neural Tube Defects by <70%)**
 - 0.4-1mg daily in low-risk women
 - 5mg daily in higher-risk women (Ie: Previous neural tube defects, on anticonvulsant meds, diabetes, or obesity)
- **Iron Supplementation**
- **Prenatal multivitamins**



**

Molloy, Anne M. et al. "Genetic Risk Factors for Folate-Responsive Neural Tube Defects." *Annual review of nutrition* 37 (2017): 269-291 .

Medication Precautions in Pregnancy:

- **Absolute Contraindications:**

Ace-Inhibitors	- Foetal renal defects - Intrauterine Growth Restriction - Oligohydramnios
Tetracyclines	- Stains infant's teeth - May affect long bone development
Retinoids (Eg: Roaccutane)	- CNS anomalies - Craniofacial Anomalies - Cardiac Anomalies - Thymic Anomalies
Misoprostol	- Mobius syndrome (congenital facial paralysis +/- limb defects) - Spontaneous abortion - Preterm labour

- **Relative Contraindications:**

Phenytoin	- 5-10% risk of Foetal hydantoin syndrome (IUGR, mental retardation, facial dysmorphogenesis, congenital anomalies)
Valproate	- Open Neural Tube Defects (1%)
Carbamazepine	- Open Neural Tube Defects (1-2%)
Lithium	- Ebstein's cardiac anomaly - Goitre - Hyponatraemia
Warfarin	- Risk of spontaneous abortion/stillbirth/prematurity - IUGR - Foetal warfarin syndrome (nasal hypoplasia, epiphyseal stippling, optic atrophy, mental retardation, intracranial haemorrhage)
Erythromycin	- Maternal acute fatty liver
Sulpha Drugs	- Anti-folate properties - Risk of kernicterus
Chloramphenicol	- Grey baby syndrome (Foetal circulatory collapse due to toxic accumulation)

Important Immunisations:

- **PRE-Conception:**

- **Any Live Vaccines should be avoided in pregnancy, so should only be given to a person known to be NOT pregnant**
 - Eg: Polio
 - Eg: Measles/Mumps/Rubella
 - Eg: Varicella
 - Eg: Oral Typhoid
- (Risk of placental and foetal infection)

- **Intra-Partum:**

- **The following vaccines are SAFE to be given during pregnancy:**
 - Tetanus
 - Diphtheria
 - Influenza
 - Hep B
 - Pertussis
- Influenza is recommended for all pregnant women in many health systems

- **Post-Partum:**

- **Rubella** recommended for all new non-immune mothers
- **Hepatitis B** recommended for all new infants (within 12hrs of birth if maternal Hep-B status is either unknown or positive)
- **Any other vaccine** desired/needed is safe post-partum

Optimizing Health Status Before & During Pregnancy:

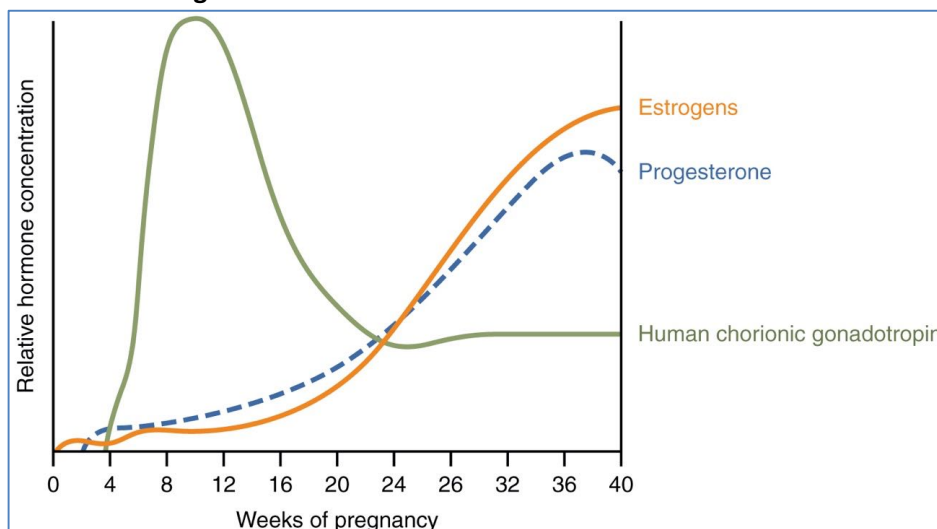
- **Regular, moderate exercise** (Eg: Walking)
 - o Helps you keep strong for the birth
 - o Lifts your mood
 - o Helps maintain a healthy weight
- **Healthy eating**
 - o keeps you feeling good
 - o gives your baby the essential nutrients he needs in utero
- **Avoid certain foods**
 - o Soft cheeses
 - o Raw fish
 - o **Limit caffeine**
- **Drugs & Alcohol:**
 - o **For prescribed drugs**, check with your doctor that they are safe
 - o **Don't smoke**
 - o **No recreational drugs**
 - o **No alcohol**



<https://www.shecares.com/pregnancy/healthy-pregnancy-tips>

Pregnancy hormones

- Between the first 6-12 weeks of pregnancy, oestrogen rises significantly → Prevents menstruation
- These hormone levels **can cause nausea and vomiting (Aka: morning sickness)**
 - o Some women get it, some don't
 - o Morning sickness *usually* abates after 1st 3-4mths, but can last <9mths
- **Can also cause mood swings**



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The Initial Prenatal Visit:

- **Timing of first visit:**
 - 8-12 wks following the first day of LMP
 - Earlier if bleeding/very nauseous/teenage/>35yrs
- **Dating the pregnancy:**
 - Use LMP (last menstrual period) to calculate 'gestational age by dates'
 - Otherwise arrange dating USS (~12wks)
- **Medical History:**
 - Medical conditions
 - Surgical history
 - Gynaecological history
 - Obstetric history
 - Current medications
 - Family history (genetic diseases/birth defects)
 - Social history (smoking/Alcohol/drugs)
- **Physical Examination:**
 - Complete physical exam
 - Baseline blood pressure
 - Baseline weight
 - Pelvic Examination
- **Investigations:**
 - Bloods – FBE, blood group, Rhesus status, Rh-D Antibody screen
 - Urinalysis – screen for bacteriuria & proteinuria
 - Urine PCR for Gonorrhoea/Chlamydia
 - Pap Smear if indicated



Pregnancy Care

Schedule of visits for routine pregnancy care

PLEASE NOTE:

The following schedule is the minimum number of appointments you can expect in a routine pregnancy. Women with more complex needs may require additional appointments. If choosing GP shared care as your model of care, please organise your appointments according to the schedule below. For all other women, these appointments will be arranged for you as your pregnancy progresses.

SCHEDULE OF VISITS	
Confirmation of pregnancy (with your GP)	Visit your GP to: - confirm pregnancy - complete referral to Barwon Health Pregnancy Care Clinic - organise referral to pathology for initial antenatal blood tests and Down Syndrome / genetic screening tests (if desired) - organise referral for 18-20 week morphology ultrasound– booked by you
12-16 weeks	At your first visit to the Pregnancy Care Clinic, you will: - see both a hospital midwife and doctor - review test results, discuss screening tests if not previously ordered - discuss options of care available – Midwife/Barwon Health (BH) Pregnancy Care Clinic/GP Shared Care - organise referral for 18-20 week morphology ultrasound if not already done by GP – this will need to be booked by you
18-20 weeks	Ultrasound of your baby by your chosen ultrasound provider
20-22 weeks	Visit Midwife/Pregnancy Care Clinic/GP shared care - review ultrasound results - organise referral to pathology for routine blood test and diabetes screen (OGTT). These tests need to be done at your chosen pathology provider from 27 weeks, before your 28 week appointment. Contact pathology provider for fasting instructions and to book.
28 weeks	Visit your Midwife/Pregnancy Care Clinic/GP shared care review blood and diabetes screen results If Rh negative blood type, first Anti D injection given
31 weeks	1st baby & previous caesarean only Visit Midwife/BH Pregnancy Care Clinic/GP shared care for standard antenatal appointment If you have had a previous caesarean section you will visit the Pregnancy Care Clinic
34 weeks	Visit with Midwife OR Barwon Health preadmission clinic if Pregnancy Care Clinic or GP Shared Care - organise referral to pathology for routine blood tests and swab to complete Group B Streptococcus (GBS) testing at 36 weeks - discuss induction of labour If Rh negative blood type, second Anti D injection given
36 weeks	1st baby only Visit Midwife/BH Pregnancy Care Clinic/GP shared care for standard antenatal appointment
38 weeks	Visit Midwife/BH Pregnancy Care Clinic/GP shared care for standard antenatal appointment
40 weeks	Visit Midwife/BH Pregnancy Care Clinic/GP shared care for standard antenatal appointment
41 weeks	Visit Midwife/BH Pregnancy Care Clinic/GP shared care for standard antenatal appointment

If you would like more detailed information on what each visit covers, please speak with a midwife or doctor or visit <https://www.barwonhealth.org.au/maternity-services/>

Pregnancy Care Clinic – Outpatients Annexe – 66 Bellerine Street, Geelong (opposite the hospital main entrance)



barwonhealth.org.au

<https://www.barwonhealth.org.au/maternity-services/pregnancy/your-pregnancy-care/schedule-of-visits>

NAUSEA & VOMITING IN PREGNANCY

Background:

- **Almost all pregnant women experience nausea of pregnancy**
- **Mostly occurs in 1st Trimester**

General Management Principles:

- **Rule out other causes:**
 - o Eg: Diabetes / infections / UTI / psychosomatic / anxiety etc
- **Non-Pharmacological:**
 - o Encourage frequent small meals
 - o Oral electrolytes (Eg: Hydralyte / powerade / Gatorade)
 - o Trial cessation of prenatal vitamins
 - o Ginger tablets
- **Pharmacological Options:**
 - o Doxylamine+B6 tablets
 - o Dimenhydrinate
 - o Pyridoxine
 - o Metoclopramide
 - o Vitamin B6 lozenges

HYPEREMESIS GRAVIDARUM:

- **What is it?**
 - o Severe, intractable nausea/vomiting
- **Epidemiology:**
 - o Typically presents in first trimester
 - o Approximately 1% of pregnancies
- **Aetiology:**
 - o Rapidly rising HCG & oestrogen levels
- **Investigations:**
 - o **Rule out other conditions:**
 - Eg: Gastritis
 - Eg: Pyelonephritis / UTI
 - Eg: Thyrotoxicosis
 - Eg: HELLP syndrome
 - o **Bloods:**
 - FBE
 - Electrolytes
 - Renal function / Creatinine
 - LFTs
 - o **Ultrasound:**
 - ?Multiple gestation
 - anatomical cause
 - gallstones
- **Management:**
 - o Consider Thiamine Supplementation
 - o Doxylamine+B6 tablets
 - o Dimenhydrinate
 - o Pyridoxine
 - o Metoclopramide
 - o If severe, consider hospital admission for fluid/electrolyte balance correction
 - o **If very severe (catabolic), total parenteral nutrition**
- **Complications:**
 - o Fluid/Electrolyte disturbance
 - o Mallory-Weiss tear
 - o Wernicke's encephalopathy (if severe and prolonged)
 - o IUGR (Intrauterine Growth Restriction)

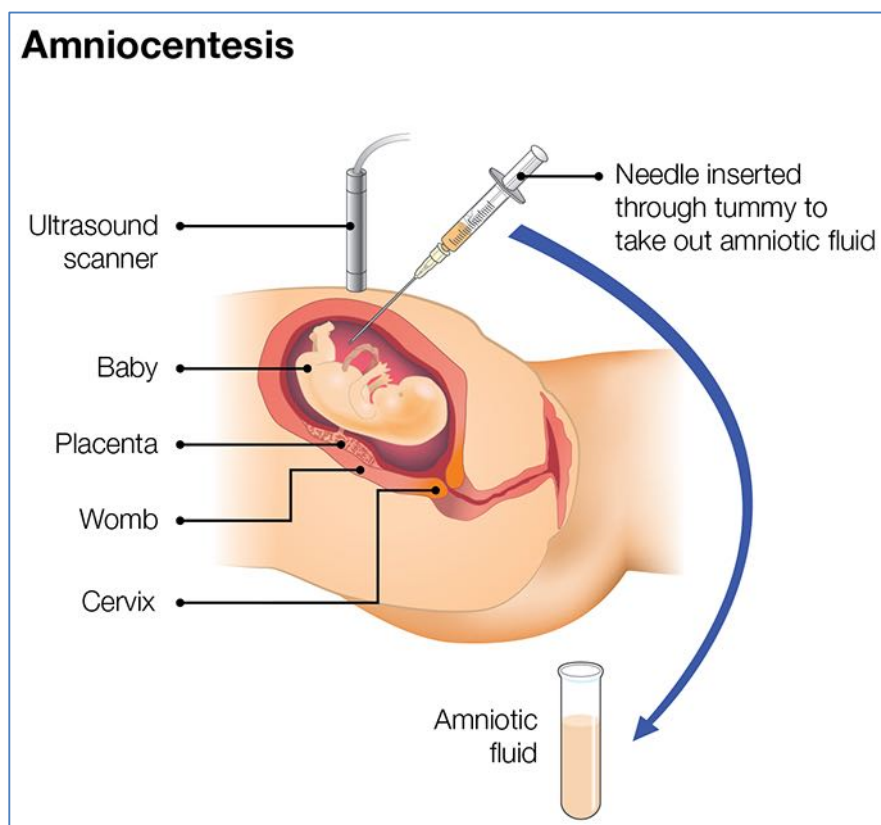
PRENATAL SCREENING TESTS:

Ultrasound Screening:

- **8-12 weeks:**
 - o Dating scan
 - o The most accurate way to estimate the gestational age of the foetus and calculate due dates
- **11-14 weeks:**
 - o Nuchal translucency scan
 - o Is an early screening test for Down's Syndrome (Trisomy 21)
 - o Can also detect cardiac defects & Turner's syndrome
- **18-20 weeks:**
 - o Growth scan
 - o Anatomy screen

Amniocentesis (15-16wks gestation):

- **What is it?**
 - o An ultrasound-guided transabdominal extraction of amniotic fluid (to pick up foetal cells)
- **Why is it done?**
 - o Detect genetic anomalies (Eg: Trisomy 21)
 - o Can also detect Open Neural Tube Defects
 - o May be to confirm a suspicion raised through ultrasound/other non-invasive testing
 - o Can also screen for foetal lung immaturity using the L/S ratio (Lecithin:Sphingomyelin)
 - If >2:1, then infant respiratory distress syndrome is less likely
- **Risks:**
 - o 1/200 risk of procedure-related pregnancy loss (1/100 if multiple gestation)



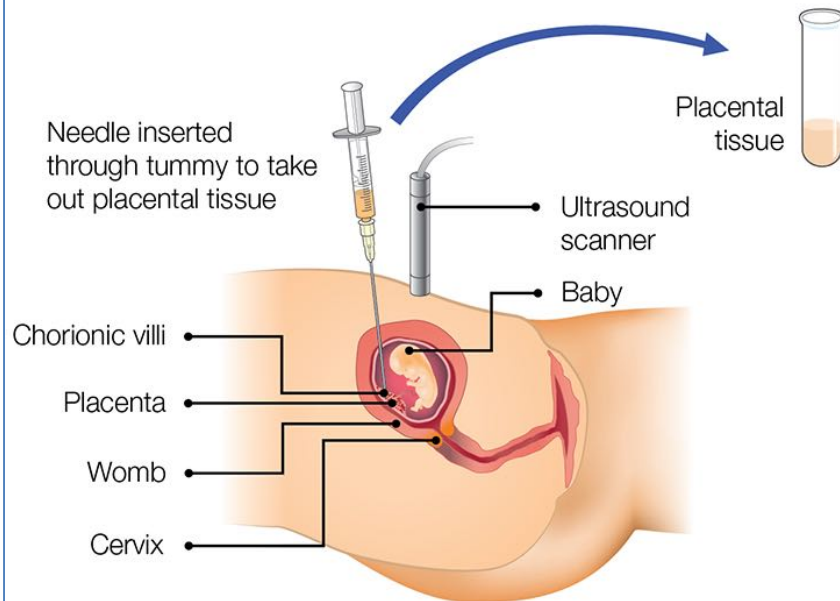
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Chorionic Villus Sampling (10-12wks gestation):

- **What is it?**
 - A biopsy of foetal chorion using a trans-cervical catheter
- **Why is it done?**
 - Same as amniocentesis (to detect genetic anomalies), but enables an earlier diagnosis
 - Earlier diagnosis enables earlier termination of pregnancy than with amniocentesis
- **Risks:**
 - 1-2% risk of procedure-related pregnancy loss
 - 1-2% chance of genetic mosaicism (which yields false negative results)

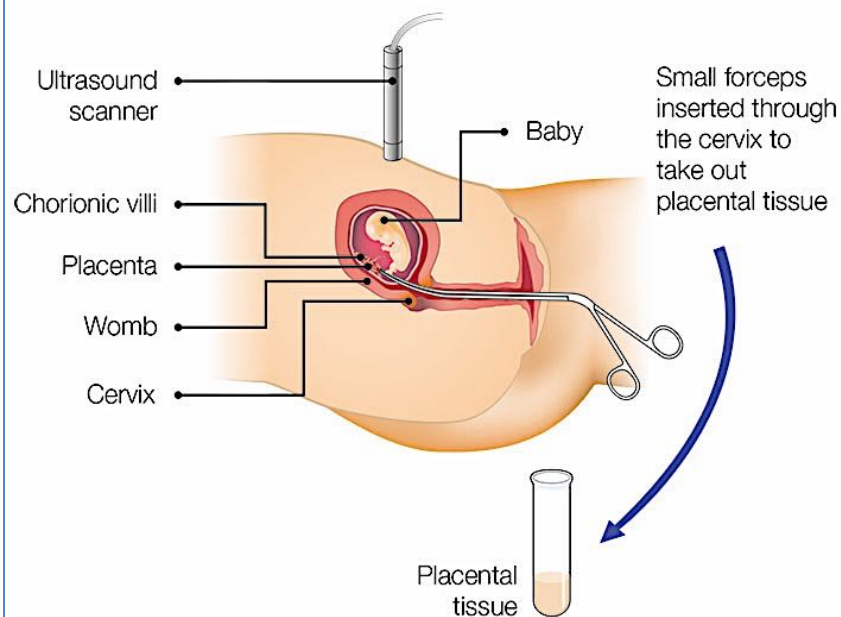
Chorionic villus sampling (CVS)

transabdominal method



Chorionic villus sampling (CVS)

transcervical method



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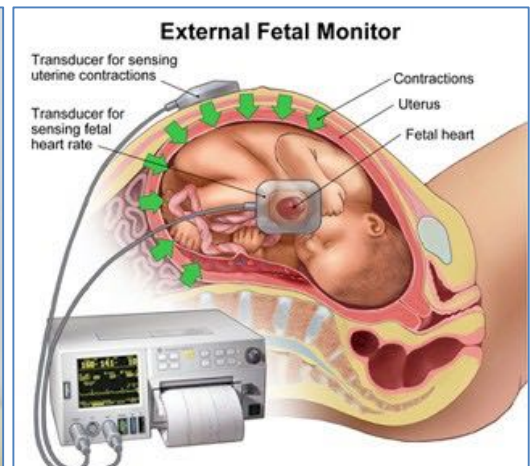
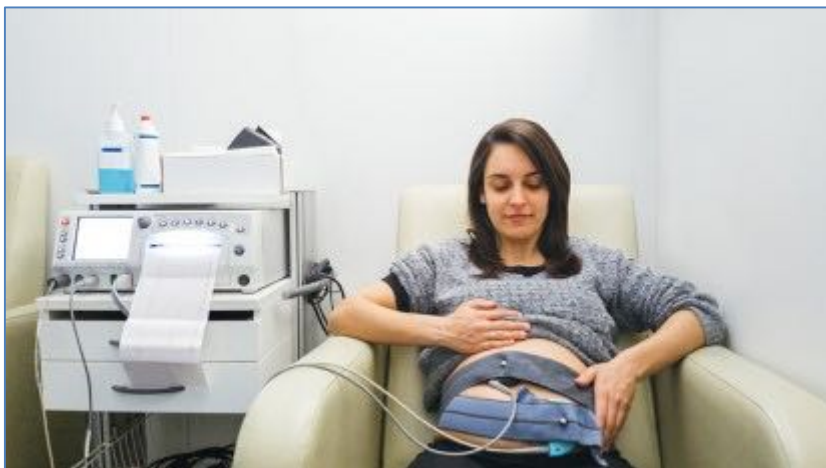
ANTENATAL FOETAL SURVEILLANCE

Foetal Movements:

- Can generally be noticed by the mother around 16-20wks gestation
- Foetal movements can be counted by the mother at a time when the foetus is normally active
- All high-risk women are recommended to do Foetal Movement Counts
 - o >6 movements in a 2hr period is expected
 - o If <6 movements in a 2hr period, a mother is advised to present to her dr/midwife for assessment

Foetal Non-Stress Test:

- **Indication:**
 - o Used to evaluate a baby's health before birth
 - o Goal is to assess the baby's oxygen supply by checking the heart rate and how it responds to the baby's movement
- **Premise:**
 - o Normally, a baby's heart rate will have spontaneous 'accelerations' if it is well oxygenated
 - o Conditions such as foetal hypoxia can disrupt the heart rate
- **Quick Interpretation:**
 - o **Normal:** >2 accelerations of >15bpm for >15s in 20min period
 - o **Abnormal:** <2 accelerations in 40min period
 - o (If nil accelerations or foetal movements in 20mins, stimulate foetus by fundal pressure/acoustic/vibratory stimulation)
- **Full Interpretation** (*Note: exact criteria may vary depending on your local health system):
 - o **Normal / 'Reactive':**
 - Shows a baseline foetal heart rate between 110-160bpm
 - Moderate variability (5-25-interbeat variability)
 - And 2 accelerations in 20mins with no decelerations
 - Each acceleration must increase the FHR by 15bpm above baseline for at least 15s for a foetus >32wks gestation; or 10bpm for >10s if <32wks gestation)
 - **Normal NST is reassuring**
 - o **Atypical / 'Non-Reactive':**
 - Low (100-110bpm) or elevated (>160bpm) FHR for no longer than 30mins
 - Absent or minimal variability for 40-80mins
 - Variable decelerations of 30-60s duration
 - **Further assessment is required**
 - o **Abnormal / 'Non-Reactive':**
 - Bradycardia (<100bpm) or Tachycardia (>160bpm) FHR for over 30mins
 - Sinusoidal variability of 25bpm for >10mins
 - Variable decelerations of >60s duration
 - **Urgent action is required >> USS or BPP is required**
 - **→ May require induction/delivery**



<https://hartfordhospital.org/services/womens-health-services/departments-services/pregnancy-childbirth/maternal-fetal-medicine/non-stress-test-nst>

OBSTETRICAL HAEMORRHAGE

OBSTETRICAL HAEMORRHAGE

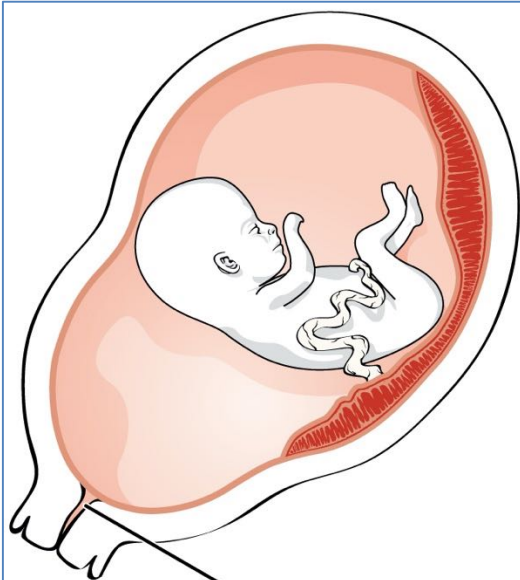
(*Vaginal bleeding from 20wks gestation to term)

Differential Diagnoses:

- 'Bloody show' (ie: Shedding of the cervical mucous plug)
- **Placenta Previa**
- **Placental Abruption** (Abruptio Placentae)
- **Vasa Previa**
- **Cervical Lesions** (Including cervicitis, cervical polyps, ectropion, cervical cancers)
- **Uterine rupture**

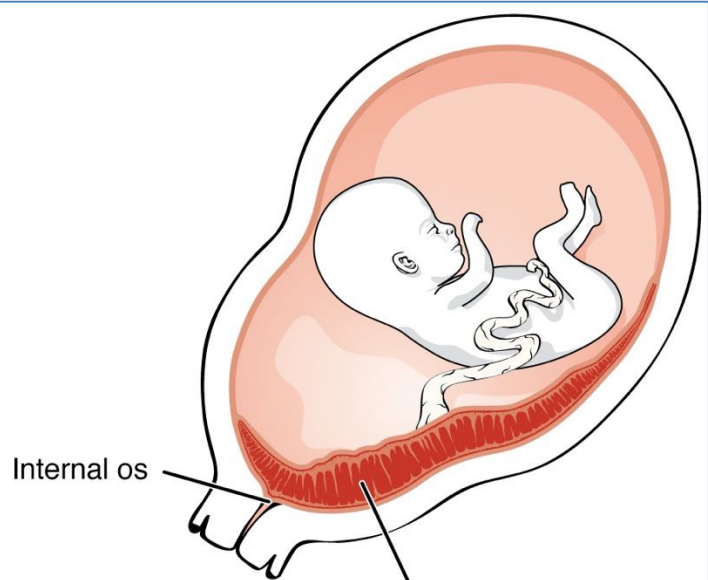
PLACENTA PREVIA:

- **What Is It?**
 - o Where the placenta is implanted in the lower segment of the uterus and presents ahead of the leading pole of the foetus
- **Clinical Features:**
 - o Painless**
 - o Bright red, recurrent vaginal bleeding
 - o Varies in degree, but can be catastrophic
- **Examination Findings:**
 - o Uterus soft/non-tender
 - o Elevated/displaced presenting foetal part
 - o Normal FHR
- **Investigations:**
 - o TVUS (transvaginal US); can also use transabdominal
 - o Repeated TVUS in third trimester if placental is between 20mm of overlap and 20mm away from internal os after 20wk mark
- **Complications:**
 - o Prematurity (as bleeding often requires early caesarean section)
 - o Intrauterine hypoxia of foetus
 - o Foetal malpresentation
 - o PPROM
 - o <1% maternal mortality (hypovolaemic shock, acute renal failure, sheehan syndrome)
 - o Anaemia
 - o Placenta accreta
 - o Hysterectomy
- **Management:**
 - o Aim to maintain pregnancy until risk of continuing pregnancy outweighs risk of delivery
 - o Haemodynamic stabilisation (IV fluids, oxygen)
 - o Anti-D immunoglobulin if mother is Rh-negative
 - o **If <37wks gestation & minimal bleeding →**
 - Admit to hospital
 - Consider corticosteroids for foetal lung maturation
 - Limit physical activity to prevent triggering further bleeds
 - o **If >37wks gestation &/or profuse bleeding →**
 - Consider immediate delivery via caesarean section



Cervix is not obstructed.

Normal location of placenta



Internal os

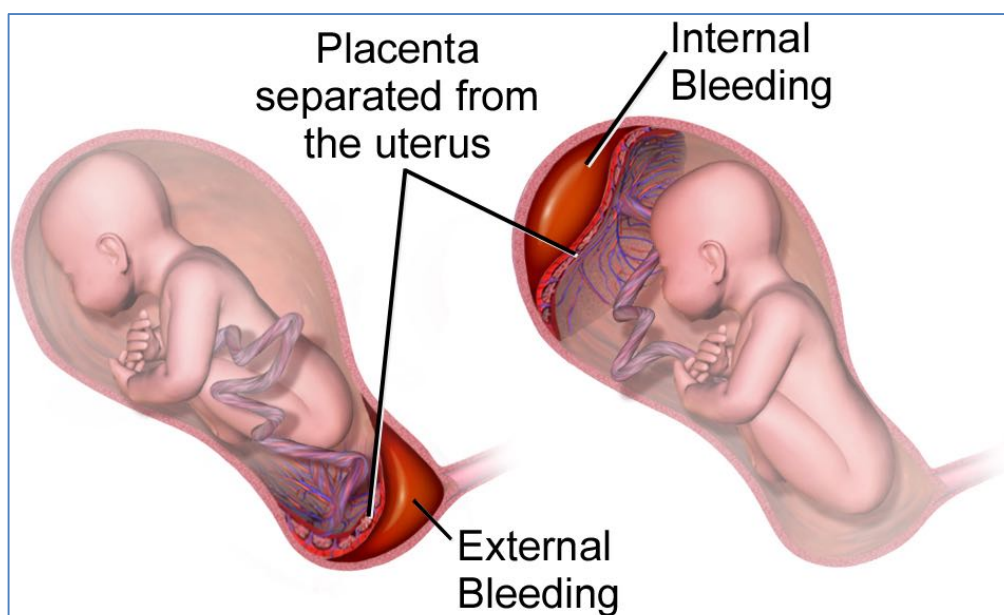
Placenta is covering cervix preventing a proper birth.

Placenta previa

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PLACENTAL ABRUPTION:

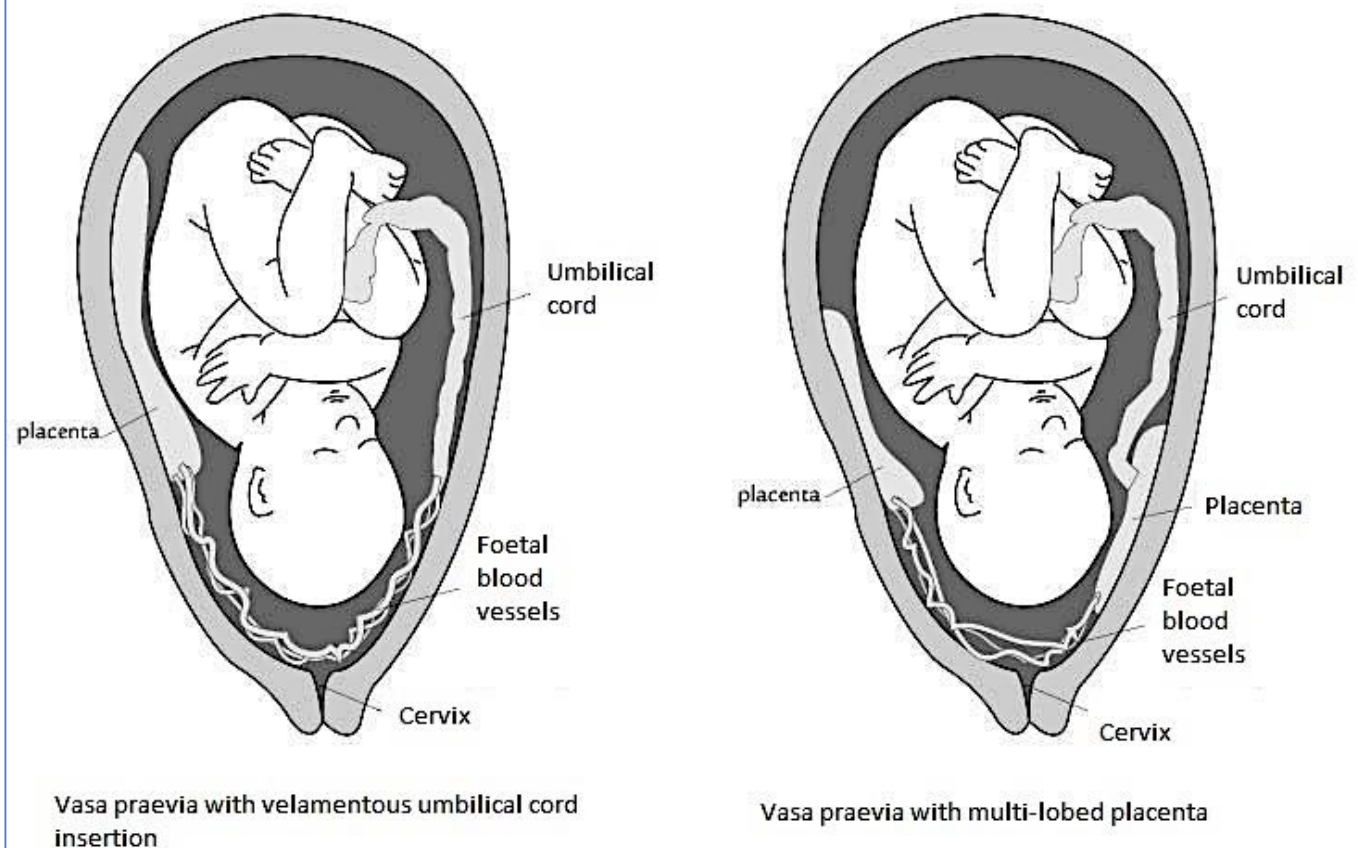
- **What Is It?**
 - Placental detachment from the uterine wall, caused by bleeding at the decidual-placental interface
- **Clinical Features:**
 - Painful** - sudden onset, constant, localised to lower back and uterus
 - Typically occurs >20wks gestation
 - **May be partial or total (Total → Foetal death is inevitable)**
 - Blood may dissect upwards towards the foetus, or downwards towards the cervix
 - Vaginal bleeding unless abruption is concealed
- **Examination Findings:**
 - Uterine tenderness
 - Uterine contractions/Hypertonic uterus
 - Shock/anaemia
 - Foetal distress
 - ~15% may present with foetal death
 - Bloody amniotic fluid
- **Investigations:**
 - Clinical diagnosis
- **Complications:**
 - Foetal mortality 25-60%
 - Foetal prematurity
 - Intrauterine hypoxia
 - Maternal mortality ~1%
 - Risk of DIC ~20%
 - Maternal shock (acute renal failure, anaemia, sheehan syndrome, amniotic fluid embolus)
- **Management:**
 - Haemodynamic stabilisation (IV fluids, oxygen)
 - Cross match and hold blood products (due to DIC risk)
 - Anti-D immunoglobulin if mother is Rh-negative
 - **Mild Abruption:**
 - **<37wks:** Monitor concealed bleeding with serial maternal haematocrits → consider delivery
 - **>37wks:** Stabilize mother and deliver
 - **Moderate to severe abruption:**
 - Vaginal delivery if no contraindication & no foetal distress
 - If foetal distress, vaginal delivery is contraindicated, or labour fails to progress → Caesarean



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VASA PREVIA:

- **What is it?**
 - o When foetal vessels pass over the cervical os
- **Clinical Features:**
 - o Approx 1 in 5000 deliveries
 - o Higher incidence in multiple gestation pregnancies
 - o Painless vaginal bleeding
 - o Foetal distress (Sinusoidal tachy-brady pattern on foetal non stress test)
- **Investigations:**
 - o 'Apt test' to determine if source of bleeding is foetal (pink colour change if foetal; yellow if maternal)
 - o Wright stain on blood smear for nucleated (foetal) RBC's
- **Complications:**
 - o 50% foetal mortality if undiagnosed
- **Management:**
 - o Emergency caesarean delivery



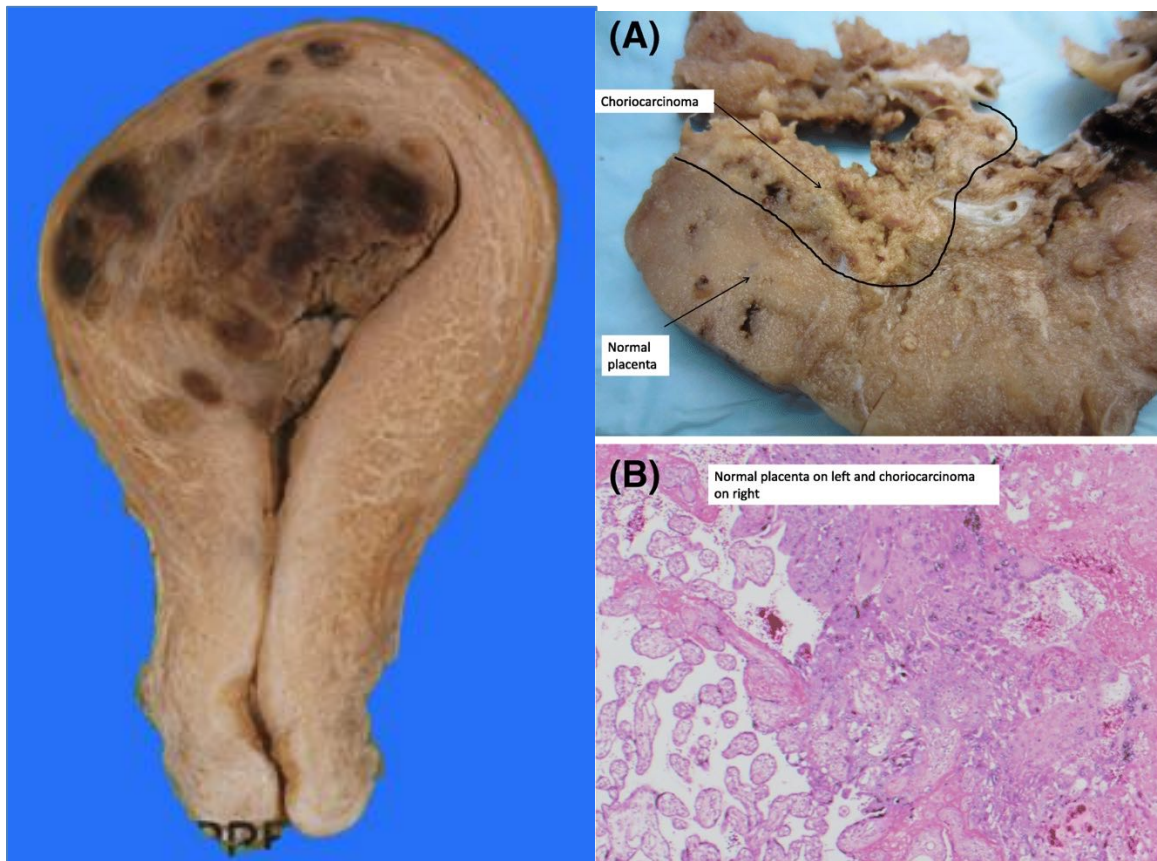
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OBSTETRIC COMPLICATIONS

OBSTETRIC COMPLICATIONS

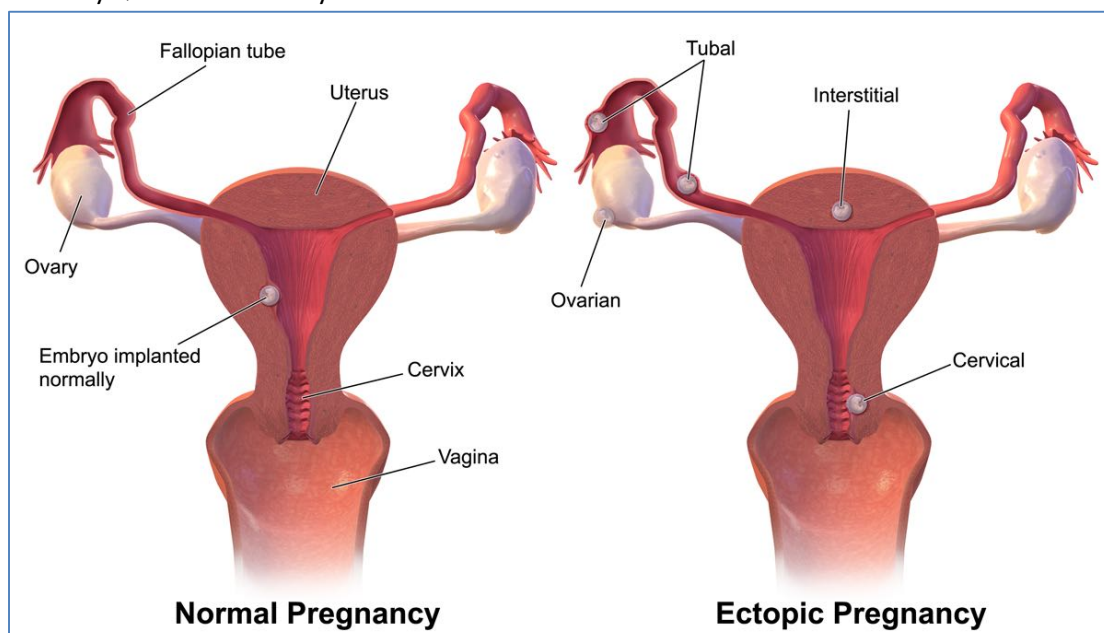
CHORIOCARCINOMA (MALIGNANT):

- **Aetiology:**
 - Risk Factors – Extremes of age <20, >40, previous abortion, abnormal gestation
 - May be De-Novo (Primary) or may progress from a Complete Mole (Secondary)
- **Pathogenesis:**
 - **May be De-Novo (Primary) or may progress from a Complete Mole (Secondary)**
 - High Grade Primary Malignancy of the Trophoblasts
 - May evolve Secondary to an Invasive/Complete Hydatidiform Mole
- **Morphology:**
 - **Macro:**
 - Invasive
 - Haemorrhagic
 - Necrosis
- **Clinical Features:**
 - Irregular Vaginal Bleeding
 - Uneven Swelling of Uterus (Mass)
 - Abdominal/Pelvic Pain
 - **Diagnosis:**
 - Rising hCG
 - Abdo US → Abdo CT
 - **Metastasis to Lungs is common → Haemoptysis**
- **Treatment:**
 - **Surgical Excision**
 - **+ Chemotherapy (Methotrexate)** – Good Prognosis
- **Prognosis:**
 - Types:
 - Gonadal (in the ovary – Not related to gestational) – Poor prognosis
 - Gestational (in the uterus – associated with pregnancy) – Good Prognosis – 100% cure rate with therapy



ECTOPIC PREGNANCY:

- **Aetiology:**
 - 50% Idiopathic
 - **Risk Factors:**
 - Obstruction
 - PID
 - Fallopian Stricture
 - IUD
 - Endometriosis
- **Pathogenesis:**
 - Implantation outside the uterus (Often within the fallopian tube wall)
- **Morphology:**
 - **Macro:**
 - 90% occur in Fallopian Tubes
 - May occur in the Abdomen
 - **Micro:**
 - Normal placental infiltration – Just in the wrong place
- **Clinical Features:**
 - 1% of pregnancies
 - **Symptoms:**
 - May mimic a normal early pregnancy – (Missed Periods, Breast Tenderness, Nausea)
 - *Sharp, Stabbing Pain (Pelvic/Abdominal)
 - *Vaginal Bleeding/Spotting
 - **Peritonitis/Shoulder Pain if Rupture = MEDICAL EMERGENCY
 - **Diagnosis:**
 - B-hCG (Pregnancy Test)
 - Abdominal Ultrasound – (If scan is –Ve, re-test hCG & re-scan every 2-3 days until foetus can be located)
 - **Complications:**
 - **Rupture → Massive Intraperitoneal Haemorrhage → Shock → **Death
 - Spontaneous Abortion
 - Chorioamnionitis
- **Treatment:**
 - If early – Medical Abortion (Methotrexate + Misoprostol)
 - If later – Surgical Abortion (Laparoscopic Salpingotomy)
- **Prognosis:**
 - Good if treated
 - May → Some infertility

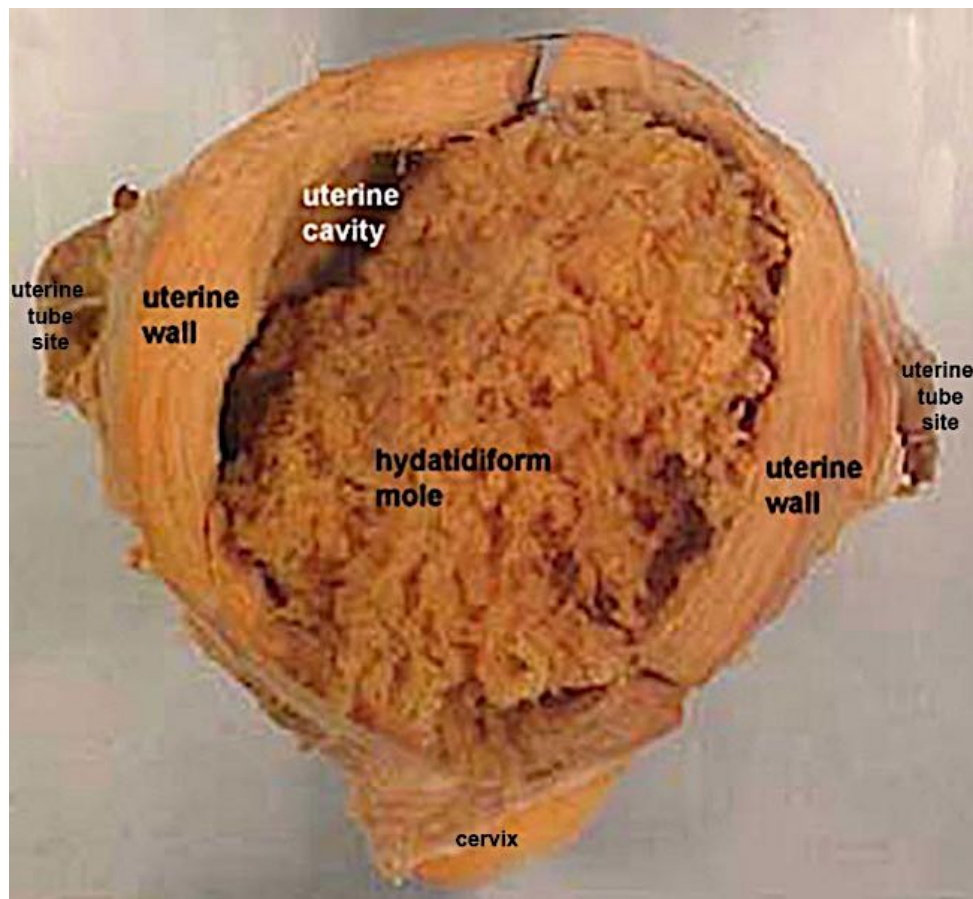


HYDATIDIFORM MOLES – (PARTIAL & COMPLETE):

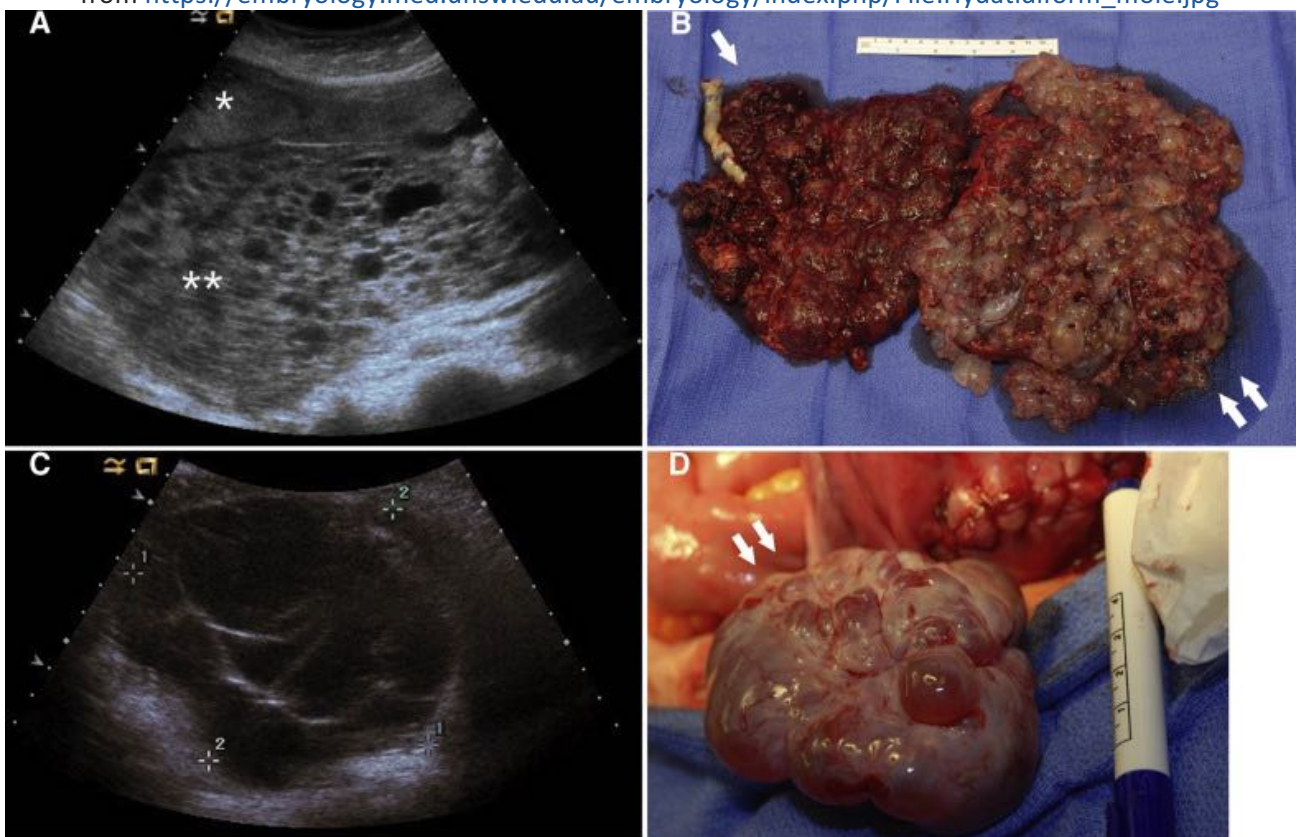
- **Aetiology:**
 - Error in Fusion of Gametes → Abnormal Karyotype → NOT compatible with life
 - **Pathogenesis:**
 - **Overproduction of Trophoblastic (Placental) Tissue** due to:
 - **1: Fusion of 2x Sperms with 1x Ovum → Triploidy → *Partial Mole***
 - → Abnormal Placenta & Some Foetal Development
 - **2: Fusion of 1x Sperm with an Ovum that has LOST its DNA; OR Fusion of 2x Sperm inside an EMPTY Ovum → Sperm Duplicates → DIPLOIDY → *Complete Mole***
 - → Abnormal Placenta, but NO Foetus
 - → **Hyperplasia of Trophoblastic Tissue + Vesicular Distension of Chorionic Villi**
 - (Note: ↑↑↑β-hCG can mimic TSH → Secondary Hyperthyroidism)
 - **Morphology:**
 - **Partial Mole:**
 - **Macro:**
 - Partially cystic & few Blood vessels
 - **ONLY SOME Chorionic Villi are Cystic (Hence “Partial”)**
 - **Micro:**
 - Focal Hyperplasia of Trophoblasts
 - **Complete Mole:**
 - **Macro:**
 - All Villi are Cystic, NO Blood Vessels within villi
 - **ALL Chorionic Villi are Cystic (Hence “Complete”)**
 - Grape-Like Appearance of Villi
 - Entire Uterine cavity is filled with swollen villi
 - **Micro:**
 - Diffuse Hyperplasia of Trophoblasts
 - **Clinical Features:**
 - Abnormal Growth of Uterus
 - Severe Morning Sickness (N/V)
 - Painless Vaginal Bleeding in 1st Trimester
 - Symptoms of Hyperthyroidism (Heat Intolerance, Diarrhoea, Tachycardia, Tremor)
 - Symptoms of Pre-Eclampsia (Hypertension, Oedema)
 - **Diagnosis:**
 - B-hCG Level (↑ in Partial; ↑↑↑ in Complete/Invasive; ↑↑↑ in Choriocarcinoma)

↑ βHCG → High grade → Poor prognosis.

 - Pregnancy Ultrasound → Abnormal Placenta (“Snowstorm”/“Grape Cluster” Uterus)
- **Treatment:**
 - Surgical Termination (D&C) + Follow-up B-HCG levels
 - (Note: Invasive/Metastatic Moles may require Chemotherapy – Methotrexate)
- **Prognosis:**
 - 80% are Benign (Partial) Moles
 - **20% may become Invasive (Complete) Moles → Choriocarcinoma**
 - Good Prognosis (~100%) with Treatment



Hill, M.A. (2022, February 23) **Embryology** *Hydatidiform mole.jpg*. Retrieved from https://embryology.med.unsw.edu.au/embryology/index.php/File:Hydatidiform_mole.jpg



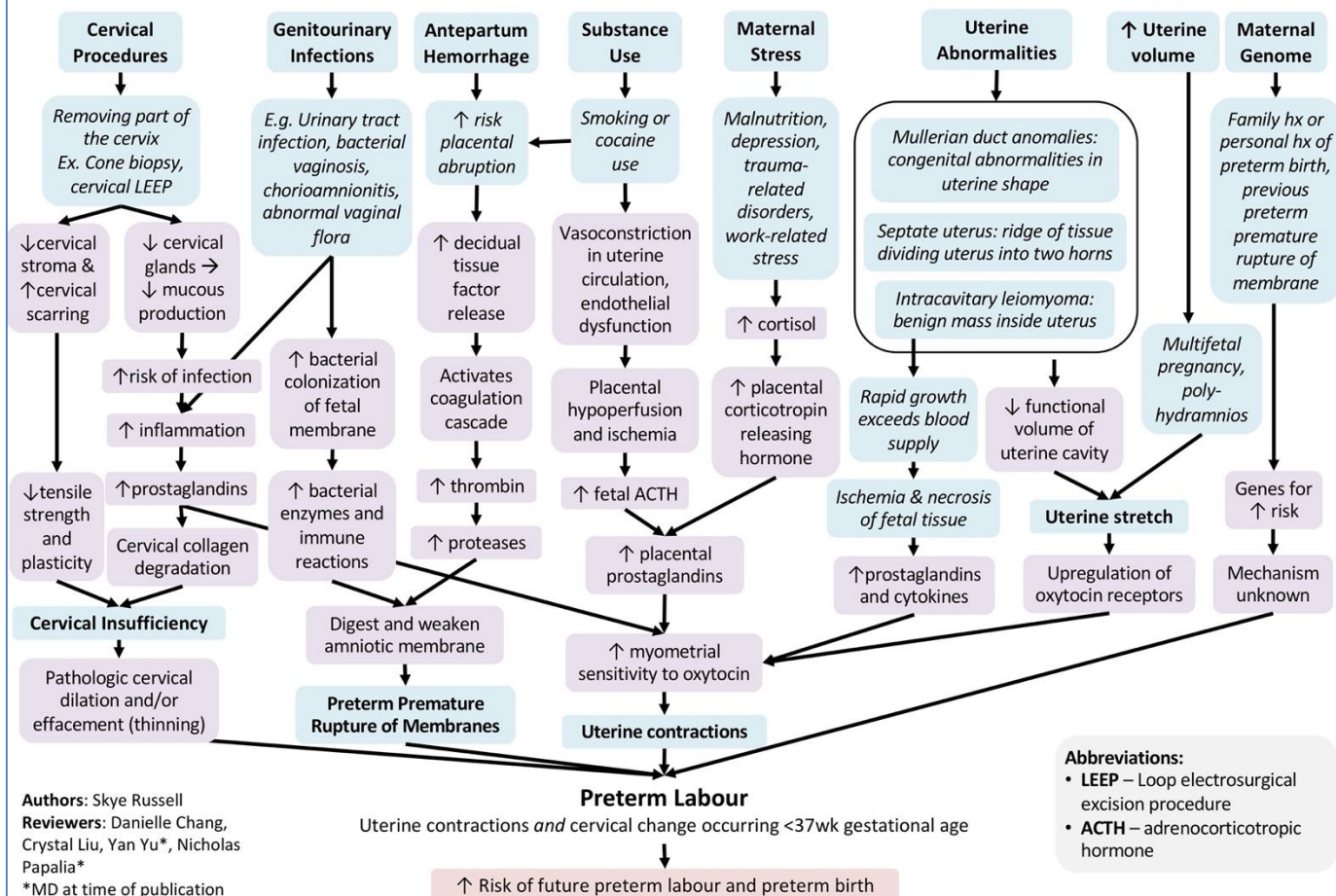
[https://www.ajog.org/article/S0002-9378\(18\)31148-7/fulltext#relatedArticles](https://www.ajog.org/article/S0002-9378(18)31148-7/fulltext#relatedArticles)

PRETERM LABOUR

- **What is it?**
 - Onset of labour between 20wks & 37wks gestation
- **Aetiology:**
 - Mostly idiopathic
 - But can also be triggered by:
 - Maternal factors: Eg: infection, HTN, diabetes, injury, poor nutrition, drugs, etoh, smoking
 - PPROM
 - Polyhydramnios
 - Placenta previa/abruption/insufficiency
 - Foetal factors: Eg: Multiple gestation, congenital abnormalities, foetal stress, hydrops
 - Uterine factors: Eg: Leiomyomas, mullerian duct abnormalities
- **Epidemiology:**
 - Accounts for approximately 10% of pregnancies
- **Risk Factors:**
 - Prior hx of Preterm Labour
 - Prior hx of cervical cone biopsies or mechanical dilation
 - Cervical length on TVUS (>30mm has reduced risk of preterm labour)
 - Smoking
 - Later maternal age
 - Multiple gestations
- **Clinical Features:**
 - Regular contractions + Cervical dilation @ >20<37wks gestation
- **Management:**
 - **Initial Things:**
 - Admit to hospital/birthing facility
 - Hydration
 - Left lateral position + bed rest
 - Foetal USS (for position, placental location)
 - **Suppression of Labour:**
 - **Aim:** Delay delivery to allow for betamethasone to aid in foetal lung maturation
 - **Prerequisites:** Live immature foetus, intact membranes, minimally dilated cervix <4cm
 - **Contraindications:** Maternal bleeding, preeclampsia/eclampsia, chorioamnionitis. Or foetal distress/demise, IUGR
 - **Agents:**
 - Calcium channel blockers (Nifedipine)
 - Prostaglandin synthesis inhibitors (Indomethacin)
 - (Magnesium sulphate – primarily for eclampsia)
 - Betamethasone (to enhance foetal lung maturity)
- **Prognosis:**
 - >30wks/1500g → 90% survival
 - >33wks/2000g → 99% survival



Preterm Labour: Pathogenesis & Maternal Complications



Authors: Skye Russell
Reviewers: Danielle Chang, Crystal Liu, Yan Yu*, Nicholas Papalia*
 *MD at time of publication

Legend: Pathophysiology Mechanism Sign/Symptom/Lab Finding Complications Published July 12, 2020 on www.thecalgaryguide.com



Source: www.thecalgaryguide.com

PREMATURE RUPTURE OF MEMBRANES (PROM):

- **Premature rupture of membranes (PROM)** = Rupture of the amniotic sac *before onset of labour*
 - o **Prolonged PROM** = >18-24 hours before labour
- **Preterm PROM (PPROM)** = PROM before 37 weeks gestation

(Note: PROM is a variation of normal; whereas PPRM is often pathological and can be dangerous)

(Note: Typically, labour begins <48hrs of PROM)

(Note: Sometimes the rupture may heal spontaneously)

- **Risk Factors:**

- o Bacterial Infection - Eg: Chorioamnionitis, Maternal Sepsis, vaginitis, cervicitis, STI, UTI
- o Multiparity
- o Cervical incompetence
- o Multiple gestation
- o Smoking
- o Anatomic Defect of:
 - the amniotic sac
 - uterus
 - cervix
 - the foetus
- o Previous PROM/PPROM

- **Clinical Features:**

- o Pregnant + Sudden Fluid Gush +/- continued leakage

- **Assessment**

- o proper medical history
- o Sterile spec gynaecological exam (look for fluid pooling or leakage from cervix)
- o Nitrazine paper test (Amniotic fluid turns paper blue)
- o Ultrasound to rule out foetal anomalies & assess presentation
- o Amniotic fluid smear cytology/microscopy (dried for 10 minutes on a slide shows a characteristic fernlike pattern)

- **Management**

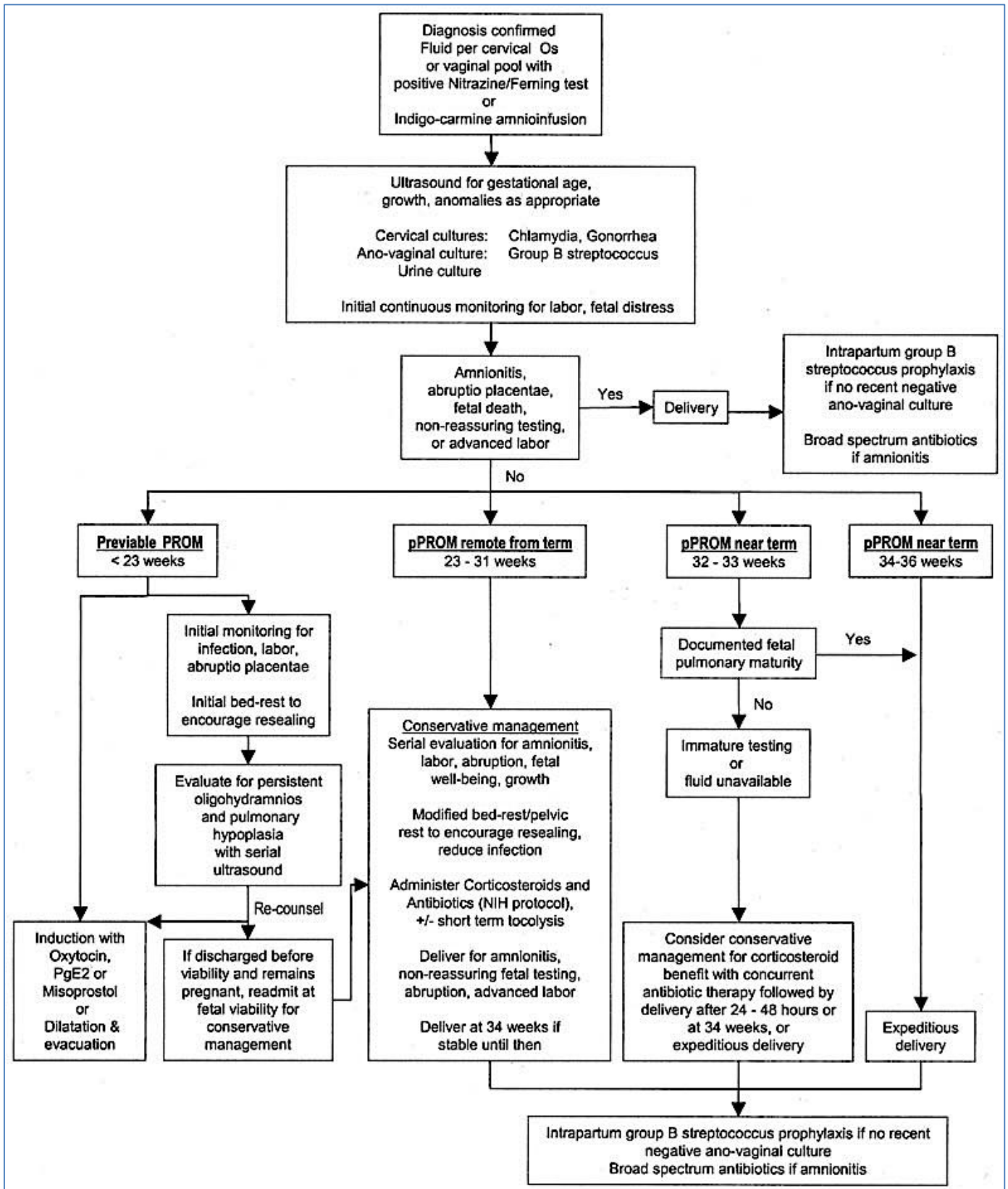
- o **PROM (>37wks):**
 - Permit spontaneous labour (up to 12hrs)
 - Induction of labour (@ 12 hours) if it has not already begun
 - Consider Group B Streptococcal prophylaxis (@ 18 hours)
- o **PPROM (<37wks):**
 - Admit mother
 - Steroids (2days)
 - Watch for Preterm Labour
 - Watch for Chorioamnionitis
 - Antibiotic prophylaxis (Ampicillin/Erythromycin) to delay delivery → ↑Foetal Maturation, & prevent sepsis
 - Avoid labour *prior to 34wks* (Tocolysis may be used)
 - Induction of labour @ 34 weeks
- o **Infection**
 - **Chorioamnionitis:**
 - Antibiotic therapy to avoid sepsis
 - Induction/Delivery is indicated
 - **Foetal:**
 - If the GBS status of the mother is unknown, Antibiotic Prophylaxis (Penicillin/other) protects against vertical transmission of Group B streptococcal infection

• **Prognosis:**

- o 90% of women with PROM at 28-34 wk GA go into spontaneous labour within 1 wk
- o 50% of women with PROM at <26 wk GA go into spontaneous labour within 1 wk

Example of a management algorithm for PROM

Follow your local clinical protocol



Mercer BM: Premature rupture of the membranes. Obstet Gynecol 101:178–193

POST-TERM PREGNANCY

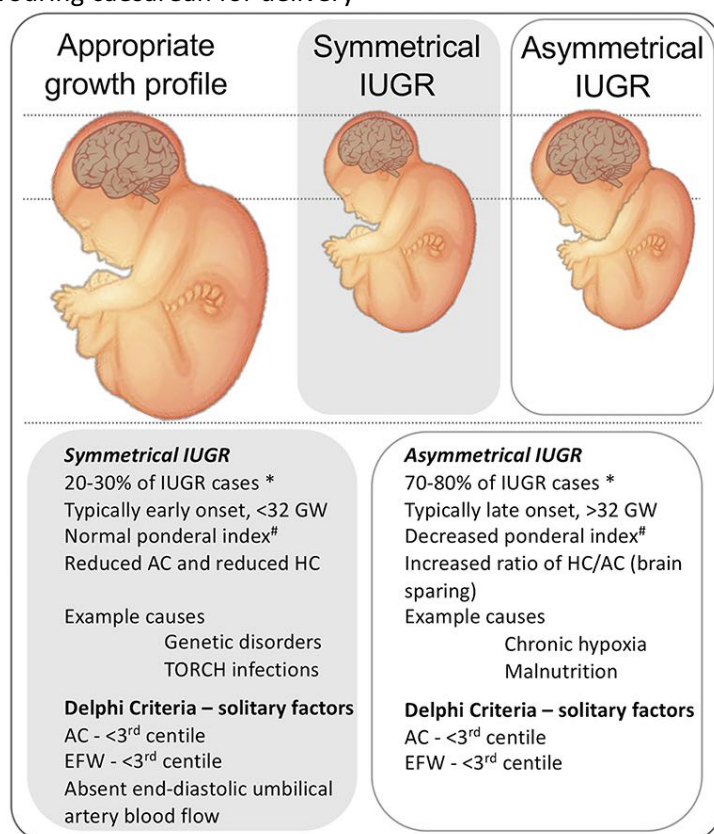
- **What is it?**
 - o Ongoing Pregnancy >42wks
- **Aetiology:**
 - o Typically idiopathic
 - o Anencephalic foetus without pituitary gland
 - o Placental sulfatase deficiency (a rare X-linked recessive condition)
- **Epidemiology:**
 - o 5.5% of pregnancies
- **Clinical Features:**
 - o Post-maturity syndrome: Foetal weight loss, reduced subcutaneous fat stores, long thin body, long nails, scaly dry skin
 - o Higher risk of placental insufficiency
 - o Higher risk of macrosomia
 - o Higher risk of dystocia
 - o Higher risk of foetal distress
- **Management:**
 - o **39wks + maternal age >40** → Consider Induction of Labour
 - o **40-41wks** → Expectant management
 - o **>41wks** → Consider/offer induction
 - o **If Amniotic Fluid Index is low** → Induce Labour
- **Prognosis:**
 - o 2-3x higher risk of perinatal mortality (due to progressive uteroplacental insufficiency)

INTRAUTERINE FOETAL DEATH

- **What is it?**
 - o Foetal death in-utero >20wks gestation (Note: <20wks = 'spontaneous abortion')
- **Aetiology:**
 - o Idiopathic
 - o Or secondary to: HTN, DM, erythroblastosis fetalis, congenital anomalies, placental failure, infection
- **Epidemiology:**
 - o Approximately 1% of pregnancies
- **Clinical Features:**
 - o Reduced foetal movements felt by mother
 - o Maternal weight not increasing
 - o Absent FHR on Doppler
 - o High maternal serum alpha-fetoprotein (AFP)
 - o No foetal heart beat on USS
- **Management:**
 - o USS for diagnosis
 - o **<12wks: D&C**
 - o **13-20wks: Dilation & evacuation**
 - o **>20wks: Induction of labour**
 - o Monitor for DIC
 - o Bereavement support

INTRAUTERINE GROWTH RESTRICTION

- **What is it?**
 - Infant weight <10th percentile for gestational age
 - Or <2500g at term
- **Aetiology:**
 - Maternal: Malnutrition, smoking, drugs, etoh, diabetes, SLE, HTN
 - Placental insufficiency
 - Prolonged gestation
 - Multiple gestation
 - Congenital/chromosomal anomalies
 - 'TORCH' infections (Toxoplasmosis, others [Including syphilis], Rubella, CMV, HSV)
- **Clinical Features:**
 - **Symmetric IUGR:**
 - Similarly reduced growth of both head and abdomen
 - Normal Head:Abdomen ratio
 - (>1 at <32wks)
 - (=1 at 32-34wks)
 - (<1 after 34wks)
 - Typically caused by congenital abnormalities or TORCH infections
 - **Asymmetric IUGR:**
 - Abdomen is disproportionately smaller than head
 - Therefore increased Head:Abdomen ratio
 - Typically caused by placental insufficiency
- **Investigations:**
 - Fundal height measurements at every antepartum visit
 - USS of Foetus (Check head & abdo circumference, femur length, foetal weight & amniotic fluid volume)
 - Umbilical cord Doppler study
- **Management:**
 - Elimination of risk factors
 - Bed rest in left lateral position
 - Delivery when benefits outweigh risks
 - Consider favouring caesarean for delivery

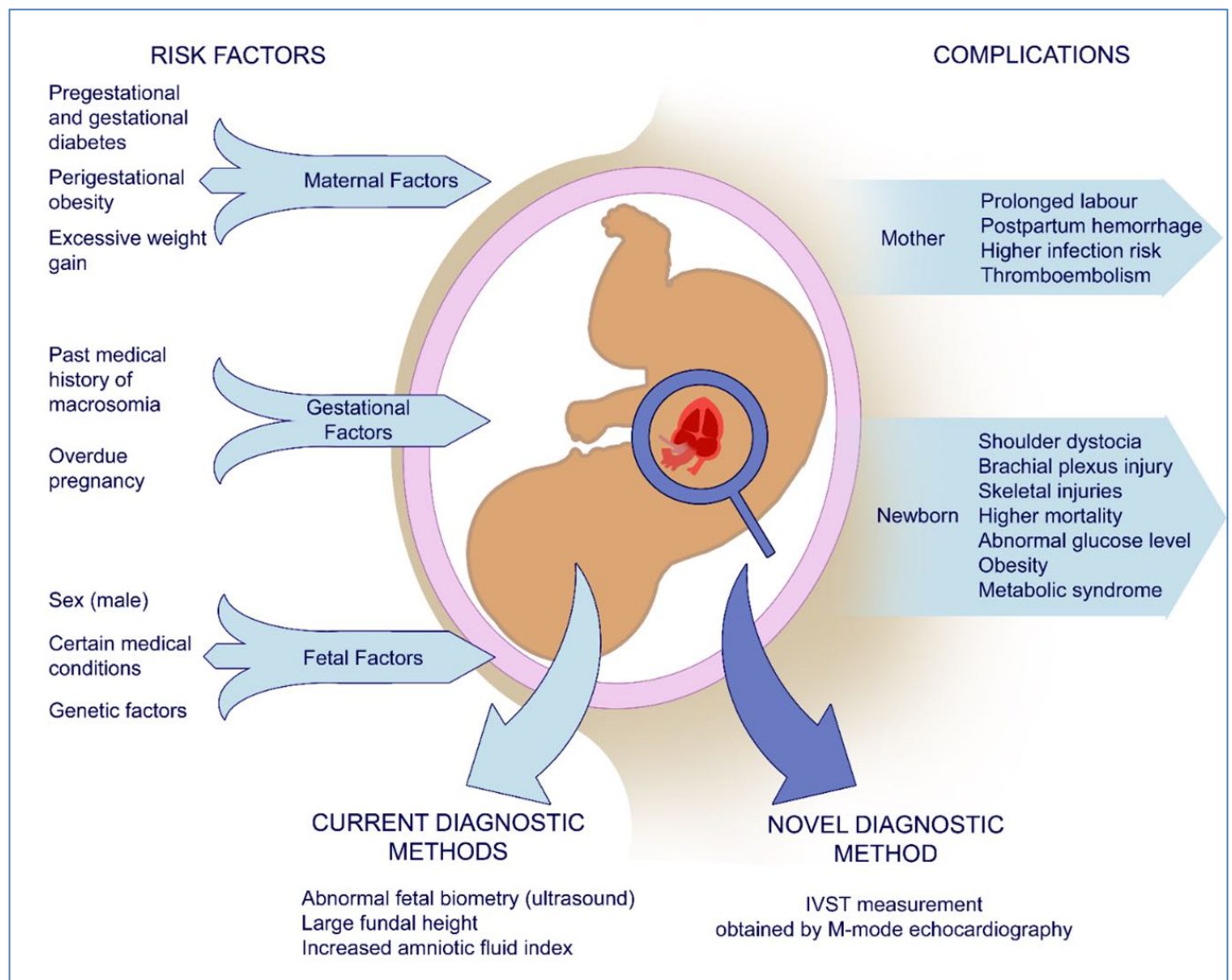


MACROSOMIA

- **What is it?**
 - Infant weight >90th percentile for gestational age
 - Or >4000g at any time
- **Aetiology:**
 - Maternal obesity
 - Gestational diabetes
 - Prolonged gestation
- **Clinical Features:**
 - Higher incidence of birth injuries (shoulder dystocia, foetal bone fracture)
 - Polyhydramnios
 - Head Circumference/Abdo Circumference ratio <10th percentile
 - Femur Length/Abdo Circumference ratio <20th percentile
- **Management:**
 - Caesarean section where foetal weight >5000g (if non-diabetic mother) or >4500g (if diabetic mum)
- **Prognosis:**
 - Increased risk of perinatal mortality



<https://www.waterloowellingtondiabetes.ca/userContent/documents/Professional-Presentations/Pregnancy%20and%20Hyperglycemic%20Emergencies%20.pdf>



Bartosz Szmyd, Małgorzata Biedrzycka, Filip Franciszek Karuga, Magdalena Rogut, Iwona Strzelecka, Maria Respondek-Liberska, CC BY 4.0 <<https://creativecommons.org/licenses/by/4.0/>>, via Wikimedia Commons

POLYHYDRAMNIOS

- **What is it?**
 - o Amniotic Fluid Index (AFI) >25cm
 - o USS: Single deepest pocket >8cm
- **Aetiology:**
 - o Mostly idiopathic
 - o Type 1 diabetes
 - o Chorioangiomas
 - o Multiple gestation
 - o Foetal chromosomal anomaly
 - o Foetal anencephaly/hydrocephalus/meningocele
 - o Foetal tracheoesophageal fistula/duodenal atresia/facial clefts
- **Epidemiology:**
 - o 0.2-1.6% of all pregnancies
- **Clinical Features:**
 - o Uterus large for dates
 - o Overdistended uterus → pressure symptoms (dyspnoea, oedema, hydronephrosis)
 - o Possible complications: Cord prolapse, placental abruption, malpresentation, preterm labour, PPH
- **Management:**
 - o Diagnose likely cause
 - o If mild-moderate; expectant management
 - o If severe → Hospitalize and consider therapeutic amniocentesis
- **Prognosis:**
 - o 2-5x risk of perinatal mortality

OLIGOHYDRAMNIOS

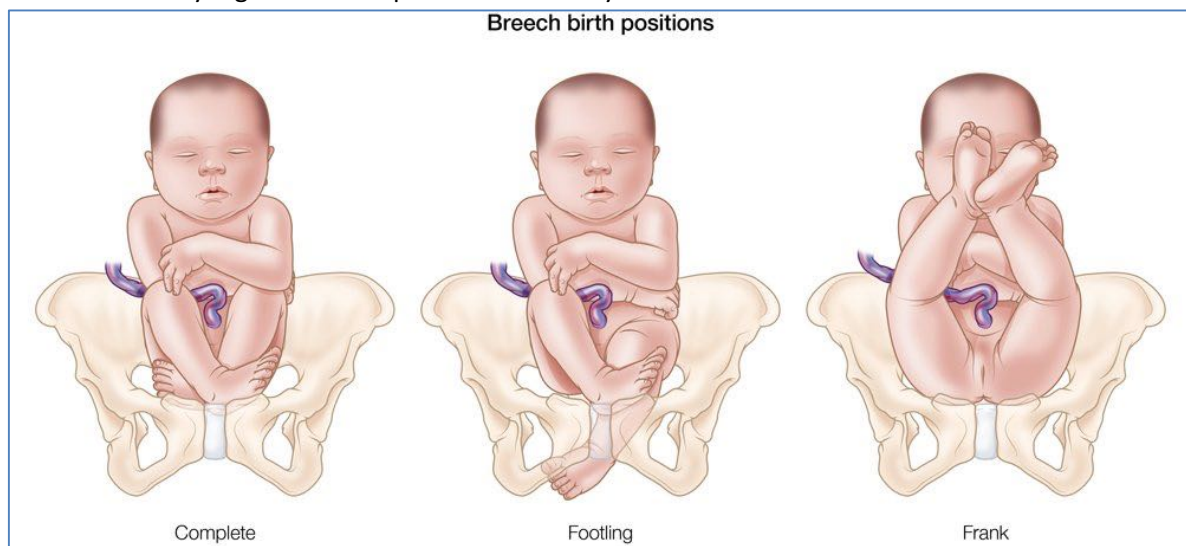
- **What is it?**
 - o Amniotic Fluid Index (AFI) <5cm
 - o USS: Single deepest pocket <2m
- **Aetiology:**
 - o Mostly idiopathic
 - o Uteroplacental insufficiency (Eg: Preeclampsia)
 - o ACE inhibitors
 - o Foetal urinary tract anomalies (Eg: Renal agenesis, renal obstruction)
 - o Foetal demise
 - o IUGR
 - o Ruptured membranes
- **Epidemiology:**
 - o ~4.5% of all pregnancies
- **Clinical Features:**
 - o Uterus small for dates
 - o 15-25% chance of foetal abnormalities
 - o Possible complications: Cord compression, pulmonary hypoplasia, higher risk labour
- **Management:**
 - o Rule out PROM
 - o Monitor foetus
 - o USS Doppler of umbilical cord & uterine artery
 - o Optimise maternal hydration
 - o Can inject fluid by amniocentesis for temporary (1wk) relief
 - o Deliver if at or near term
- **Prognosis:**
 - o High foetal mortality if associated with congenital malformations / pulmonary hypoplasia

MALPRESENTATION

MALPRESENTATION

BREECH PRESENTATION

- **What is it?**
 - Foetal buttocks/lower extremity is the presenting part
 - **10% - 'Complete breech':** Hips and knees both flexed
 - **60% - 'Frank breech':** Hips flexed, knees extended, buttocks against cervix
 - **30% - 'Incomplete / footling':** Both or one hip flexed, and both or one knee present below buttocks; feet or knees presenting first
- **Risk Factors:**
 - Contracted maternal pelvis
 - Abnormally shaped uterus (Eg: Fibroids, uterine tumours)
 - Pelvic tumours
 - Placenta previa
 - Poly/oligohydramnios
 - Prematurity of foetus
 - Multiple gestation
 - Congenital malformations
- **Epidemiology:**
 - 3-4% of pregnancies
- **Management:**
 - **External Cephalic Version (ECV):**
 - → Reposition a single foetus under USS guidance
 - ~65% success rate
 - **Prerequisites:** >36wks, single foetus, unengaged, reactive Non-Stress test & not in labour
 - **Contraindications:** previous caesarean, oligohydramnios, PROM, placenta previa, IUGR
 - **Anti-D-Ig administration if mother Rh negative**
 - **If delivering vaginally:**
 - Must be active; avoid foetal traction
 - Encourage maternal pushes
 - **Caesarean recommended if:**
 - Breech hasn't yet descended into perineum at 2hrs into labour, or if vaginal delivery is not imminent
 - Cord presentation
 - Foetal factors not compatible with vaginal delivery (Eg: hydrocephalus, macrosomia, foetal growth restriction)
- **Prognosis:**
 - Complications of ECV: Abruption, cord compression, ROM, labour, foetal bradycardia, rhesus sensitisation
 - Generally higher rates of perinatal mortality



HYPERTENSION IN PREGNANCY

HYPERTENSION IN PREGNANCY

General Info:

- **Definition:**
 - = “BP $\geq$ 140/90 measured on >2 Separate Occasions”
- **Types – In Order of Severity:**
 - **Chronic (Pre-existing) Hypertension:**
 - = “HTN Before Pregnancy OR Within 1st 20wks of gestation”
 - (No associated problems; But $\uparrow$ Risk of $\rightarrow$ Gestational HTN & Pre-Eclampsia)
 - **Gestational Hypertension:**
 - = “HTN occurring After 20wks”
 - (No associated problems; But $\uparrow$ Risk of $\rightarrow$ Pre-Eclampsia)
 - ***Pre-Eclampsia (Including Chronic HTN with Superimposed Pre-Eclampsia):**
 - = “HTN in Pregnancy + ANY Sign of Organ Dysfunction”
 - Eg: Kidney – (Proteinuria, $\uparrow$ Creatinine)
 - Eg: Liver – ($\uparrow$ AST/ALT, RUQ Pain, Hepatitis)
 - Eg: Neuro – (Headaches, Visual Disturbance, Hyperreflexia, Clonus)
 - Eg: Haem – (DIC, Consumptive Thrombocytopenia, Haemolysis)
 - Eg: Placental – (Foetal Growth Restriction)
 - ****Eclampsia:**
 - = “Pre-Eclampsia Complicated by a Generalised Tonic-Clonic Seizure”
 - (ACUTE & LIFE-THREATENING: Can $\rightarrow$ Maternal & Foetal Complications/Death)
- **General Management:**
 - **ANY HTN IN PREGNANCY NEEDS INVESTIGATION & CLOSE MONITORING!!!**

GESTATIONAL HYPERTENSION:

- **What is it?**
 - BP >140/90 developing after 20th wks gestation (in the absence of proteinuria in a woman known to be normotensive before pregnancy)
- **Risk Factors:**
 - Primigravidas account for >80% of gestational HTN
 - First conception with new partner
 - Diabetes
 - Renal insufficiency
 - Obesity
 - extremes of maternal age (<18 or >35 yr)
 - IUGR / oligohydramnios
 - Multiple gestation
- **Investigations:**
 - Clinical examination (check BMI, oedema, nervous system exam)
 - Foetal non-stress test / USS for growth / Doppler flow studies
 - Bloods + 24hr urine collection (for protein/albumin : creatinine ratio)
- **Management:**
 - Pharmaceutical options: Labetolol, Nifedipine XR, or a-methyldopa
 - (IV Hydralazine if severe)
 - AVOID: ACE-inhibitors, Angiotensin receptor blockers, diuretics, prazosin or atenolol
- **Complications:**
 - Maternal liver/renal dysfunction
 - Eclampsia
 - Placental abruption
 - LVF/pulmonary oedema
 - DIC
 - HELLP syndrome
 - Haemorrhagic stroke
 - IUGR
 - Foetal prematurity

PRE-ECCLAMPSIA / ECCLAMPSIA:

- **Aetiology:**
 - **Defective Placentation**
 - **(Risk Factors – Primigravid, Older Mums, FamHx, Chronic HTN, Diabetes, Twins, Molar Pregnancy)**
- **Pathogenesis:**
 - **Insufficient Placental Invasion into Spiral Arterioles → ↓ Placental Blood Flow:**
 - **Pre-Eclampsia:** Placental Ischaemia → Vasoconstrictors → HTN
 - **Eclampsia:** Placental Infarction → Severe HTN → Seizures & Organ Failure
- **Clinical Features:**
 - **Common:** 5-10% pregnancies
 - **Symptoms:**
 - ****Headaches, Visual Disturbances** (Neuro Complications)
 - *** Abdo Pain** (Hepatitis)
 - ***Pitting Oedema** (Renal Failure)
 - **!!Purpura & DIC** (HELLP Syndrome)
 - **!!Seizures** (IF Eclampsia)
- **Diagnosis:**
 - **Symptom Inquiry:**
 - Headaches/Visual Disturbances?
 - Epigastric Pain?
 - Oedema? (Seen as rapid weight gain)
 - Rashes?
 - **Take Blood Pressure (>140/90) →**
 - **Do Urine Dipstick/Urinalysis (Proteinuria) →**
- **Management:**
 - **Admit to BS for 4hrly Monitoring:**
 - Urinalysis (Protein++)
 - Serial BP's (Seated) every 4hrs
 - Daily UECs, FBC, LFTs
 - Daily USS for Foetal Growth & Amniotic Fluid Volume
 - **Drugs:**
 - **Antenatal Corticosteroids – (Betamethasone)**
 - **CaChBlockers – (24hr Magnesium Sulfate Infusion or Nifedipine)**
 - **B-Blockers – (Labetalol)**
 - ****Definitive = Delivery (Early Induction of Labour)**
 - *****If → ECCLAMPSIA (Ie: Seizures):**
 - **1: Stabilize with Magnesium Sulfate** (Note: Do NOT use Anticonvulsants!)
 - **2: Immediate Delivery**
 - **3: Recovery in HDU/ICU for >4days AFTER BP HAS NORMALISED**
 - ***(CONTRAINDICATED – ACEi's/ARBs & Diuretics)**
- **Complications:**
 - **Foetal Growth Restriction**
 - **Liver Failure**
 - **Acute Renal Failure**
 - **HELLP Syndrome – (Haemolysis, Elevated Liver Eenzymes, Low Platelets)**
 - → Jaundice, Epigastric Pain, Vomiting
 - **DIC**
 - **Eclampsia → Seizures →**
 - **Placental Abruption**
 - **Cerebral Haemorrhage**
 - **Aspiration Pneumonia**
 - **Death**
- **Prognosis:**
 - **Eclampsia is rare with proper treatment; BUT has 20% Mortality!!**

NORMAL LABOUR AND DELIVERY

NORMAL LABOUR AND DELIVERY

Time of birth:

- 280 Days after last menstrual period +/- 15 days
 - o Preterm (>20 to <36+6 wk GA)
 - o Term (37-41+6 wk GA)
 - o Post term (>42 wk GA)

Foetal Maturation of Organ Systems:

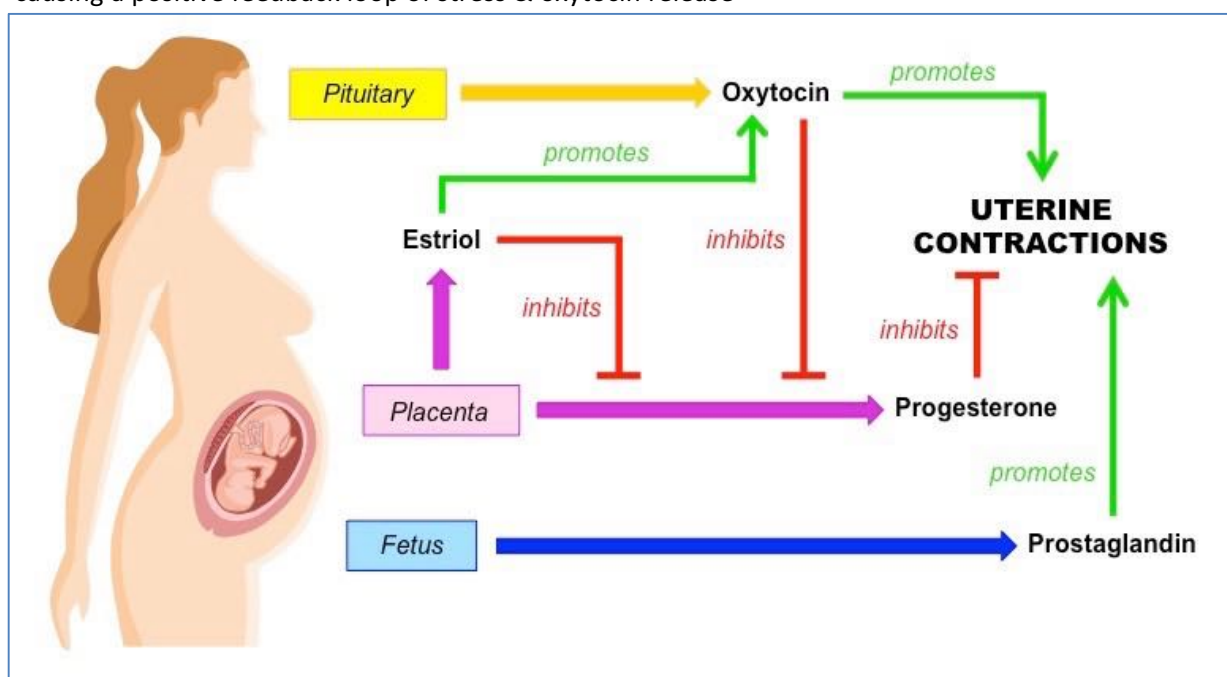
- **Essential systems for extrauterine life:**
 - o Interface organs – lungs, gut, immune system
 - o Homeostatic organs – endocrine (hypothalamo-pituitary-adrenal axis), kidneys, liver, pancreas
- **Promoted by maternal adrenal glucocorticoids (steroid hormones). Causes:**
 - o Surfactant production in lungs
 - o Activity of enzyme systems in gut, retina, pancreas, thyroid gland, brain
 - o Deposition of glycogen in liver

Definition of Labour:

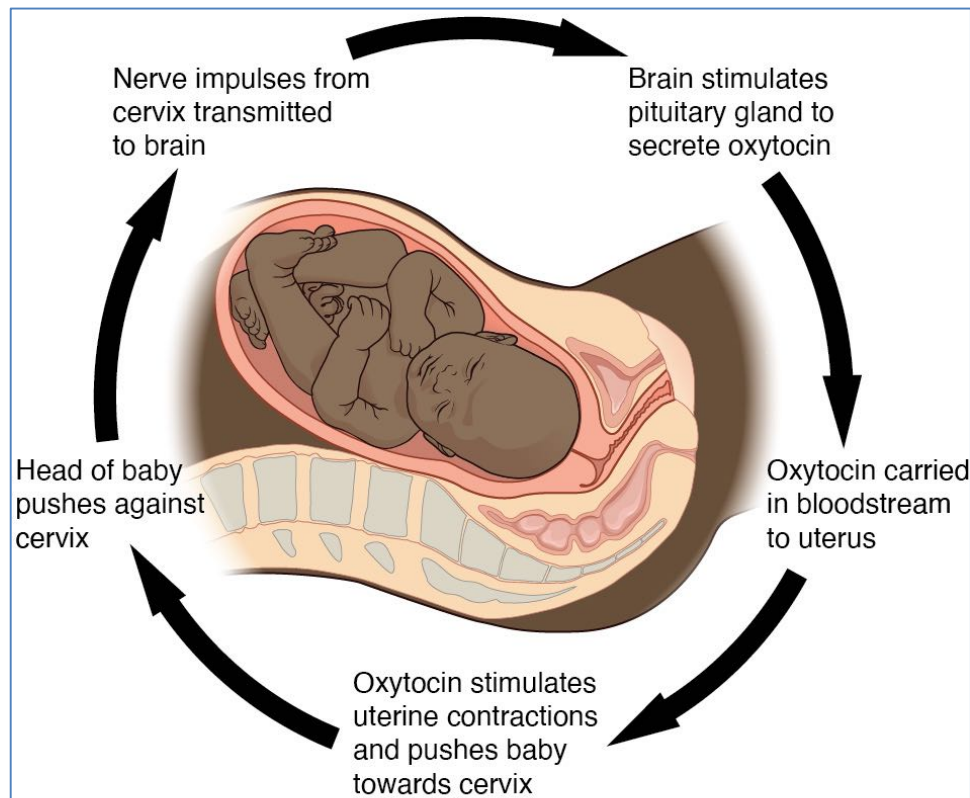
- **“True Labour”:**
 - o = ‘Regular, painful contractions of increasing intensity associated with progressive dilatation and effacement of cervix and descent of presenting part, or progression of station’
- **“False Labour” (Braxton-Hicks contractions):**
 - o = ‘Irregular contractions, with unchanged intensity and long intervals, occur throughout pregnancy and not associated with any cervical dilatation, effacement, or descent’

Initiation of Labour / Parturition

- **Triggered by a combination of hormonal changes**
- **Oestrogen levels peak in late pregnancy, resulting in:**
 - o Increased uterotonic receptors in uterus – oxytocins + prostaglandins
 - o Decreased effect of progesterone → Myometrium becomes irritable → weak, irregular contractions = **false labour**
- Shortly before parturition (**true labour**), foetal cells produce **oxytocin** – cause placenta to release **prostaglandins**
 - o Both hormones (**oxytocin + prostaglandins**) are powerful **uterine muscle stimulants**
 - o The highly sensitive uterus responds – contractions become more frequent & vigorous
- Then, increasing stresses activate mother’s **hypothalamus** – causes **Posterior Pituitary** to release **oxytocin** – causing a positive feedback loop of stress & oxytocin release



Unattributable



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Cervical Variables During Labour:

- **Dilation:** Latent phase (0-4 cm, variable time); active phase (4-10 cm)
- **Effacement:** Thinning of the cervix by percentage or length of cervix (cm)
- **Consistency:** Firm vs soft
- **Position:** Posterior, mid, or anterior
- **Application:** Contact between the cervix and presenting part (Ie: Well or poorly applied)

Foetal Variables During Labour:

- **Foetal lie:** The orientation of the long axis of the foetus with respect to the long axis of the uterus
 - Longitudinal (Vertex)
 - transverse
 - oblique



<https://www.pregnancybirthbaby.org.au/malpresentation>

- **Foetal presentation:** The foetal body part closest to the birth canal:
 - Breech (complete, frank, incomplete)
 - Cephalic (vertex/occiput, face, asynclitic, brow)
 - Transverse (shoulder)
 - Compound (foetal extremity prolapses along with presenting part)
 - All except vertex are considered malpresentations)
- **Foetal position:** position of presenting part of the foetus relative to the maternal pelvis
 - **Occiput Anterior:** most common presentation ("normal")
 - **Occiput Posterior:** most rotate spontaneously to OA; may cause prolonged second stage of labour
 - **Occiput Transverse:** leads to arrest of dilatation
 - (normally, foetal head enters maternal pelvis and engages in OT position)
 - Subsequently rotates to OA position (or OP in a small percentage of cases)

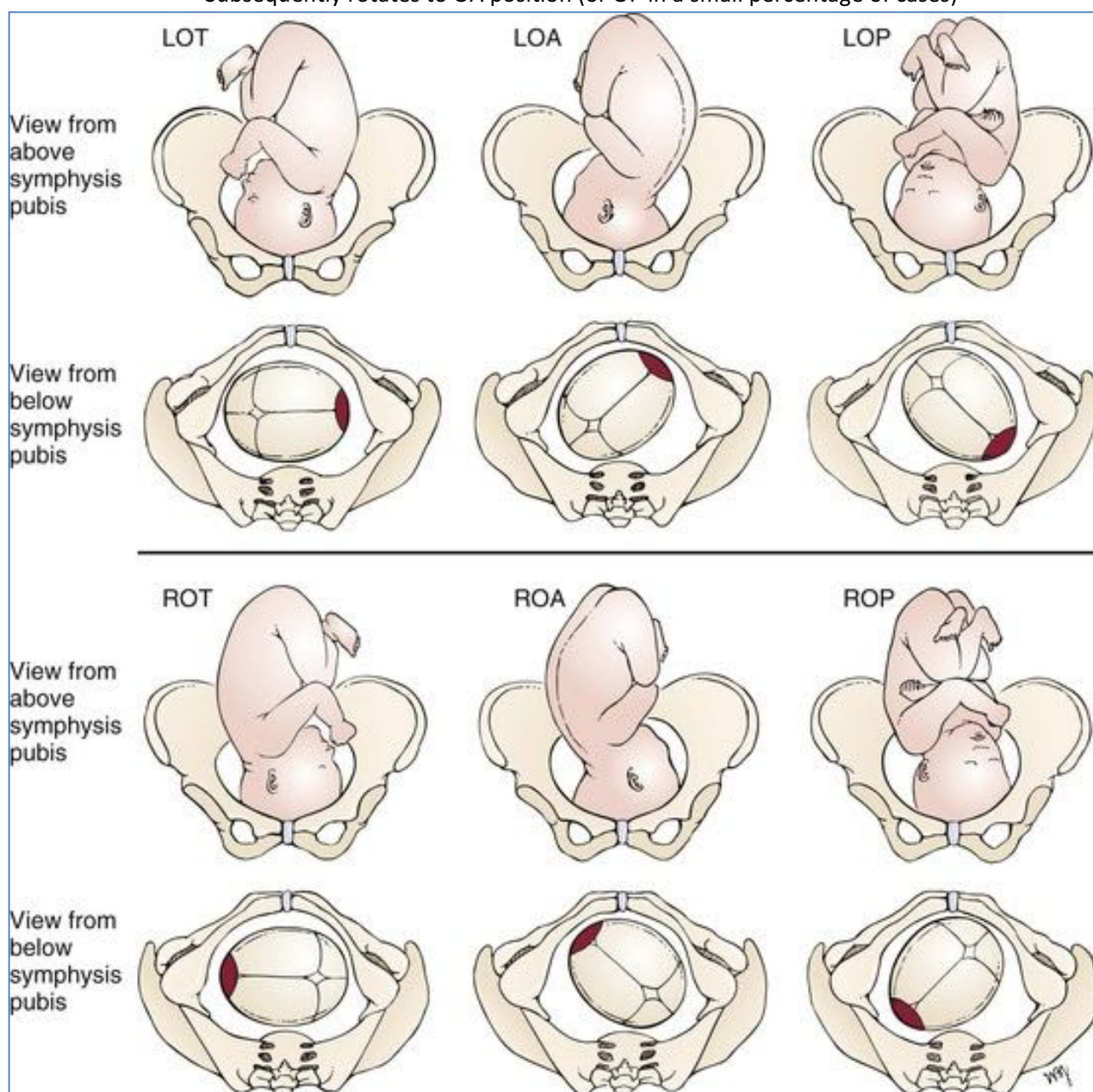


Image credit: <https://nursekey.com/the-pregnant-patient/>

- **Attitude:** Flexion/extension of foetal head relative to shoulders
 - **Brow presentation:** head partially extended (requires caesarean)
 - **Face presentation:** head fully extended (Requires caesarean if mentum posterior; mentum anterior can deliver vaginally)

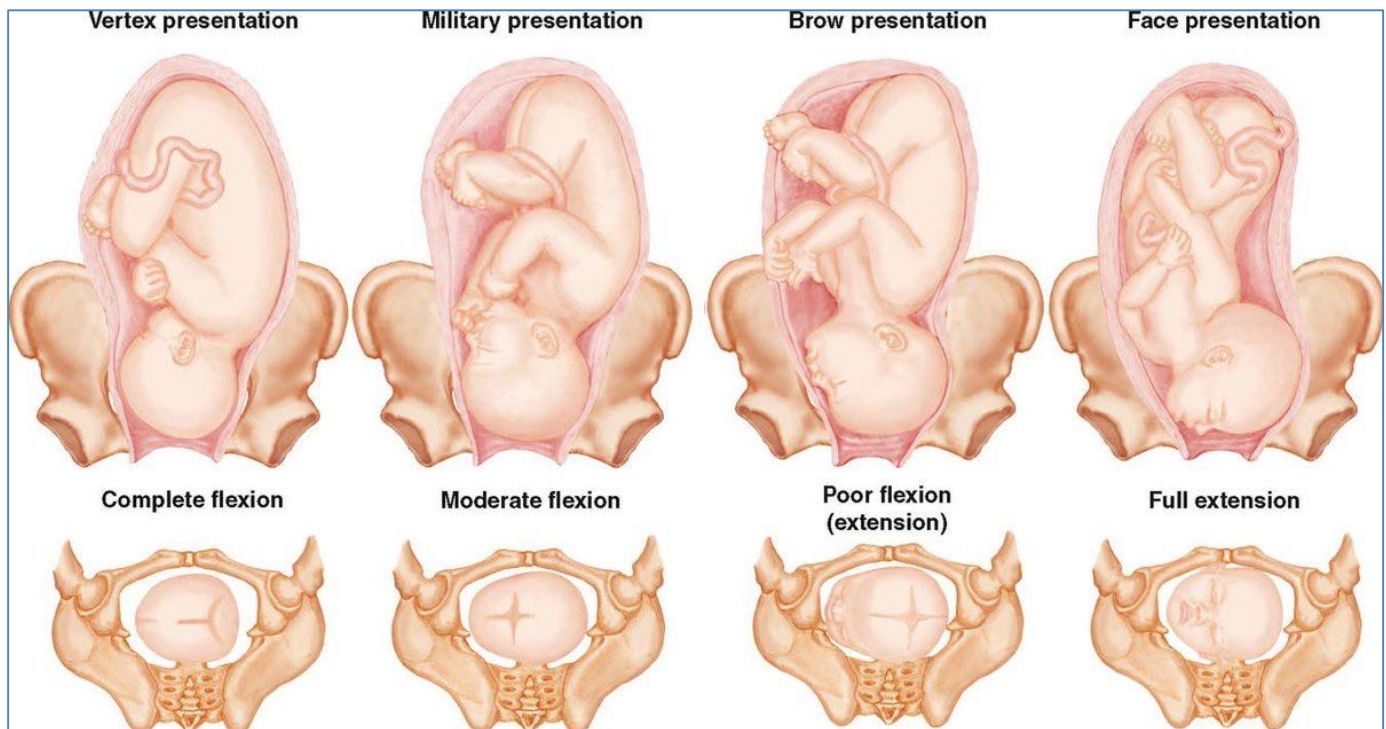


Image credit: <https://www.nursingawareness.com/message.php?id=90>

- **Station:** Position of presenting bony part relative to ischial spines – determined by vaginal exam:
 - **At ischial spines = station 0 = engaged**
 - **-5 to -1 cm above ischial spines or**
 - **+1 to +5 cm below ischial spines**

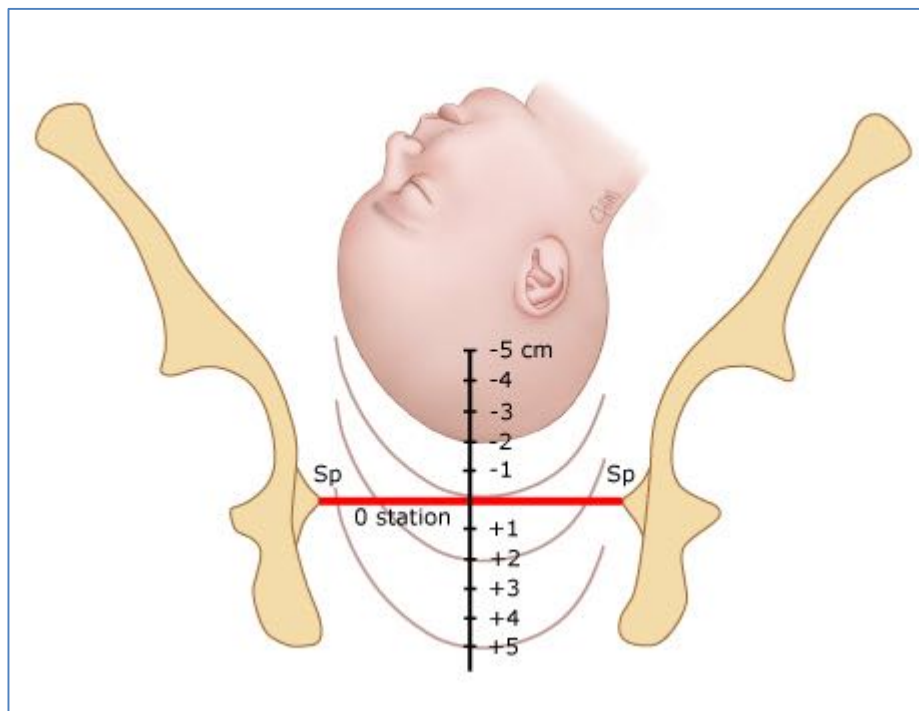
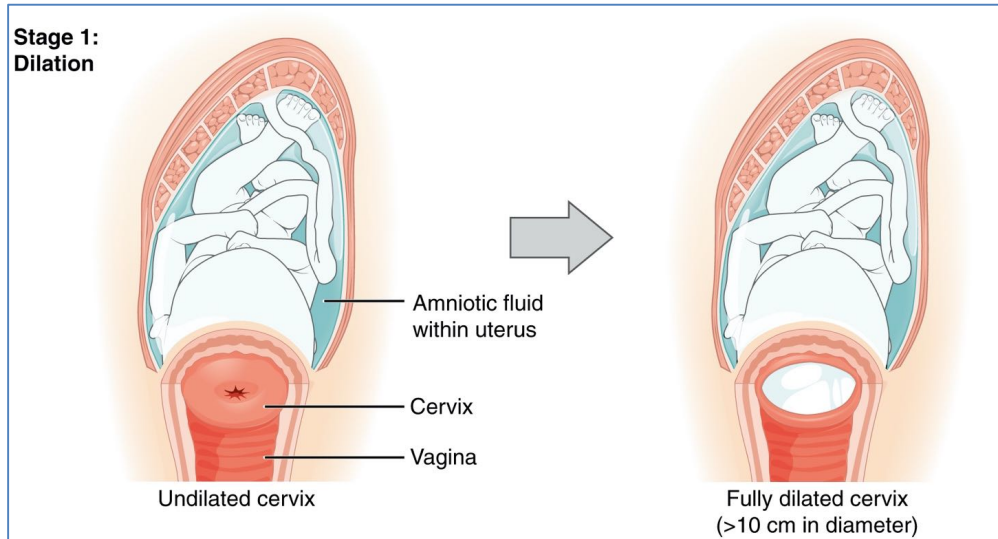


Image credit: <https://www.uptodate.com/contents/image/print?imageKey=OBGYN%2F67068>

STAGES OF LABOUR:

Stage 1: Dilation Stage:

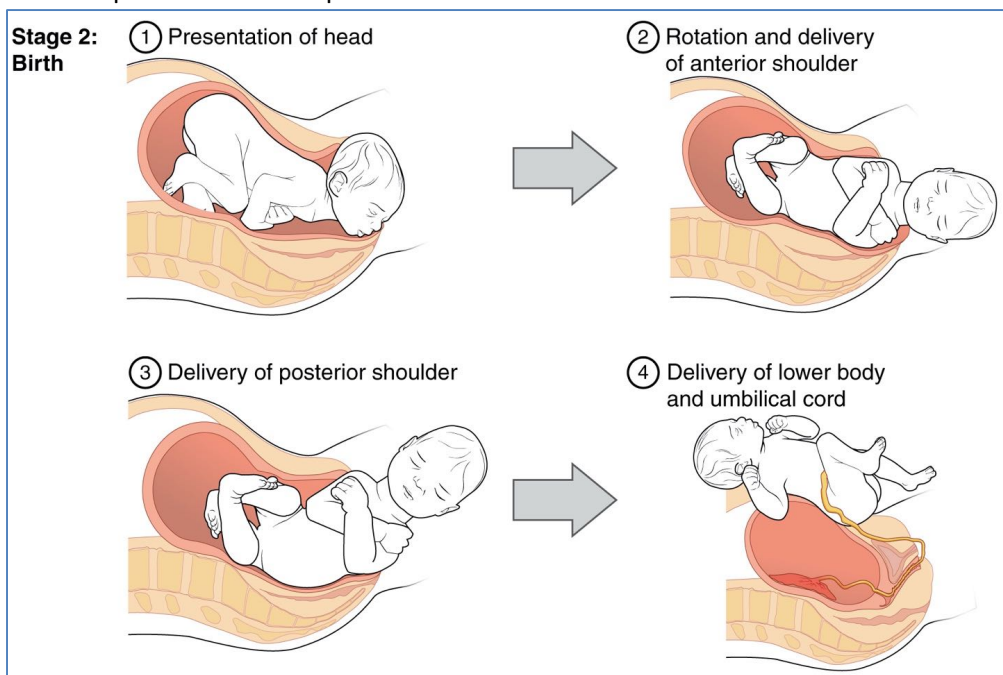
- **Time between initiation of labour and full cervical dilation (10 cm)**
- Amniotic sac usually breaks just before this stage
 - At first, contractions begin in the fundus, then spread down towards vagina
 - As the infant's head is forced against cervix, the cervix softens and thins – begins to dilate
 - Takes approx 6-12 hrs
 - Baby's head rotates to minimise its profile



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Stage 2: Expulsion Stage

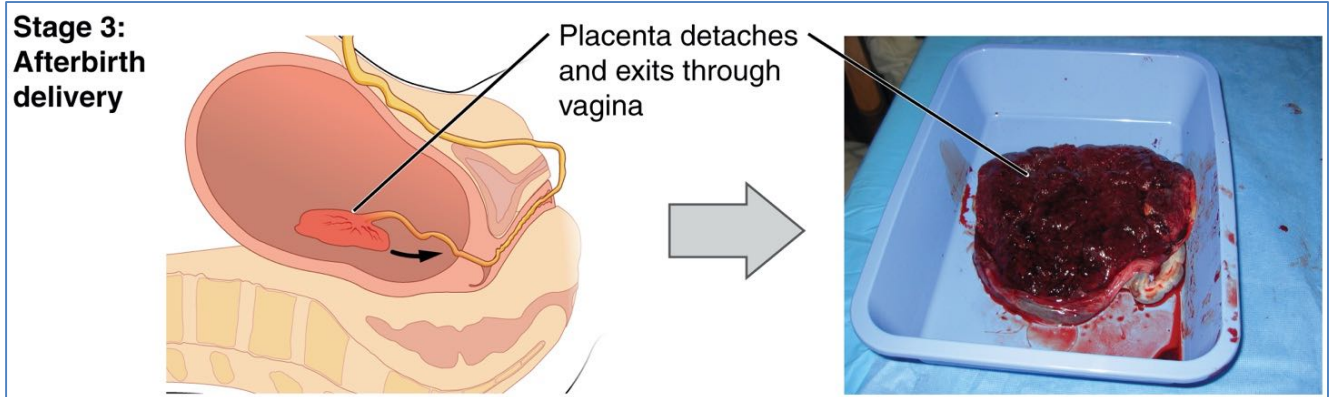
- **Time between full dilation to delivery of infant**
 - By the time cervix is fully dilated, contractions are strong and regular – 2-3 min intervals
 - Takes on average 1hr
- **Vertex position (usual – head first)**
 - **Crowning** occurs – baby's head is halfway distended from the vulva
 - Baby's head is delivered, then the rest of the body
 - Umbilical cord is clamped and cut
- **Breech position (unusual – bum first)**
 - Much more difficult
 - Often requires use of forceps or C-section



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Stage 3: Placental Stage:

- **Delivery of the placenta (approx 30 mins)**
- Baby should be breast fed directly after 2nd stage → stimulates oxytocin release → causes uterine contractions:
 - Uterine contractions compress uterine blood vessels, sheering placenta from uterine wall
- Note: Routine Oxytocin administration can reduce risk of **postpartum bleeding** by >40%



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INDUCTION OF LABOUR (IOL)

What is IOL?

- = Artificial initiation of labour prior to spontaneous initiation with the intent of delivering the foetus

Requirements for Induction of Labour:

- Availability of C/S if necessary
- Inducible/ripe cervix (short, thin, soft, anterior cervix with open os)
- If cervix is not ripe, use prostaglandin vaginal insert/prostaglandin/misoprostol/ or Foley catheter
- Normal foetal heart tracing
- Cephalic presentation
- Adequate foetal monitoring available
- Likelihood of success determined by Bishop score
 - o Cervix considered unfavourable if <6
 - o Cervix favourable if ≥6
 - o Score of 9-13 associated with high likelihood of vaginal delivery

	Score			
	0	1	2	3
Position	Posterior	Middle	Anterior	--
Consistency	Firm	Medium	Soft	--
Effacement	0-30%	40-50%	60-70%	80%+
Dilation	Closed	1-2cm	3-4cm	5+cm
Station	-3	-2	-1/0	+1/+2

Additional factors: +1 point for each previous vaginal birth, -1 point for first time birth givers

Add the score for each factor.

Scores lower than 5 suggest labour will not begin without induction.
Scores 9 and higher indicate labour will likely begin spontaneously.
Scores 3 and lower may indicate that an induction would not be successful

Indications for Induction of Labour:

- Post dates (>41 wks)
- Maternal Diabetes
- Gestational HTN
- Maternal age >40
- PROM
- Chorioamnionitis
- Suspected foetal jeopardy
- Foetal demise
- IUGR
- Oligo/polyhydramnios
- Twins

Risks Associated with Induction of Labour:

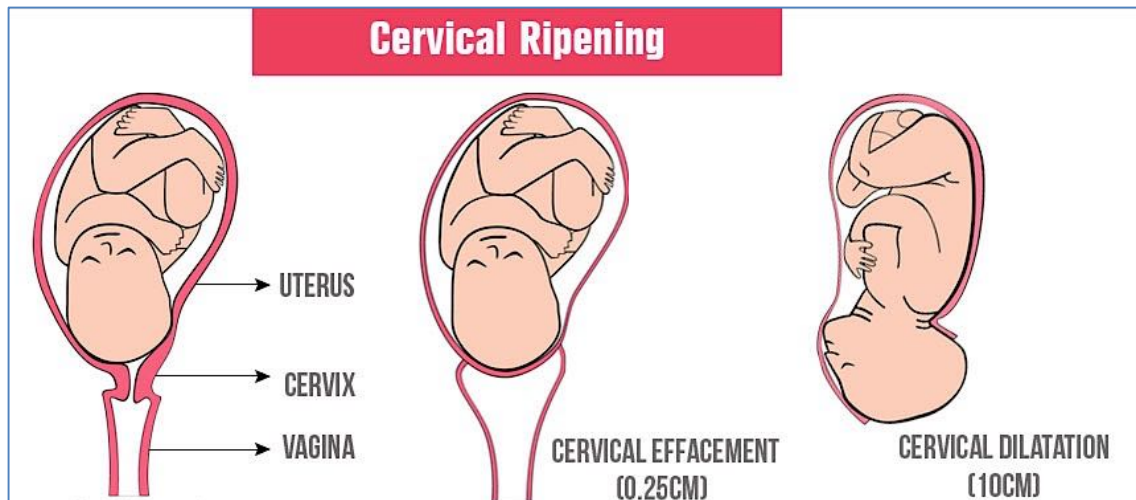
- Failure to achieve labour and/or vaginal birth
- Uterine hyperstimulation with foetal compromise or uterine rupture
- Maternal side effects to medications
- Uterine atony and PPH

Contraindications to Induction of Labour:

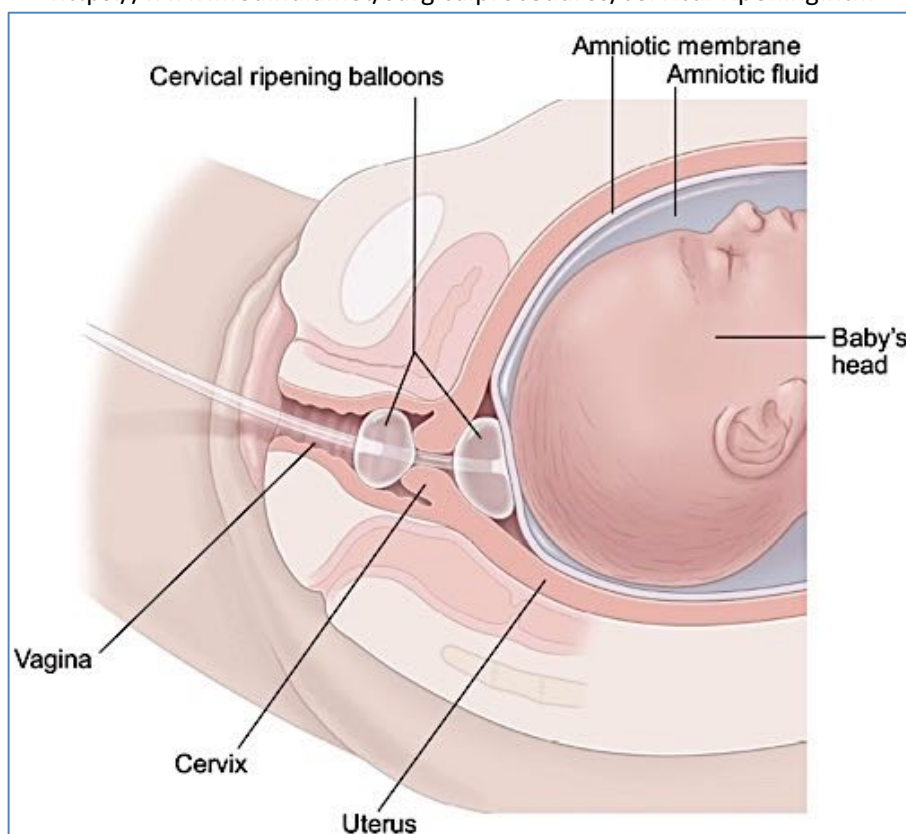
- Prior LUCS or uterine surgery
- Unstable maternal condition
- Active maternal genital herpes
- Invasive cervical carcinoma
- Pelvic structure deformities
- Placenta previa / Vasa previa
- Cord presentation
- Foetal distress / malpresentation
- Preterm foetus without lung maturity

Cervical 'Ripening':

- Softening/effacement/dilation of the cervix in order to increase chances of successful induction
- May be done via medications or other means
- Indicated if cervix is unfavourable (Bishop score <6) prior to labour
- **Methods:**
 - o Intravaginal Prostaglandin Gel
 - o Intravaginal Prostaglandin Insert
 - o Intravaginal Misoprostol
 - o Mechanical dilation of cervix via Foley catheter



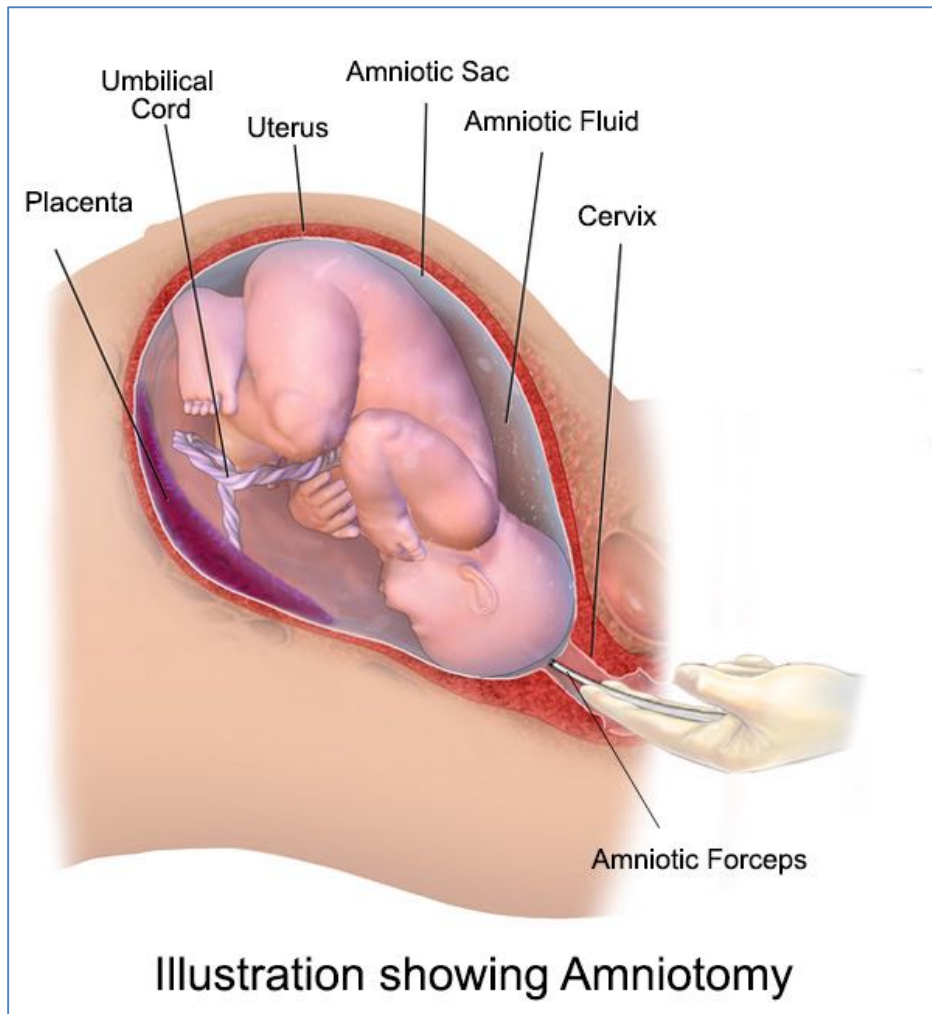
<https://www.medindia.net/surgicalprocedures/cervical-ripening.htm>



https://www.babycenter.com/pregnancy/your-body/inducing-labor_173

Methods of Inducing Labour:

- **Amniotomy:**
 - The artificial rupture of membranes to stimulate prostaglandin synthesis and secretion
 - Can be attempted if cervix is open & soft, membranes can be felt, and head is present at the cervix
 - Addition of IV oxytocin increases chance of successful induction of labour
- **Oxytocin:**
 - Administered via IV infusion to stimulate uterine contractions
 - Proven to reduce rate of unsuccessful vaginal deliveries
 - Potential complications:
 - Hyperstimulation/tetanic contraction of uterus → Foetal distress / uterine rupture
 - Uterine muscle fatigue → Uterine Atony → increased risk of PPH



Unattributable

NEONATAL ADAPTATIONS TO EXTRAUTERINE LIFE

NEONATAL ADAPTATIONS TO EXTRAUTERINE LIFE

After Birth:

- Baby is cast out of its watery, warm environment
- Placental life supports are severed
- Now baby must independently breathe, obtain nutrients, excrete waste, and maintain its body temperature

Respiratory Changes:

- Cortisol stimulates surfactant production in foetal lungs - last months of pregnancy
- After birth, CO₂ accumulates in blood → acidosis
 - Acidosis → excites respiratory centres → triggers 1st breath
 - Adrenal Medulla → Adrenaline (from stress of birthing) also supports surfactant secretion
 - Surfactant in alveolar fluid → reduces surface tension in alveoli → Aids inspiration
- Surfactant reduces surface tension of alveolar fluid – makes initial breathing easier
- Premature babies (no surfactant) are treated with Cortisol OR ACTH (Adrenocorticotrophic Hormone)
 - Cortisol → Surfactant
 - ACTH → Anterior Pituitary → Adrenal Cortex → Glucocorticoids (Eg: cortisol) → surfactant

Circulatory Changes:

- Foetal circulation is different from neonatal circulation
 - Has to integrate circulation of placenta
 - Blood flow to non-functional lungs & liver are partially bypassed
- “Bypasses” / “Shunts” of foetal circulatory system:
 - **Ductus Venosus**
 - Directs the oxygenated blood from the placental vein into inferior vena cava → heart
 - Partially bypasses the liver sinusoids
 - **Foramen Ovale**
 - An opening in the **interatrial septum** loosely closed by a flap of tissue
 - Directs some of blood entering the right atrium into the left atrium → Aorta
 - Partially bypasses the lungs
 - **Ductus Arteriosus**
 - Directs most blood from right atrium of the heart directly into aorta
 - Partially bypasses the lungs
 - **All of these “shunts” are occluded at birth due to pressure changes

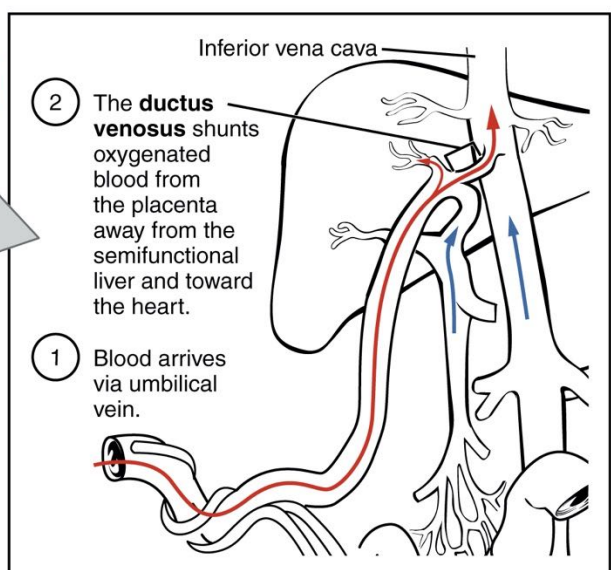
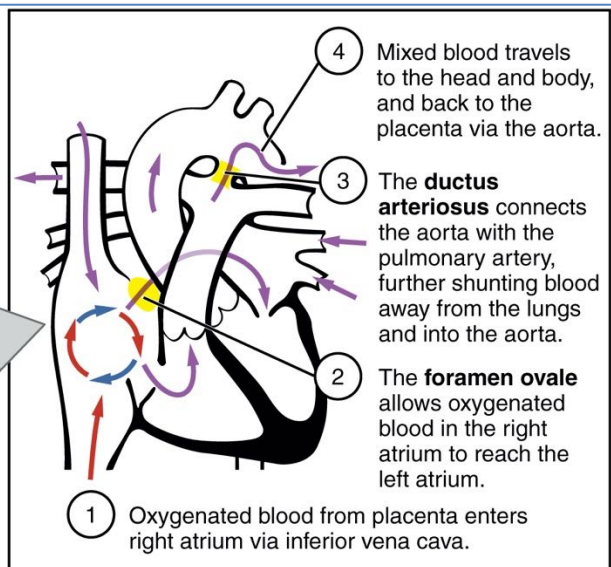
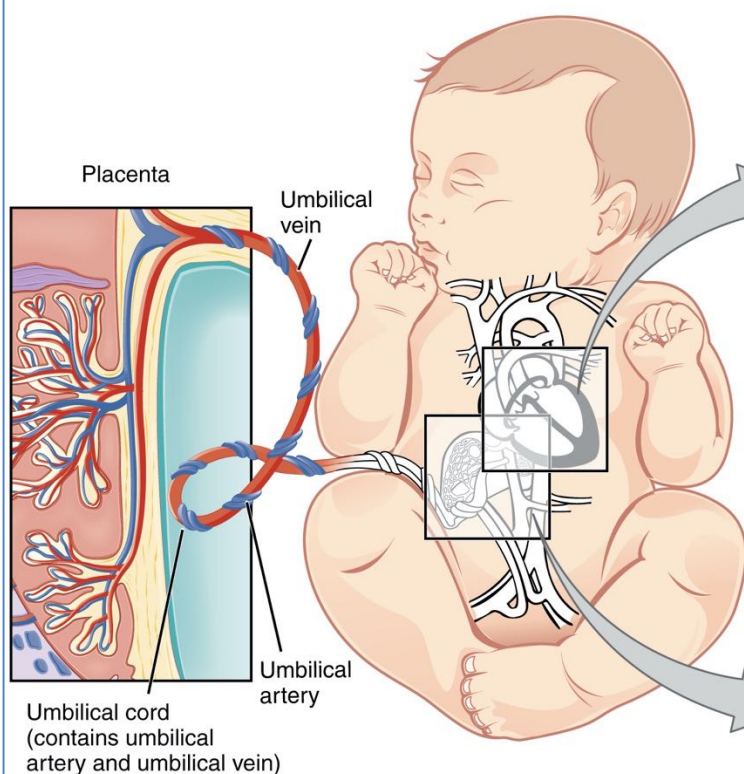
Thermoregulatory Changes:

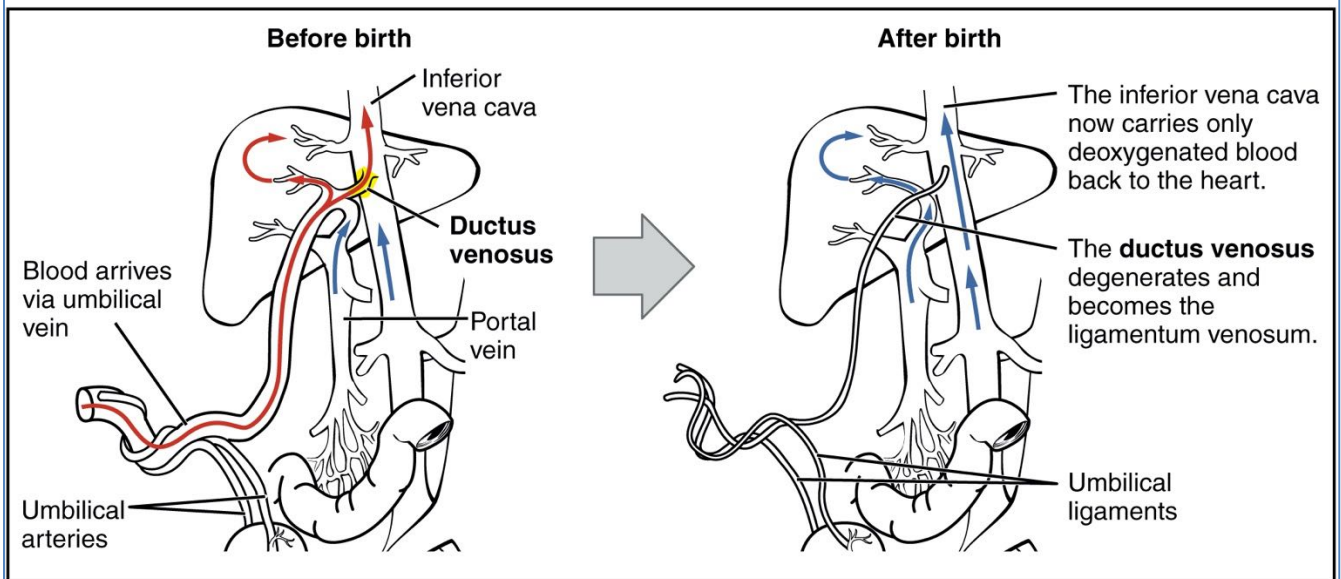
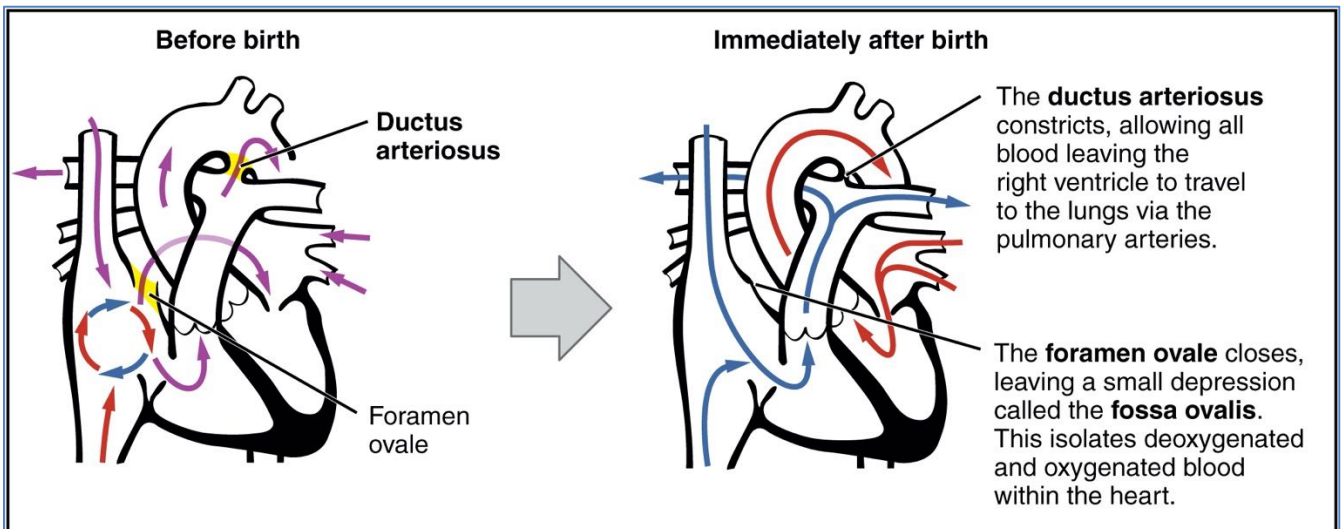
- Newborns have high “Surface Area – Body Mass” ratio → potentiates heat loss
 - Especially head
- Newborns cannot increase heat production by shivering
 - Instead, heat is produced by **uncouplers** in the mitochondria of **brown adipose tissue**

Apgar Score:

- Measures a baby's health against several criteria (expressed as #/10)

SCORE	0 points	1 point	2 points
Appearance - Skin colour	Cyanotic/ Pale all over	Peripheral cyanosis only	Pink
Pulse (Heart rate)	0	<100	100-140
Grimace - Reflex irritability)	No response to stimulation	Grimace (facial movement)/ weak cry when stimulated	Cry when stimulated
Activity - Tone	Floppy	Some flexion	Well flexed and resisting extension
Respiration	Apnoeic	Slow, irregular breathing	Strong cry





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BABY CHECKS: THE PROCESS

Background History:

- **Neonatal History**
- **Mum's/Bub's Blood Group** (*Beware the Rh-Negative Mother with a Rh-Positive Baby)
- **Obstetric History** (Gestational Diabetes, Pre-Eclampsia)
- **Method Delivery** (Natural/Forceps/Suction, Caesarean)
- **Risk Assessment for Sepsis**

Vital Signs:

- **P: **Bradycardia** (Often responds to ventilation) → Resuscitate
- **R: **Signs of Respiratory Distress** (Treat cause) → Resuscitate
- **T: **Fever** could = Infection (Beware *Group B Strep*, and *Chlamydia*) → Investigate & ABs

General Inspection:

- **Colour** (Pink = Good; Yellow/Jaundice = Often Normal; Blue/Cyanotic = Bad)
- **Distress** (Respiratory/Other)
- **Red Reflex of Retina** (Ensure the baby has a retina)
- **10 Fingers, 10 Toes**
- **Patent Anus** (Note: Presence of Meconium ≠ Patent Anus; Beware any Ano-Vaginal Fistulas)
- **Normal Genitalia**

Palpation:

- **Fontanelles** (Ant, Post & 2x Laterals; Beware Bulging = Hydrocephalus, & Sunken = Dehydration)
- **Cleft Palate**
- **High-Arched Palate** (Marfan's)
- **Rooting Reflex** (Stroke Cheek → Baby Turns Head to That Side)
- **Palmar Grasp Reflex**
- **Femoral Pulses** (120-180bpm)
- **Hip Examination** (Push down onto hips through the femur with 90° Hip Flexion, then abduct both hips. Feel for clicking/dislocation)
 - o "Barlow's Hip" = Hips dislocate with the above manoeuvre
 - o "Ortolana Hip" = Hip/s are *already* dislocated
- **Muscle Tone** (Beware a Floppy Baby)
- **Spine – Neural Tube Defects** (Spina Bifida)

Auscultation:

- **Heart Sounds:**
 - o *Rate
 - o Murmurs are Normal
 - o ?Dextrocardia
- **Lung Sounds:**
 - o Crackles
 - o Wheezes
 - o Stridor
- **Bowel Sounds:**
 - o Present
 - o Absent

Measurements:

- **Head Circumference** → Record on Growth Chart (*Bad if >90th/ $<10^{\text{th}}$ Centile)
- **Length** → Record on Growth Chart (*Bad if >90th/ $<10^{\text{th}}$ Centile)
- **Weight** → Record on Growth Chart (*Bad if >90th/ $<10^{\text{th}}$ Centile)
- **BSL**

INFANT RESPIRATORY DISTRESS

Note: Background Info:

- <37 wks = Pre-Term
- 37-42 wks = Term
- >42 wks = Post-Term

- (>23 wks & <500g = NON-Viable. I.e: A Miscarriage [Not obligated to resuscitate])
- (>24wks & >500g = Viable. I.e: A Birth [Obligated to resuscitate])

Causes for Respiratory Distress:

General Causes	Primary Causes: Surfactant Deficiency (Hyaline Membrane Disease) Infection (Pneumonia) with Group B Strep, or Chlamydia Meconium Aspiration (Meconium is produced later once the gut is developed) Other Causes: Transient Tachypnoea of the Newborn (Not pathological) Aspiration (Meconium, Milk, Blood) Pneumothorax (Overventilation) → Requires pleural tap @ 2 nd ICS Pleural Effusion (Heart failure, Anaemia, Congenital Heart Failure) Rarer Causes: Cystic Fibrosis Congenital Cyanotic Heart Diseases Muscle Weaknesses (Myasthenia Gravis, Muscular Dystrophies, etc)
(#1 Pre-Term Cause)	Surfactant Deficiency (Hyaline Membrane Disease)
(#1 Term Cause)	Infection (Pneumonia) with Group B Strep, or Chlamydia
(#1 Post-Term Cause)	Meconium Aspiration (Meconium is produced later once the gut is developed)

Signs of Respiratory Distress:

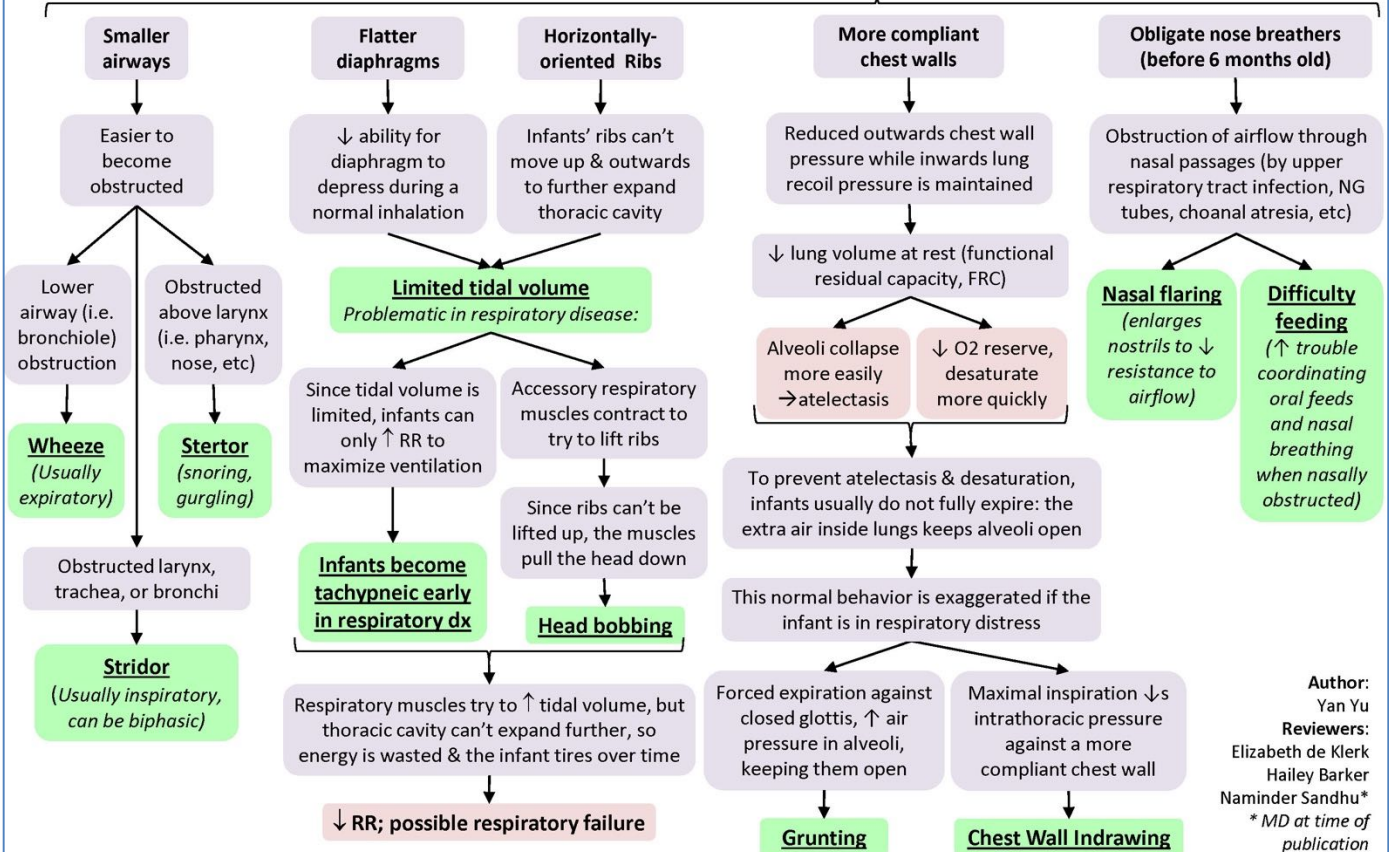
- **Tachypnoea** (Rate Compensation. A baby's diaphragm is already flat, so breathing deeper isn't an option)
- **Tachycardia** Initially → **Bradycardia** (Stroke Volume Compensation – Via Frank Starling Mechanism)
- **Accessory Muscle Usage**
 - o "Head Bobbing" (Sternocleidomastoid Muscle Usage)
 - o Intercostal Recession
 - o Tracheal Tugging
 - o Supraclavicular Recession
 - o "Harrison's Sulcus" (Subcostal Recession)
- **Grunting** (An autonomous *Valsalva Manoeuvre* to ↑Positive Airway Pressure)
- **Nasal Flaring**

Signs of Decompensation:

- Bradycardia
- Cyanosis
- Desaturation

Infant Respiratory Distress: Clinical findings

Infants respond to Respiratory Distress differently than adults,
due to anatomical and physiological differences:



Author:
Yan Yu
Reviewers:
Elizabeth de Klerk
Hailey Barker
Naminder Sandhu*
*MD at time of publication

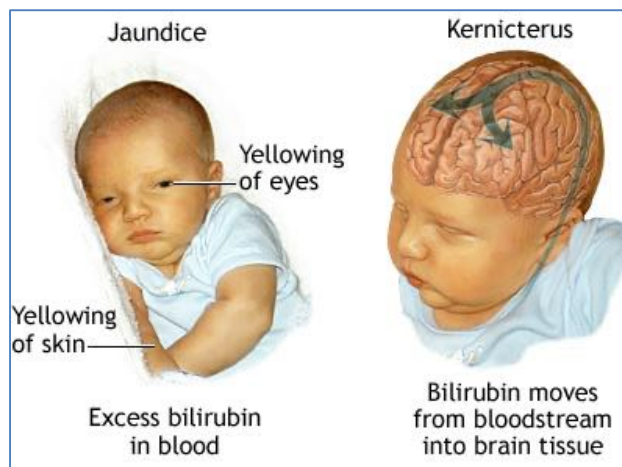
Legend: Pathophysiology Mechanism Sign/Symptom/Lab Finding Complications Published January 15, 2013 on www.thecalgaryguide.com



Source: thecalgaryguide.com

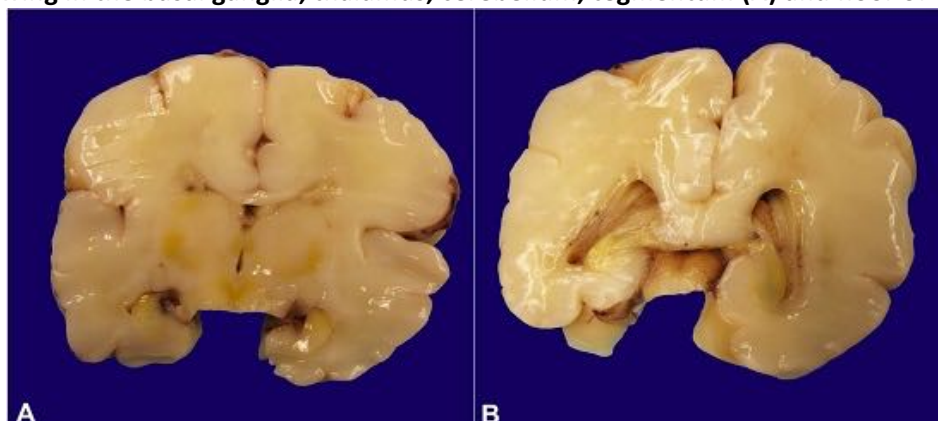
NEONATAL JAUNDICE & KERNICTERUS

- **Aetiology:**
 - Neonatal Jaundice due to:
 - ↑RBC Breakdown
 - & ↓Ability of Liver to Conjugate Bilirubin
- **Pathogenesis:**
 - Neonates have shorter-living RBCs (ie: ↑RBC Breakdown) & Immature Livers with limited Conjugation Capacity → → Hyperbilirubinemia_(Unconjugated)
 - Note: If Bilirubin levels are Extreme, it can collect in Brain Tissue
 - →→ Brain Damage & Deafness
- **Morphology:**
 - **Macro:**
 - “Kern”-“Icterus” = “Yellow Nuclei” = Yellowing of the Basal Ganglia & Thalamus
- **Clinical Features:**
 - Jaundice Within 1st week of life
 - Poor Feeding
 - Hypersomnolence
 - Absent Startle Reflex
 - Bulging Fontanel
- **Complications:**
 - High-Frequency Deafness
 - Mental Retardation, Speech Difficulties
 - Seizures
- **Treatment:**
 - Phototherapy
 - Exchange Transfusions
 - Vitamin K



A.D.A.M. Medical Encyclopedia [Internet]. <https://medlineplus.gov/ency/imagepages/19875.htm>

Extensive yellowing in the basal ganglia, thalamus, cerebellum, tegmentum (A) and floor of the ventricles (B)



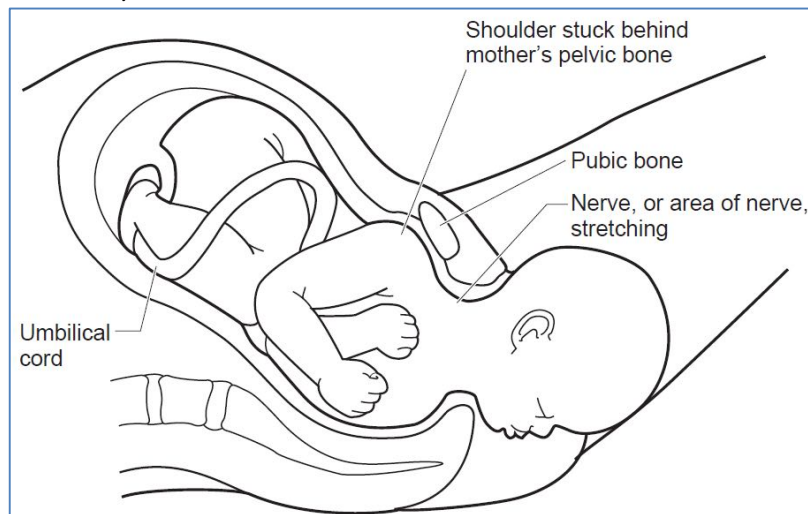
Kernicterus; Autopsy and case reports, vol. 9, no. 1, e2018057, 2019, DOI: <https://doi.org/10.4322/acr.2018.057>

COMPLICATIONS OF LABOUR AND DELIVERY

COMPLICATIONS OF LABOUR AND DELIVERY

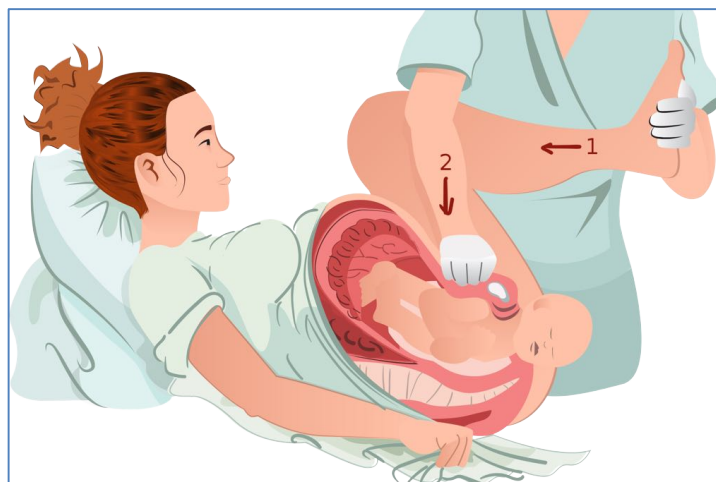
SHOULDER DYSTOCIA:

- **What is it?**
 - Foetal shoulder impacted above pubic symphysis after foetal head has been delivered
- **Aetiology:**
 - Occurs when width of shoulders is greater than the biparietal diameter of the head
 - **Risk Factors:**
 - Maternal Obesity / Diabetes
 - Prolonged gestation
 - Macrosomia
 - Prolonged 2nd stage of labour
- **Presentation:**
 - **Turtle sign:** Head is delivered but the retracts against the pubic symphysis
 - **Complications:**
 - Foetal hypoxic encephalopathy
 - Brachial plexus injury
 - Fractures (Clavicle/humerus/c-spine)
 - Death
 - PPH
 - Uterine rupture
- **Management:**
 - Set of manoeuvres designed to displace anterior shoulder from behind pubic symphysis
 - (Other options include: Cleidotomy, Zavanelli manoeuvre, or symphysiotomy)
- **Prognosis:**
 - 1% risk of disability for infant



<https://www.rcog.org.uk/en/patients/patient-leaflets/shoulder-dystocia/>

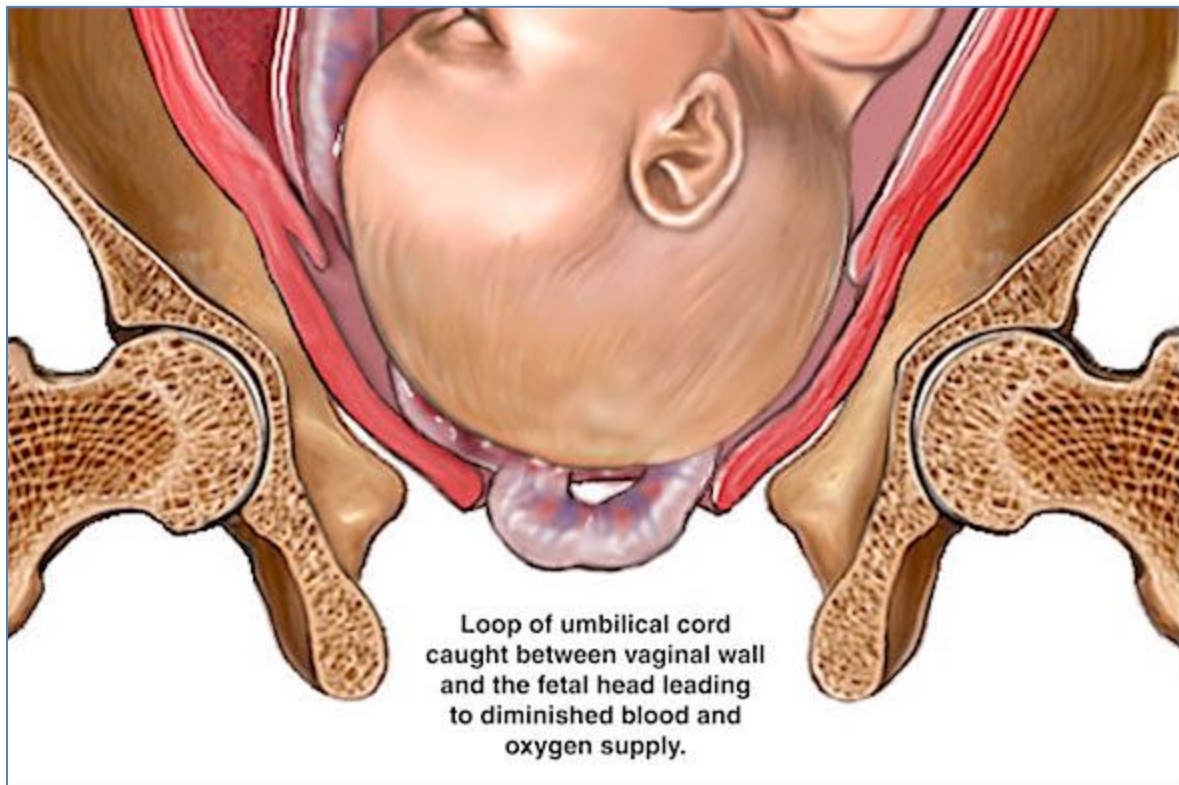
McRobert's Manoeuvre:



geraldbaeck, CC0, via Wikimedia Commons

UMBILICAL CORD PROLAPSE:

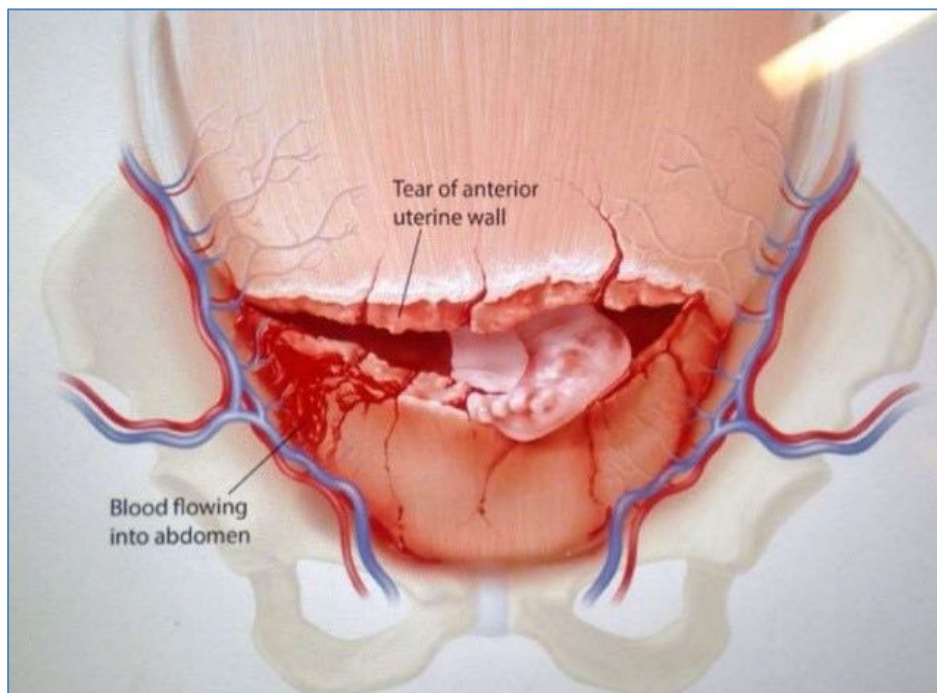
- **What is it?**
 - Descent of the cord to the level of the presenting part
 - → Cord compression between presenting part and the pelvis
- **Risk Factors:**
 - Prematurity
 - PROM
 - Foetal malpresentation
 - Low-lying placenta
 - Polyhydramnios
 - Multiple gestation
- **Presentation:**
 - Visible or palpable cord
 - Foetal heart rate changes (decelerations, bradycardia)
- **Management:**
 - Emergency caesarean section
 - Supplemental oxygen
 - Manually decompress cord by displacing presenting part via pelvic exam until caesarean
 - Position mother on all fours
- **Prognosis:**
 - May lead to foetal demise



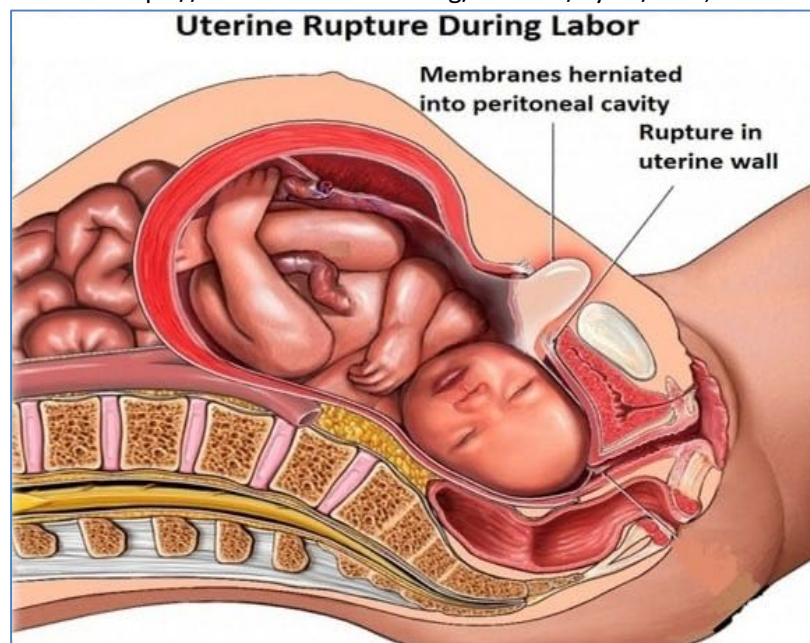
<https://www.birthinjuryhelpcenter.org/umbilical-cord-prolapse.html>

UTERINE RUPTURE:

- **Aetiology:**
 - 40% - Previous uterine scar ruptures
 - Also, hyperstimulation with oxytocin / grand multiparity
- **Presentation:**
 - Prolonged Foetal Bradycardia
 - Acute, lower abdominal pain
 - Vaginal bleeding
 - Intra-abdominal haemorrhage
 - Sudden loss of foetal descent
- **Management:**
 - Immediate delivery of foetus is required for survival
 - May require hysterectomy
 - Treat shock/hypovolaemia
- **Prognosis:**
 - 1-10% maternal mortality (haemorrhage, shock, DIC, amniotic fluid embolus)
 - Up to 50% foetal mortality



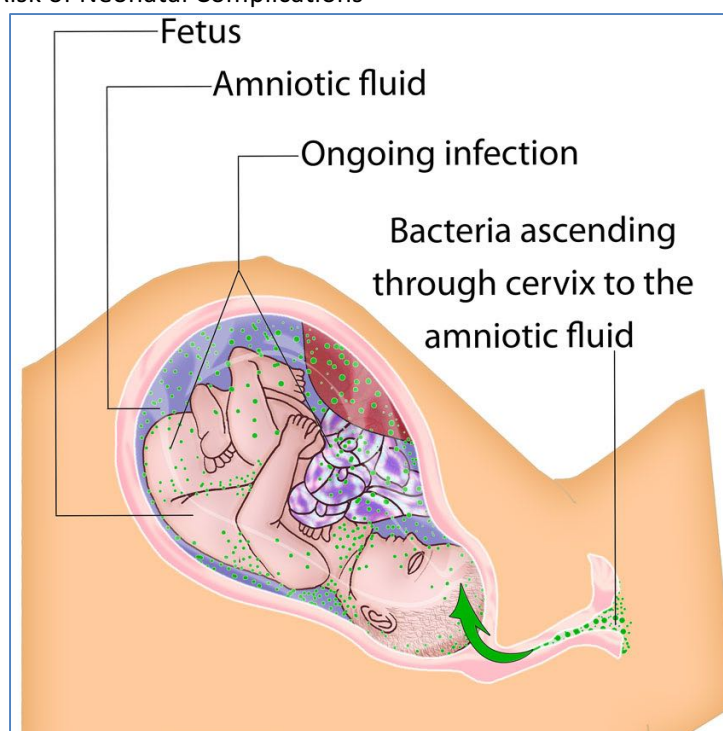
Khcnrc01, CC BY-SA 4.0 <<https://creativecommons.org/licenses/by-sa/4.0/>>, via Wikimedia Commons



Unattributable

CHORIOAMNIONITIS:

- **Aetiology:**
 - Placental Infection
 - **Risk Factors:**
 - Premature Birth
 - PPROM
 - PROM
 - Prolonged Labour
- **Pathogenesis:**
 - Infection & inflammation of the Chorionic Membrane & Villi due to:
 - Ascending Infection from Vagina (Vaginal Flora, Candida, etc)
 - Blood-Spread from Systemic Infection (HSV, Syphilis, Toxoplasmosis, Rubella, CMV)
- **Morphology:**
 - **Macro:**
 - May have Abscess Formation
 - **Micro:**
 - Inflammation of Chorionic Plate (WBCs)
 - Vasculitis of Umbilical Vessels
 - Infarctions
- **Clinical Features:**
 - **Maternal Symptoms:**
 - Fever
 - Uterine Tenderness
 - ****Neonatal Complications – (TORCHS Syndrome: Toxoplasmosis, Rubella, CMV, Herpes, Syphilis):**
 - Neonatal Sepsis
 - Neonatal Asphyxia
 - Microcephaly
 - Brain Damage/Hearing Impairment
 - Neonatal Organomegaly
 - Miscarriage/Death
- **Treatment:**
 - **IV Antibiotics (Ampicillin + Gentamicin) + (Clindamycin)**
 - **+ Induction of Labour**
- **Prognosis:**
 - Low maternal mortality if treated
 - Significant Risk of Neonatal Complications



OPERATIVE OBSTETRICS

OPERATIVE OBSTETRICS

What is an 'Operative' Vaginal Delivery?

- = A delivery via forceps or vacuum extraction

What are the Indications?

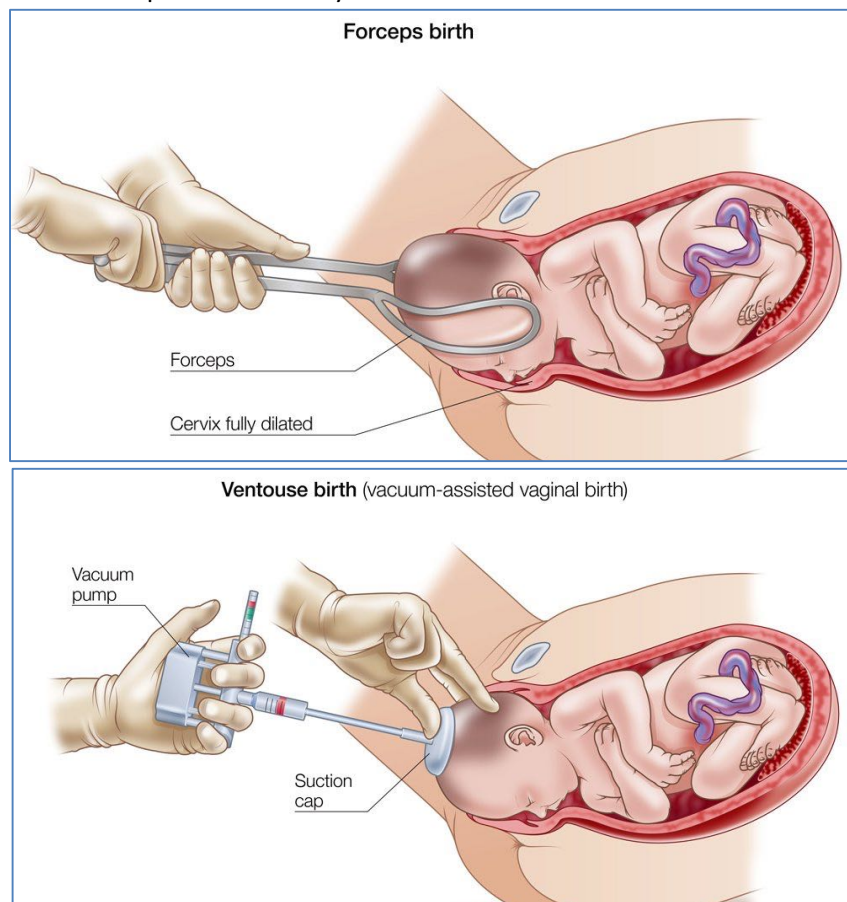
- **Foetal:**
 - o Atypical / abnormal foetal heart rate
 - o Foetal compromise
 - o Prolonged second stage labour
- **Maternal:**
 - o If there is a contraindication to maternal voluntary effort (Eg: Cerebrovascular disease/IHD/aneurysms etc)
 - o Maternal Exhaustion / sedation

Contraindications:

- Non-vertex cephalic presentations (Eg: Brow or face presentations)
- Non-engaged head
- Incompletely dilated cervix

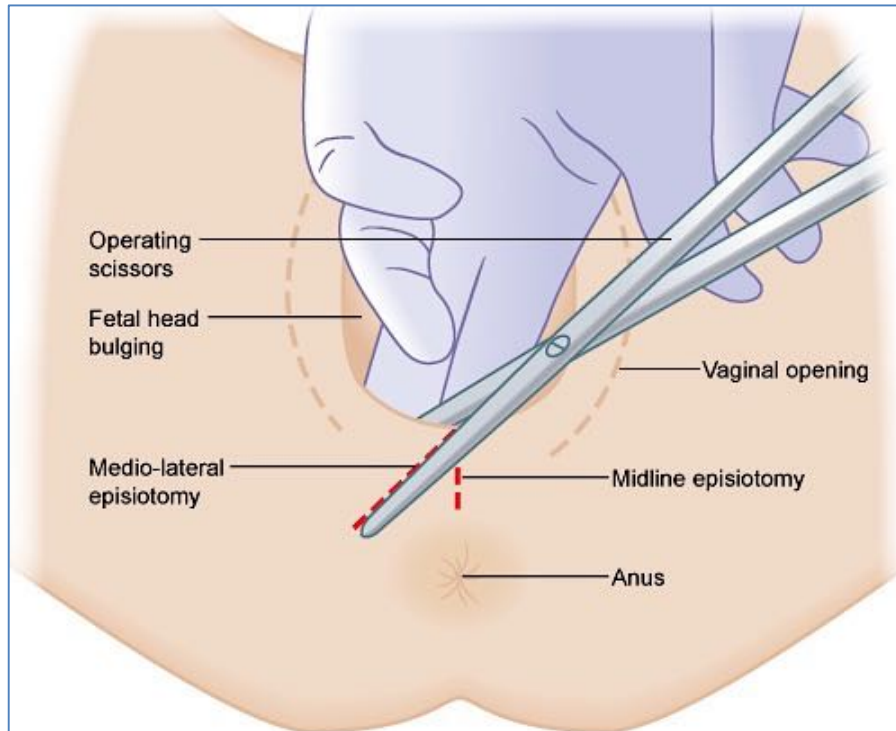
Pros & Cons:

- **Forceps:**
 - o + Higher success rate
 - o + Reduced foetal morbidity
 - o – Higher rate of maternal injury
- **Vacuum:**
 - o + easier
 - o + requires less anaesthesia
 - o + reduced rate of maternal injury
 - o – Maternal effort required
 - o – Can't be used in preterm delivery



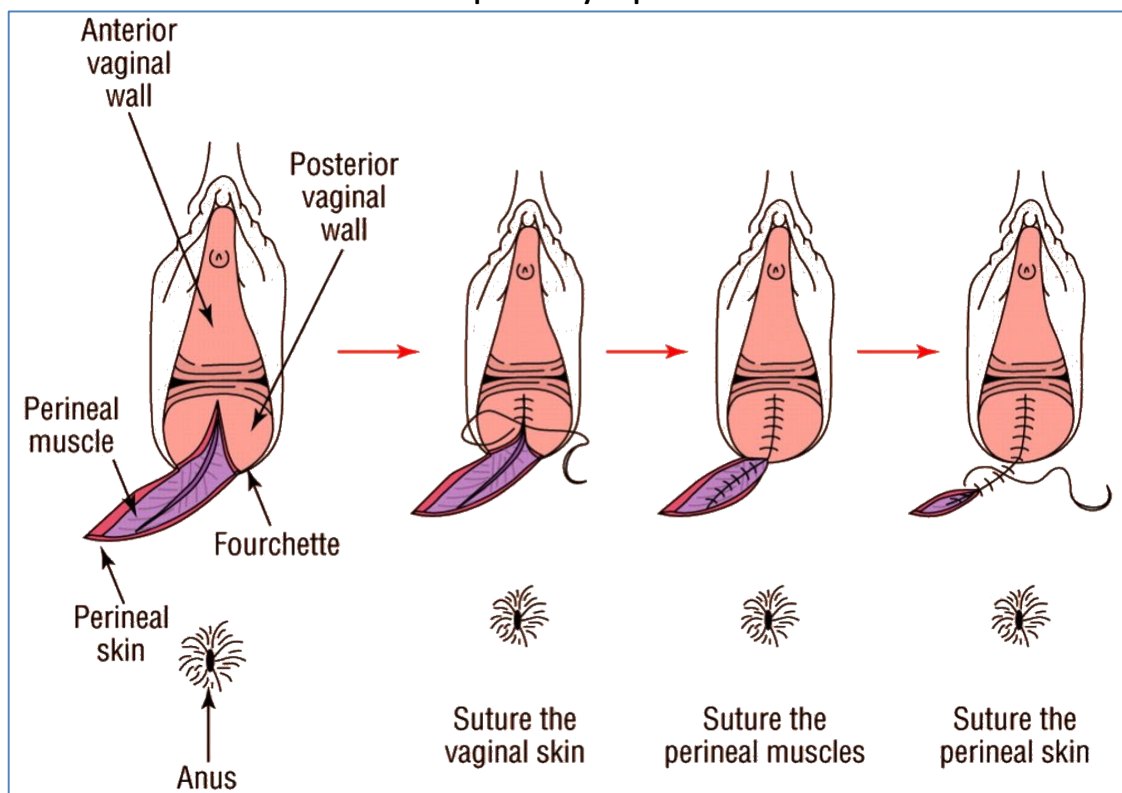
EPISIOTOMY:

- **What is it?**
 - Incision to the perineum at the time of delivery
- **Indications:**
 - Goal is to relieve obstruction of the perineum
 - Or to expedite delivery (Eg: Foetal distress)
- **Complications:**
 - Infection
 - Haematoma
 - Rectovaginal fistula formation
 - Incontinence



<https://elearning.rcog.org.uk//easi-resource/vacuum-extraction/delivery/assessing-need-episiotomy>

Episiotomy Repair

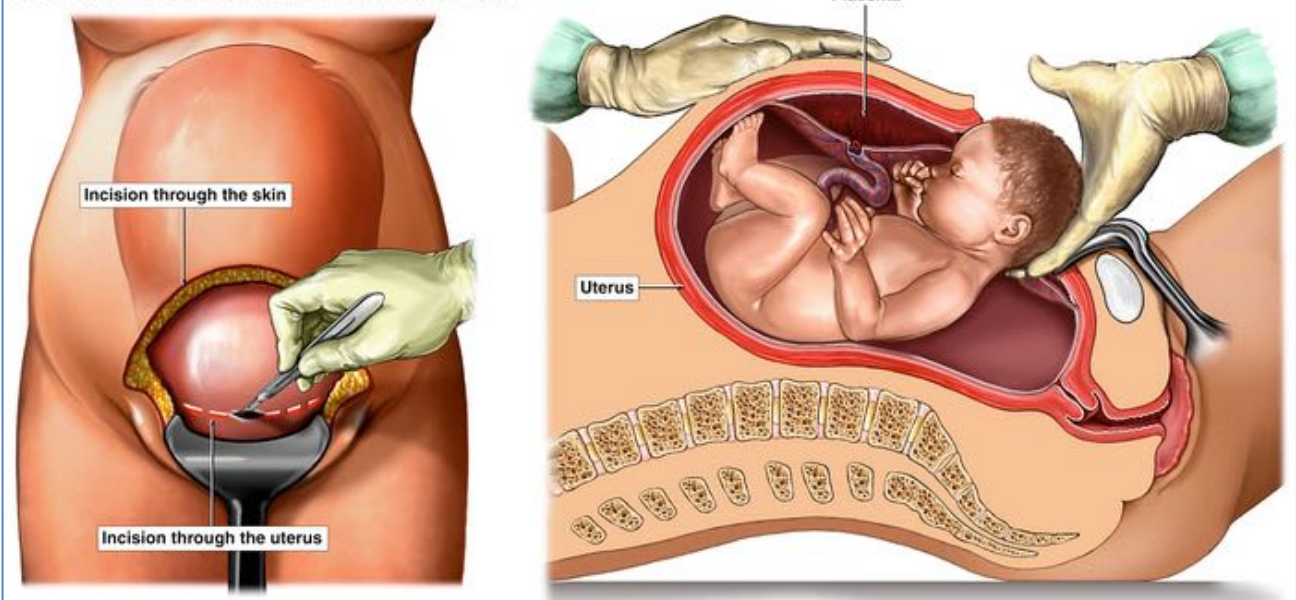


Adapted from: Geoffrey Chamberlain, Philip Steer; DOI: 10.1136/bmj.318.7193.1260

CAESAREAN DELIVERY:

- **What is it?**
 - Percutaneous delivery of a foetus via surgical means through abdominal wall
 - Accounts for 20-25% of births
- **Indications:**
 - Maternal obstruction
 - Active maternal herpes on vulva
 - Invasive cervical cancer
 - Previous uterine surgery (Including previous caesarean)
 - Eclampsia
 - HELLP syndrome
 - Failure to progress
 - Placental abruption/previa
 - Vasa previa
 - Malpresentation of the foetus
 - Cord prolapse
- **Complications:**
 - Anaesthetic risks
 - Haemorrhage
 - Infection (Including endometritis) – Covered with prophylactic antibiotics
 - Injury to collateral structures
 - DVT/PE
 - Increased recovery time

Incisions for caesarean section



Source: <https://www.pregnancybirthbaby.org.au/caesarean>

PUERPERAL COMPLICATIONS

PUERPERAL COMPLICATIONS

Puerperium: (4-6 weeks post-partum)

- Mother's body readjusts to the loss of the foeto-placental unit
- Reversal of numerous pregnancy-induced anatomical and physiologic changes

POST PARTUM HAEMORRHAGE (PPH):

- Definitions:

- **PPH = >500mls Blood Loss**
- **Primary Vs Secondary:**
 - Primary PPH = <24hrs during/after labour
 - Secondary PPH = <6-12wks after labour
- **Minor Vs Major:**
 - Minor PPH: 500-1000mL Blood Loss (5-15% Prevalence)
 - Major PPH: >1000mL Blood Loss (1-3% Prevalence)

- Importance:

- A major cause of Mortality:
 - 1:100000 women in Australia
 - 1:1000 women in Developing Countries

- Pathophysiology of PPH:

- **(Normally Haemostasis is Achieved by 2 things):**
 - 1: Myometrial Contraction → Constricts the Placenta Bed
 - 2: Normal ↑ in Thrombin during Pregnancy → Hypercoagulable State
- **PPH is Due to the "4x 'T's":**
 - **Tone** (Ie: Atonic Uterus/Insufficient Contraction)
 - **Trauma** (Ie: Uterine Tear/Perineal Tear/Vaginal Tear/Instrumental)
 - **Tissue** (Ie: RPOC – Retained Products preventing uterine contraction)
 - **Thrombin** (Ie: Bleeding Disorders)

- General Management:

- **ABC's + Call for assistance**
- **2x large bore cannulae → Stat crystalloids**
- **FBC + Coags**
- **Blood group and x-match**
- **Treat underlying cause**
- **Catheter to monitor urine output**

- Medical Therapy:

- **Oxytocin IV**
- **Ergotamine**
- **Carboprost (synthetic prostaglandin analogue)**
- **Misoprostol**
- **Tranexamic Acid**

- Complications:

- Hypovolaemia
- Organ Failure (Renal/Hepatic/etc)
- Postpartum Pituitary Failure (Aka: "Sheehan's Syndrome")
- Shock
- Death
- ARDS (in this case due to acute haemorrhagic anaemia)
- DIC

- Notes:

- **Uterotonic agents** (Eg: Oxytocics – Syntocinon/Syntometrin/Ergometrin) **in the 3rd stage of labour** enhances haemostasis via uterine contraction → **↓Risk by 2x fold**
- **"An intact, empty and contracted uterus WILL NOT BLEED"**
- **1x Resuscitation should be enough; if you need to do it twice, TAKE HER TO THEATRE!!**

PRIMARY PPH:

(From beginning of labour to within 24hrs after)

Risk Factors:

- **Intrapartum:**
 - Prolonged Labour (→ Tired Uterus)
 - Trauma (Instrumental/Tears)
 - Caesarean Section
 - Tocolytics during labour (→ ↓ Contractility of Uterus)
 - Dystocia (→ Prolonged Labour / ↑ Risk of Tearing)
- **Postpartum:**
 - Previous Hx of PPH
 - Bleeding Disorders (Including Anticoagulation)
 - Multiparity (→ Stretched Uterus)
 - Including Multiple Pregnancy (Eg: Twins/Triplets/etc)
 - Including Polyhydramnios
 - Including Macrosomia
 - Chorioamnionitis
 - Placental Abnormalities
 - Accreta (Abnormally Deep Placenta – Into Myometrium; 1/500 pregnancies)
 - Praevia (Placenta Close/Covering Cervical Canal)
 - Placental Abruption
 - RPOC - Retained Products of Conception

Prevention & Avoiding Complications:

- Detect Risk Factors
- IV Access
- G&H
- X-Match
- Syntocinon @ the ready
- Active Mx of 3rd Stage Labour (Ie: Oxytocics - Syntocinon/Syntometrin/Ergometrin – Usually mixed with Hartmann's Solution)

Management:

- **Minor PPH (500-1000mLs) (5-15% Incidence)**
 - Assess
 - 1x Large Bore IVC
 - Resuscitate with Fluids (Crystalloids)
 - Call for help (Registrar/Consultant)
 - Bloods: Group & Hold, X-Match, Rhesus status, FBC (Baseline Hb)
 - Give Syntocinon
 - Massage Fundus (of Uterus) → Expels Clots/RPOCs
 - Manual Removal of RPOCs
 - Bimanual Compression if bad
 - +Treat Cause
- **Major PPH (>1000mLs) (1-3% Incidence)**
 - All of Above; PLUS:
 - Call Operating Theatre
 - Call Anaesthetist
 - Additional Large Bore IVC
 - IDC – Monitor Urine Output
 - Additional Oxytocics (Ie: Syntocinon/Syntometrin/Ergometrin)
 - Blood Transfusion/s:
 - Packed Red Cells
 - FFP
 - Cryoprecipitate
 - + Treat Cause

Goals of Treatments:

- **1: Stop Bleeding:**
 - **Tone**
 - **Oxytotics:** (Syntocinon/Syntometrin/Ergometrin)
 - **Prostaglandins:** (Misoprostol, PGF2a Injection)
 - **Bimanual Compression**
 - **Theatre:**
 - Balloon Tamponade
 - “B-Lynch Suture”
 - Uterine Artery Ligation
 - Hysterectomy
 - **Tissue** (Manual/Surgical Removal of Incomplete RPOCs)
 - **Trauma** (Repair tears/Uterine Ruptures/Cervical Rupture/etc)
 - **Thrombin** (Give FFP infusion)
- **2: Transfuse (Usually by Anaesthetist):**
 - Packed RBCs (4-6 Units)
 - FFP (4 Units [per 6units of RBCs])
 - Platelets
 - Cryoprecipitate/Recombinant Factor-VII
- **3: Close Monitoring in ICU for Organ Failure:**
 - BP
 - Urine Output

SECONDARY PPH

(>24hrs after labour → 6-12mths later)

Causes:

- Infection
- RPOC
- Gestational Trophoblastic Diseases (Rare)
 - Eg: Molar Pregnancy
 - Eg: Invasive Trophoblastic Disease
 - Eg: Choriocarcinoma
 - Eg: Placental Site Tumour

Presentation:

- Typically slower bleed (A Trickle)

Management:

- Curette for RPOC
- Antibiotics
- B-HCG (check for molar pregnancy)
- Pelvic USS

POST-PARTUM HAEMORRHAGE

1^o Post-Partum Haemorrhage

>500mls of blood in first 24 hours post-delivery

TONE (very common): uterine atony +/- distended bladder

TISSUE (common): retained placenta or clots

TRAUMA (sometimes): lacerations of uterus, cervix or vagina

THROMBIN (rare): pre-existing or acquired coagulopathy

ED MANAGEMENT

Call for help

ABC resuscitation

Large bore IV access x2

Bloods + Group and Cross-Match

Low threshold for major haemorrhage protocol

Fluid/blood product resuscitation

Examine placenta for missing parts

Abdominal Massage

To stimulate contraction

Medications

Oxytocin (Syntocinon) 10IU Slow IV

Onset by 2mins and peak effect by 5mins

Feel for uterine change when massaging and/or reduced PV bleeding

There is less duration of action if retained placental part

and/or

Ergometrine 0.5mg IM

(both uterotonics kept in fridge in resus)

Tranexamic Acid 1-2g/hourly IV if above failed or thought due to trauma

Ensure bladder is empty

Catheterise

Temporising Measures

If drugs fail or not available

INTERNAL BIMANUAL COMPRESSION

Insert gloved fist into vagina and push up against uterus

Press other hand on uterine fundus and compress between hands

Maintain for at least 8-10mins

EXTERNAL AORTIC COMPRESSION

Push hard with fist onto abdominal aorta (upper abd + left of midline)

Successful compression is when femoral pulses no longer palpable

RETAINED PLACENTA:

- **What is it?**
 - Placenta still not delivered by 30mins post-partum
- **Aetiology:**
 - Placenta may have abnormal implantation:
 - Eg: Placenta accreta
 - Eg: Placenta increta
 - Eg: Placenta percreta
- **Risk Factors:**
 - Placenta previa
 - Prior caesarean
 - Uterine infection
- **Clinical Features:**
 - Increased Risk of PPH
 - Increased risk of infection
- **Investigations:**
 - Uterus exploration
- **Management:**
 - 2x large bore cannulae
 - Blood group and X-match
 - 'Brant manoeuvre' (firm umbilical traction + suprapubic pressure to avoid uterine inversion)
 - Oxytocin into umbilical vein
 - Manual removal if otherwise unsuccessful
 - D&C if required

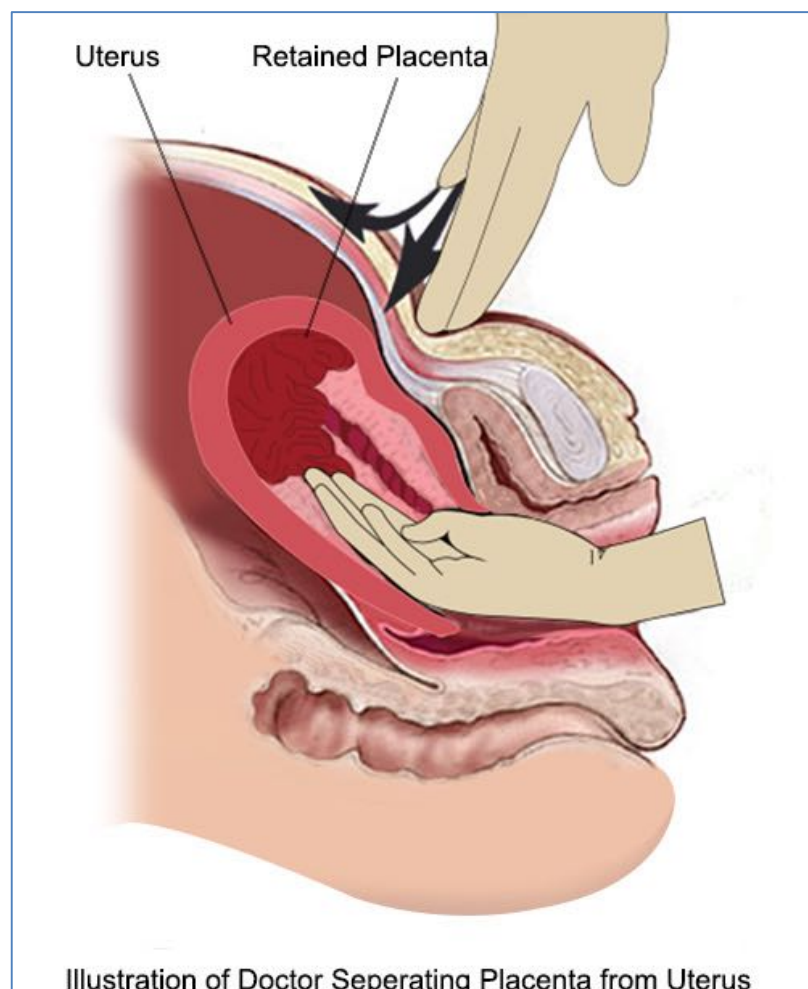


Illustration of Doctor Separating Placenta from Uterus

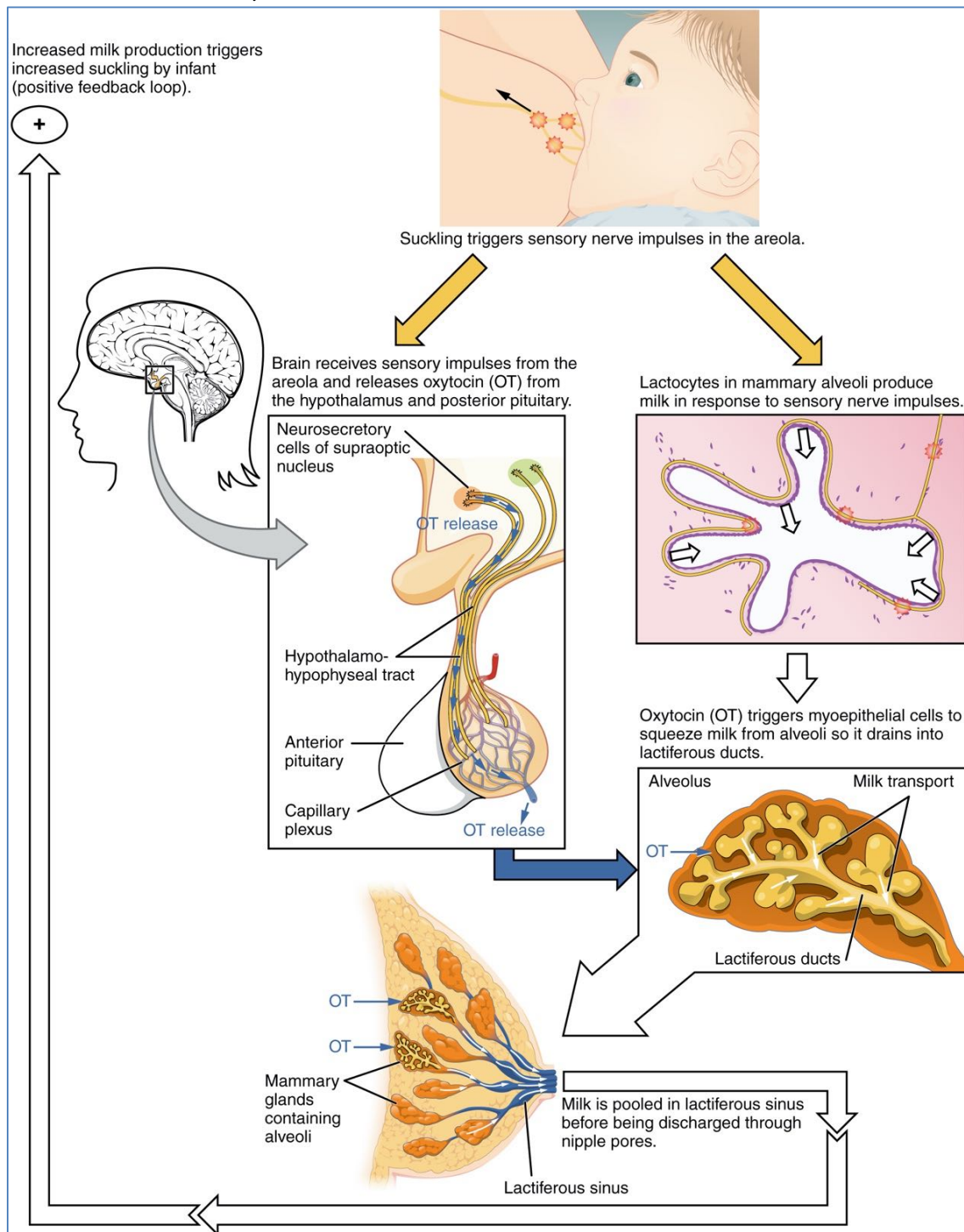
<https://www.aafp.org/aafp/2017/0401/p442.html>

POSTPARTUM CARE

POSTPARTUM CARE

Breast Feeding

- **Lactation** = production of milk by mammary glands
- During late pregnancy, high levels of estrogens, progesterone & lactogen → stimulates hypothalamus – secrete **prolactin-releasing hormone (PRH)** → Anterior Pituitary secretes **prolactin**
- **For first 3 days** – colostrum is secreted (less lactose than milk & minimal fat – but contains more protein, Vit-A & minerals) – Rich in IgA antibodies
- **After first 3 days** – true milk production begins
 - **Milk production depends on mechanical stimulation of nipples – sucking infant**
 - Stimulates hypothalamus → secretes **PRH** → burst of **prolactin** from Anterior Pituitary → stimulates milk production for the next feeding
 - Also stimulates hypothalamic → posterior pituitary release of **oxytocin (OT)** – causes the **let-down reflex**, the actual **ejection of milk** from alveoli in both mammary glands
 - Oxytocin also causes uterus to contract → to return to normal size



ACUTE MASTITIS:

- **Aetiology:**
 - Acute Breast Infection (Typically Bacterial Skin Flora – **Staph aureus**/Strep pyogenes)
- **Pathogenesis:**
 - **99.9% - Lactational (First few weeks post-partum)** → Crack in Nipple = Entry Point → Bacterial Infection (*Staph aureus*, Strep Pyogenes) → Inflammation + Pain
- **Morphology:**
 - Acute Inflammation, Swelling, Erythema & Pus
 - May → Single/Multiple Abscesses
- **Clinical Features:**
 - Initial Weeks Post-Partum
 - Unilateral, Painful, Erythematous, & Swollen Breast
 - + Fever, Inflammation, Flu-Like Symptoms
 - (+/- Pus Discharge)
 - (+/- Nipple Cracks/Fissures)
- **Diagnosis:**
 - **Clinical Diagnosis** (Hard, Tender, Red, Swollen Area of one breast + Fever in a Nursing Mother)
 - (Note: Distinguishable from Engorgement which is Bilateral)
 - (Note: Breast USS can distinguish between Mastitis & Abscess)
 - (+/- Breastmilk Culture if Infection is Severe/Hospital-Acquired)
- **Management:**
 - **Analgesia** (*Ibuprofen*)
 - **Cold Compresses**
 - **Improve Breast-Feeding Techniques** (Eg: Nipple Shields to stop Chapping)
 - (Note: Breastfeeding can continue during treatment)
 - **Antibiotics** (Anti-Staphylococcal; *Cephalexin*/*Dicloxacillin*/*Clindamycin*)



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POSTNATAL MENSTRUATION & CONTRACEPTION:

- **Amenorrhoea while Breastfeeding:**
 - Breastfeeding can cause amenorrhoea for between 2-12mths (Highly variable)
 - (If not breastfeeding, periods usually return within a few weeks)
- **Postnatal Menstruation:**
 - The 1st period after a baby may be heavier and more uncomfortable than usual
 - Remember conception is still possible before this period, so contraception is advised
- **Contraception**
 - **Lactational Amenorrhoea Method:**
 - Oxytocin inhibits ovulation for the first few months
 - This is only effective (But not guaranteed) if:
 - Exclusively breastfeeding **day and night**
 - Haven't had the 1st period yet
 - Baby is less than six months old
 - **The Minipill (Progesterone Only OCP):**
 - Safe to use while breastfeeding
 - **Condoms:**
 - Safe to use while breastfeeding
 - **Postpartum IUD**
 - **Note:** The **Combined oral contraceptive pill should NOT be used when breastfeeding** because it decreases the milk supply

TABLE 3	Patient-centered immediate postpartum contraceptive counseling
GOAL: The goal of contraceptive counseling is to educate women about contraception, discuss and address current and future needs, and devise a contraceptive plan that best meets those needs.	
Step	Key points
Establish and maintain rapport	Use open-ended questions, active listening, demonstrate empathy, and encourage questions
Obtain clinical and social information	■ Medical history ■ Reproductive goals and timing desires ■ Past experiences with contraceptives and assessment of patient preference
Work interactively to select the most effective and appropriate contraceptive method	■ Present all contraceptive methods along with their advantages and disadvantages ■ Provide information on side effects and risks ■ Address specific concerns and misconceptions without judgment • Immediate postpartum IUD insertion Communicate the advantages and disadvantages of IPPI versus interval placement of an IUD • Discuss increased risk of IUD expulsion with IPPI of an IUD
Conduct a physical assessment, as needed	Consider any contraindications
Provide the contraceptive method and help establish a follow-up plan	■ Follow the protocol for immediate postpartum insertion • Insert device • Counsel on signs and symptoms of IUD expulsion • Develop a follow-up plan: - String-check visit in 2 weeks - Standard 4- to 8-week postpartum follow-up visit

Abbreviations: IUD = intrauterine device; IPPI = immediate postplacental insertion

THE 3 POSTNATAL MOOD DISORDERS:

1. "THE BABY BLUES" (MILD)

- a. Affects <80% of new mothers
- b. Typical onset between Day 3-10 after birth
- c. Symptoms:
 - i. Tearfulness
 - ii. Anxiety
 - iii. mood swings
 - iv. irritability
- d. Treatment:
 - i. "The Baby Blues" are transient and pass with supportive therapy

2. "POSTNATAL DEPRESSION" (MODERATE)

- a. Affects 15% of women and 10% of men
- b. Onset anywhere from 24 hours-several months after delivery
- c. Symptoms:
 - i. Insomnia
 - ii. Anorexia
 - iii. Crying
 - iv. Acopia with daily tasks
 - v. Exhaustion
 - vi. Irritability
 - vii. Anxiety
 - viii. Fear of social contact
 - ix. Fear of being alone
 - x. Guilt
 - xi. Low confidence
 - xii. Suicidal thoughts
 - xiii. **"There is no joy in anything anymore"**
- d. Treatment:
 - i. **Emotional support from family and friends**
 - ii. Antidepressants are also effective

3. "POSTNATAL PSYCHOSIS" (SEVERE)

- a. Affects 1 in 500 mothers
- b. Onset within the 1st month of delivery
- c. Symptoms:
 - i. The mother may be unaware she is ill (Due to psychosis)
 - ii. Severe mood disturbance (Manic and/or Depressive)
 - iii. Thought disturbance (either processing or bizarre thoughts)
 - iv. Insomnia
 - v. Inappropriate responses to the baby
 - vi. Can be LIFE THREATENING for both mum & bub if undiagnosed
- d. Treatment:
 - i. **Requires hospitalisation**
 - ii. Anti-psychotics and/or Antidepressants

POSTNATAL/POSTPARTUM DEPRESSION

General Overview

- = **Non-psychotic depression occurring within 4-wks following delivery**
- **Typically lasts 2-6mths**
- **If severe, can lead to aversion to baby, suicidal and infanticidal ideation**
- Affects 12-15% of mothers
- <50% recurrence rate

Presentation:

- Anxiety/Depression
- Referral from external source (Midwives, community health)
- Inability to cope
- Husband/family member presents with concerns
- Somatic Symptoms
- Issues with children
- Irritability & Tearfulness
- Avoiding personal discussion
- Denial
- Delayed attachment
- Negative feelings to infant

Risk Factors:

- Minority groups
- Lower Socio-economic status
- Younger age
- Absence of partner
- Medical complications
- Marital problems
- History of abuse
- Not breast-feeding
- No job to return to
- Problematic births
- Reluctance to seek help

Protective Factors:

- Optimism & Self esteem
- Higher education
- Good SES
- Strong relationship with partner

Screening:

- **K10 Score** (previously discussed)
- **EPDS - Edinburgh postnatal depression scale** (10qs, 5mins, responses graded)

Management:

- **Psychotherapy** (Eg: CBT)
- **Pharmacological** (Eg: SSRI's)
- **If severe/psychotic, consider ECT**

Effect on Infant:

- Insecure infant – lack of trust, poor interaction with caregiver
- Attachment issues – discipline, behaviour & aggression problems
- Infant withdrawn, passive
- Slow to reach milestones
- High risk of mental health issues in child

Other Postpartum Distresses:

- **Post-partum Anxiety**
- **Postpartum OCD**
- **Postpartum Psychosis (Hallucinations & Delusions)**
- **Baby Blues**



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